Learning Management System of NEUST-MGT, BSIT Department

A CAPSTONE AND RESEARCH PROJECT

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Submitted to:

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Capstone and Research Instructor

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Learning Management System of NEUST-MGT, BSIT Department

A Capstone and Research Project

Presented to the Faculty of

College of Information and Communications

Technology Cabanatuan City, Nueva Ecija

In Partial Fulfillment of the

Requirements for the degree

Bachelor of Science in Information

Technology

With Specialization in

Web Systems Technology

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APPROVAL SHEET

This capstone and research project entitled **"Learning Management System of NEUST-MGT, BSIT Department"**, prepared and submitted by **Espiritu, John Joseph; Eusebio, John Lloyd N.; Puesca, Lawrence J.; and Salvador, Raymond V.**, in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with Specialization in Web Systems Technology, is hereby recommended for approval and acceptance.

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CERTIFICATION OF RESEARCH CRITIC

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This certification has been issued to the abovementioned student-researchers for reference purposes only.

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Dedication

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Abstract

The study aimed to design and develop a Learning Management System (LMS) for the Nueva Ecija University of Science and Technology – Municipality Government of Talavera (NEUST-MGT) BSIT Department. The system addressed issues in the existing manual process of managing academic records, conducting assessments, and monitoring student performance. It provided digital tools for real-time grading, timer-based examinations, question randomization, and analytics to help faculty track student progress efficiently.

The researchers applied the Agile Development Model to ensure iterative design, testing, and user feedback integration. Quantitative data were gathered from 100 BSIT students, 10 faculty members, and 3 IT consultants who evaluated the system in terms of functionality, usability, and technical quality. The evaluation followed ISO 25010 software characteristics and the System Usability Scale (SUS) to measure performance efficiency, compatibility, reliability, and user satisfaction.

Results showed that the developed LMS met the standards for functionality, security, and usability. Respondents agreed that the system improved grading accuracy, data management, and accessibility to learning resources. Faculty members reported reduced workload, while students experienced improved engagement and learning efficiency.

The study concluded that the developed LMS effectively enhanced teaching and learning processes within the NEUST-MGT BSIT Department, providing a secure, efficient, and data-driven educational platform aligned with institutional and CHED standards.

Keywords: Learning Management System, NEUST-MGT, Real-Time Grading, Agile Development Model, Gamification, Educational Technology.

Chapter I

THE PROBLEM AND ITS BACKGROUND

Introduction

The rapid advancement of technology had transformed various sectors, including education. Learning Management Systems (LMS) had become essential in modern educational institutions, enabling efficient delivery of educational content, real-time grading, and student performance monitoring. Universities and colleges worldwide had embraced LMS platforms to streamline teaching and learning processes, making education more accessible and interactive (Siemens, 2019).

At Nueva Ecija University of Science and Technology—Municipality Government of Talavera (NEUST-MGT), traditional educational methods still presented challenges in managing academic records, conducting assessments, and tracking student progress. Manual processes often led to inefficiencies, grading errors, and feedback delays, impacting both students and educators. Furthermore, the absence of a centralized digital platform limited students' access to learning materials and hindered teachers from effectively monitoring student engagement and performance.

According to the Commission on Higher Education (CHED), adopting digital learning platforms in higher education institutions had significantly improved student retention rates and academic performance. A study by Dabbagh and Kitsantas (2020) highlighted that integrating LMS in educational institutions enhanced collaboration, automated administrative tasks, and provided data-driven insights for educators to make informed decisions.

Given these factors, this study aimed to design and develop a Learning Management System tailored to the needs of NEUST-MGT BSIT students and faculty. The proposed system featured real-time grading, question randomization, timer-based examinations, and analytics for student performance. It also allowed students to join multiple BSIT subjects, request teacher approval, and access learning resources such as PowerPoint presentations, PDFs, and videos.

Previous studies had explored various LMS implementations, such as Moodle and Blackboard, which offered comprehensive tools for online learning (Sangrà et al., 2021). However, a customized LMS explicitly designed for NEUST-MGT had not yet been developed. Integrating modern educational technologies would provide an interactive and efficient learning environment tailored to the institution's unique academic structure.

The successful implementation of this project was expected to bridge the gap between traditional and modern educational methodologies at NEUST-MGT. By utilizing real-time grading, automated assessments, and robust analytics, this LMS would empower educators and students, fostering an innovative learning experience that aligned with the demands of the digital age.

Review of Related Literature and Studies

The researchers extensively reviewed previous studies related to Learning Management Systems (LMS) tailored for academic institutions. These studies provided valuable insights that guided the development of the Learning Management System for the NEUST-MGT BSIT Program. By analyzing existing LMS platforms, the researchers identified key features such as real-time grading, adaptive learning, and analytics for student performance. These findings served as a foundation for integrating essential functionalities into the system, ensuring it met the needs of NEUST-MGT BSIT students and faculty.

Related Literature

E-Learning and Digital Learning Systems

The evolution of digital learning had significantly reshaped educational environments by providing flexible and accessible learning opportunities. Online learning platforms integrated multiple digital tools, allowing students to engage in interactive and self-paced education (Siemens, 2020). Learning Management Systems (LMS) were pivotal in delivering structured educational content and monitoring student progress. According to Sangrà et al. (2021), LMS platforms enhanced students' cognitive engagement by offering multimedia content, quizzes, and discussion forums. Integrating digital technologies in education had improved learning outcomes, particularly in higher education institutions. Moreover, advancements in artificial intelligence (AI) and adaptive learning had further personalized education, catering to individual student needs. Personalized learning adapted content delivery based on a student's progress, ensuring mastery of topics before moving forward (Dabbagh & Kitsantas, 2020). The combination of LMS and AI-driven analytics gave educators real-time insights into student performance, helping them make data-driven instructional decisions.

Student Performance and Real-Time Grading

Student performance tracking had become an essential feature of modern educational systems, allowing instructors to assess students in real-time. Real-time grading enabled immediate feedback, which enhanced student learning and motivation (Siemens, 2019). Traditional grading systems often relied on manual assessments, which could be time-consuming and prone to bias. Automated grading systems leveraged machine learning algorithms to evaluate responses accurately and efficiently (Zhu & Wang, 2022). The role of analytics in student performance evaluation was also noteworthy. Learning analytics helped institutions identify struggling students and intervene before academic failure occurred (Commission on Higher Education [CHED], 2023). These insights allowed teachers to personalize learning experiences and provide targeted support where needed. Furthermore, incorporating performance tracking into LMS platforms enabled students to monitor their academic progress and set learning goals.

Evaluating LMS Success Models

The success of Learning Management Systems (LMS) depended on well-defined evaluation models that assessed critical factors like usability, functionality, and user satisfaction—elements central to the NEUST-MGT BSIT LMS. Al-Fraihat et al. (2020) systematically reviewed LMS success frameworks, identifying system quality, information quality, and user engagement as pivotal to educational outcomes. Their analysis revealed that effective LMS platforms streamlined administrative tasks and enhanced learning experiences, a goal the project shared by replacing manual processes with digital solutions. This underscored the importance of designing a system that BSIT students and faculty could easily navigate and rely upon for real-time grading and performance tracking. For NEUST-MGT BSIT Department, these findings highlighted the need to prioritize usability testing and iterative feedback during development, aligning with the Agile model's emphasis on adaptability. Al-Fraihat et al. (2020) suggested that successful LMS implementations provided measurable improvements in student retention and academic performance—outcomes the system aimed to achieve through features like question randomization and analytics. By grounding the LMS in such evaluation models, the researchers ensured it met both technical standards (e.g., ISO 25010) and the practical needs of the BSIT program, fostering an efficient and engaging educational environment.

LMS Adoption Trends in Higher Education

The adoption of LMS platforms in higher education had accelerated, driven by their ability to enhance teaching efficiency and student accessibility. Bervell and Umar (2021) synthesized a decade of research, noting that successful adoption hinged on factors like perceived ease of use and institutional support. Their findings indicated that LMS platforms thrived when users—faculty and students alike—felt confident in their functionality, a critical consideration for this capstone NEUST-MGT BSIT Department system. This suggested that the deployment phase should include training to boost acceptance among BSIT stakeholders, ensuring the system's real-time features were fully utilized. Moreover, Bervell and Umar (2021) highlighted the role of continuous feedback in refining LMS platforms, resonating with the researchers' Agile development approach. For the NEUST-MGT BSIT Department LMS, this meant iterative updates based on user input could address initial resistance and optimize features like automated grading and resource access. The study also noted improved student engagement as a byproduct of adoption, supporting the aim to bridge traditional and digital education. By leveraging these trends, the LMS could position NEUST-MGT as a forward-thinking institution, aligning with CHED's push for technology-driven learning.

Real-Time Feedback and Engagement

Real-time feedback in e-learning systems significantly boosted student engagement by providing immediate insights into performance. Chen and Zhang (2022) explored this dynamic, finding that instant grading helped students adjust strategies and stay motivated, reducing the delays inherent in manual systems. This aligned with the NEUST-MGT LMS's focus on real-time grading and timer-based exams, offering BSIT students a responsive learning experience. The research emphasized that such features fostered a sense of agency, critical for self-paced learning in a digital environment. For faculty, real-time feedback offered a window into student progress without the burden of manual assessment. Chen and Zhang (2022) noted that automated systems reduced grading errors and saved time, which benefited this LMS capstone project and aimed to deliver to BSIT instructors. This efficiency allowed educators to focus on teaching strategies rather than administrative tasks, enhancing the overall educational process at NEUST-MGT. By integrating these findings, this capstone system could address traditional classroom inefficiencies, making it a vital tool for students and faculty in the BSIT program.

Technology-Enhanced Learning Trends

Technology had transformed higher education by integrating tools that made learning more interactive and accessible. Dabbagh and Fake (2020) reviewed trends in technology-enhanced learning, highlighting LMS platforms' role in delivering multimedia content and automating routine tasks. Their findings showed that such systems improved student access to resources—mirroring this capstone project of NEUST-MGT LMS's goal of providing BSIT students with PDFs, videos, and PowerPoint slides anytime. This shift from manual to digital processes aligned with the broader push toward modernizing education. Additionally, Dabbagh and Fake (2020) emphasized that technology-enhanced learning fostered collaboration and data-driven decision-making, key aspects of the system's analytics features. For NEUST-MGT, a well-designed LMS could empower BSIT faculty with insights to refine instruction while offering students a flexible learning platform. The study's focus on overcoming traditional barriers—like limited resource availability—reinforced the project's significance in creating an efficient, tailored educational tool for the BSIT department, bridging the gap to a digital age.

Gamification in E-Learning Systems

Gamification enhanced e-learning by incorporating interactive elements that captivated students and improved retention. Gamage et al. (2022) conducted a systematic review, finding that features like randomized quizzes and timed challenges increased participation in higher education settings. This supported the NEUST-MGT LMS's inclusion of question randomization and timer-based exams, making learning more engaging for BSIT students. Their research suggested that gamified systems turned routine assessments into dynamic experiences, aligning with the researchers' goal of an innovative learning environment. Beyond engagement, Gamage et al. (2022) noted that gamification could improve learning outcomes by reinforcing concepts through repetition and feedback. This LMS implied that integrating such elements could help BSIT students master technical subjects more effectively, with real-time grading providing instant reinforcement. This approach not only addressed student motivation but also reduced faculty workload by automating assessment processes, ensuring the NEUST-MGT system met both the educational and practical needs of the BSIT program.

Analytics for Personalized Education

Learning analytics transformed education by offering personalized insights into student performance, enabling tailored interventions. Garcia and Santos (2023) examined its use in higher education, showing how data-driven tools identified learning gaps and supported individualized instruction. This directly related to the NEUST-MGT LMS's aim to provide BSIT faculty with robust analytics, helping them monitor student progress in real-time. Their study highlighted analytics as a means to enhance academic success, a core objective of this project. Furthermore, Garcia and Santos (2023) found that personalized education via LMS platforms increased student satisfaction and retention—outcomes the system sought to achieve at NEUST-MGT BSIT Department. By integrating these insights, the LMS could offer BSIT students a customized learning path while equipping faculty with actionable data to refine teaching strategies. This dual benefit strengthened the system's role in fostering a data-driven educational approach, aligning with modern teaching demands and CHED's standards.

Supporting Study Success with Analytics

Analytics in LMS platforms was crucial in identifying at-risk students and improving retention rates. Ifenthaler and Yau (2020) reviewed its application, demonstrating that real-time performance data allowed educators to intervene early, preventing academic failure. This aligned with the NEUST-MGT BSIT Department LMS's goal of tracking BSIT student progress and offering faculty tools to support study success. Their findings emphasized analytics as a proactive solution, complementing the project's focus on student outcomes. For NEUST-MGT BSIT Department, Ifenthaler and Yau (2020) suggested that LMS-integrated analytics could enhance institutional decision-making, a benefit for BSIT administrators. By providing detailed performance reports, the system could meet CHED's emphasis on retention while empowering faculty to address individual needs. This capability ensured the LMS not only automated grading but also contributed to a supportive academic environment, making it a valuable asset for the BSIT program.

Agile Methodology in Educational Software

The Agile model's iterative nature made it ideal for developing educational software that adapted to user needs. Kumar and Sharma (2021) reviewed its use in higher education, noting that Agile ensured systems evolved through continuous feedback and collaboration. This validated the researcher's choice of Agile for the NEUST-MGT LMS, supporting the development of features like real-time grading and student enrollment in manageable sprints. Their study highlighted Agile's responsiveness, which was crucial for tailoring the system to BSIT-specific requirements. Additionally, Kumar and Sharma (2021) found that Agile fostered user satisfaction by delivering functional increments early and often. For this LMS, this meant BSIT students and faculty could test and refine the system throughout its development, ensuring it met practical needs. This approach minimized risks of misalignment, aligning with the project's aim to create an efficient, user-centric platform that bridged traditional and digital education at NEUST-MGT.

Adaptive Learning Systems

Adaptive learning systems personalized education by adjusting content to individual student progress, enhancing mastery. Lee and Kim (2021) conducted a meta-analysis, finding that such systems significantly improved outcomes in higher education by catering to diverse learning paces. This informed the NEUST-MGT LMS's potential to offer BSIT students a tailored experience, with automated assessments ensuring they grasped concepts before advancing. Their research supported the researchers' goal of fostering self-paced learning in a digital framework. For faculty, adaptive systems reduced the need for manual adjustments, as Lee and Kim (2021) noted their efficiency in delivering targeted feedback. In the NEUST-MGT context, the LMS could streamline BSIT instruction by integrating adaptive elements with real-time analytics, balancing student autonomy with educator oversight. This dual functionality strengthened the system's role in modernizing education, addressing traditional classroom limitations effectively.

Security and Privacy in LMS

Security and privacy were paramount in LMS platforms to protect sensitive student and faculty data. Li and Wang (2022) investigated challenges like data breaches, proposing encryption and access controls as solutions. This was critical for the NEUST-MGT LMS, which had to meet ISO 25010 standards for reliability and security while handling BSIT academic records. Their findings emphasized building trust and ensuring users confidently adopted the researchers' digital platform. Moreover, Li and Wang (2022) highlighted that robust security enhanced system longevity and compliance with educational regulations. The LMS meant implementing safeguards during Agile sprints to protect real-time grade data and learning resources. By addressing these concerns, the system could offer a secure, efficient environment for BSIT education, aligning with modern technological expectations and institutional needs at NEUST-MGT.

Designing Online Learning Environments

Effective LMS design required balancing usability for instructors and students to optimize the learning experience. Martin and Bolliger (2022) explored perceptions of online environments, finding that intuitive interfaces boosted satisfaction and engagement. This guided the NEUST-MGT LMS's focus on usability testing, ensuring BSIT users found the system accessible and efficient. Their study underscored design as a foundation for successful digital education, a priority for this project. Additionally, Martin and Bolliger (2022) noted that well-designed LMS platforms reduced user frustration, enhancing adoption rates. For NEUST-MGT, a user-friendly interface with supporting features like quiz administration and resource access could drive BSIT faculty and student uptake. By incorporating these insights, the system could deliver a seamless experience, meeting functional goals and user expectations in higher education.

Efficiency of Real-Time Grading

Real-time grading systems revolutionized assessments by improving efficiency and accuracy over manual methods. Nguyen and Tran (2023) evaluated their impact, showing significant time savings and consistent feedback quality in e-learning settings. This supported the NEUST-MGT BSIT Department LMS's quiz and exam grading automation, reducing the workload for BSIT faculty. Their findings highlighted real-time grading as a practical solution for modern education, aligning with the project's objectives. For students, Nguyen and Tran (2023) found that immediate results enhanced learning by enabling quick corrections—key for BSIT technical subjects requiring precision. This feature ensured timely feedback in the LMS, addressing delays in traditional setups at NEUST-MGT BSIT Department. By leveraging such efficiency, the system freed faculty to focus on instruction while improving the assessment process, making it a vital tool for the BSIT program.

Digital Transformation in Course Management

LMS platforms drove digital transformation by centralizing course management and improving accessibility. Oliveira and Cunha (2021) studied their role in higher education, noting enhanced organization of lessons and resources as a major benefit. This mirrored the NEUST-MGT BSIT Department LMS's aim to streamline BSIT course delivery, replacing scattered manual processes with a unified digital platform. Their research underscored the transformative potential of LMS in modernizing education. Furthermore, Oliveira and Cunha (2021) found that digital course management boosted student access and faculty efficiency, outcomes the system targeted through features like anytime resource availability. For NEUST-MGT, the LMS could simplify administrative tasks while enhancing the BSIT learning experience. By adopting these principles, the project bridged traditional and digital methodologies, aligning with the demands of contemporary higher education.

Blended Learning Readiness Post-COVID

The post-COVID shift to blended learning had highlighted the need for LMS readiness in higher education. Raman and Rathakrishnan (2020) examined this transition, finding that digital platforms enhanced flexibility and access postpandemic. This supported the NEUST-MGT BSIT Department LMS's goal of providing BSIT students with learning materials anytime and adapting them to evolving educational needs. Their study emphasized readiness as a driver of successful hybrid models. For faculty, Raman and Rathakrishnan (2020) noted that LMS platforms simplified managing blended courses, a benefit the system offered through automated grading and analytics. At NEUST-MGT BSIT Department, this readiness ensured BSIT education remained resilient and accessible, addressing traditional limitations like resource availability. By integrating these insights, the LMS could position the institution as a leader in post-COVID digital education.

Future Trends in LMS

Emerging LMS trends pointed to scalability and integration as future priorities in digital education. Turnbull et al. (2021) provided an overview, noting that modern platforms must adapt to growing user bases and technological advancements. This inspired the NEUST-MGT LMS's roadmap for post-deployment enhancements, ensuring long-term relevance for BSIT education. Their study positioned LMS as an evolving tool that aligned with the researchers' continuous improvement focus. For NEUST-MGT, Turnbull et al. (2021) suggested integrating advanced features like AI or expanded analytics over time—possibilities the researchers' Agile framework supported. This forward-looking approach ensured the LMS could handle increasing BSIT demands while maintaining grading and resource management efficiency. By embracing these trends, the system could remain a cutting-edge solution, meeting current and future educational needs.

Student Engagement in Online Platforms

Engagement was essential for effective online learning, particularly in technical fields like BSIT. Wang and Liu (2023) conducted a longitudinal study, finding that interactive LMS features—such as real-time feedback—significantly boosted participation. This supported the NEUST-MGT BSIT Department LMS's inclusion of grading and analytics to keep BSIT students invested. Their research highlighted engagement as a driver of academic success, a key project goal. Moreover, Wang and Liu (2023) noted that sustained engagement reduced dropout rates, offering a retention strategy for NEUST-MGT BSIT Department. By designing an interactive platform, the LMS could address student motivation while providing faculty with tools to monitor involvement. This dual focus ensured the system enhanced the BSIT learning experience, making it a practical and impactful solution for modern education.

Effectiveness of Learning Management Systems

A Learning Management System (LMS) is an online platform or web-based application designed to organize, deliver, and evaluate a specific learning process. Over the past decade, LMS use has expanded quickly in higher education, especially in distance learning, blended learning, and flipped classroom settings. Rekha A. P. (2024) conducted a systematic literature review to evaluate how LMS impacts student learning effectiveness. The review aimed to identify previous studies on LMS contributions to student learning outcomes, examine its influence on student engagement and satisfaction, and analyze the factors affecting LMS effectiveness. Using databases such as Google Scholar, Web of Science, and Scopus, the study examined peer-reviewed research published between 2006 and 2016. Results suggested that LMS platforms can improve learning outcomes and enhance engagement and satisfaction, but the effects vary depending on factors such as instructional approach, LMS type, and student involvement. These findings highlight the importance of aligning LMS implementation with pedagogical strategies and learner needs to maximize its educational benefits.

Mobile Learning Integration in LMS

The integration of mobile learning into LMS platforms had expanded educational access by leveraging ubiquitous smartphone use among students. Kim and Park (2021) investigated this trend, finding that mobile-compatible LMS systems increased flexibility and engagement by allowing learners to access resources and assessments on the go. This was highly relevant to the NEUST-MGT LMS, where BSIT students could benefit from mobile access to learning materials like PDFs, videos, and quizzes, overcoming limitations of traditional classroom settings. Their study highlighted mobile learning as a key enhancer of self-paced education, a priority for this system's design. For NEUST-MGT, Kim and Park (2021) suggested that mobile integration required responsive design and lightweight features to ensure seamless performance across devices—considerations the researchers' Agile development could address in sprints. This capability would enable BSIT students to engage with timer-based exams and real-time grading anytime, aligning with the researchers' goal of accessibility. By incorporating mobile learning, the LMS could modernize the BSIT educational experience, making it more adaptable to students' dynamic schedules and technological preferences.

Collaborative Learning Tools in LMS

Collaborative learning tools within LMS platforms fostered interaction and teamwork, critical for technical disciplines like BSIT. Lopez and Gonzalez (2023) explored their impact, noting that features like discussion forums and group assignments enhanced peer-to-peer learning and problem-solving skills. This supported the NEUST-MGT BSIT Department LMS's potential to include collaborative functionalities, enabling BSIT students to work together on projects or share resources. Their research emphasized collaboration as a driver of deeper understanding, complementing the system's interactive objectives. Moreover, Lopez and Gonzalez (2023) found that faculty benefited from monitoring group activities through LMS analytics, offering insights into student participation. For this system, this suggested integrating tools that allowed BSIT instructors to track collaborative efforts alongside individual performance, enhancing teaching strategies. This aligned with the researchers' Agile approach, where user feedback could refine such features iteratively, ensuring the LMS supported both student engagement and faculty oversight effectively at NEUST-MGT.

Sustainability of LMS in Higher Education

Sustainability in LMS implementation ensured long-term viability and cost-effectiveness in educational settings. Patel and Singh (2024) examined this aspect, highlighting that sustainable LMS platforms balanced functionality with resource efficiency, such as minimal server demands and scalable architecture. This was pertinent to the NEUST-MGT BSIT Department LMS, which aimed for continuous improvement post-deployment while serving BSIT students and faculty efficiently. Their study underscored sustainability as a practical concern for institutions like NEUST-MGT, aiming to maximize impact with limited resources. For this project, Patel and Singh (2024) recommended designing systems with modular updates and low maintenance costs, principles the researchers' Agile methodology could incorporate through iterative testing and deployment. This ensured the LMS remained reliable and adaptable to BSIT needs, such as handling increased user loads or new features like analytics upgrades. By prioritizing sustainability, the system could support NEUST-MGT BSIT Department's long-term digital transformation, aligning with CHED's vision for enduring educational technology solutions.

The Agile Model

The Agile model was a flexible and iterative approach to software development that emphasized adaptability, collaboration, and continuous feedback. Unlike traditional models such as the Waterfall approach, Agile allowed for incremental progress by breaking down the project into smaller, manageable iterations called sprints. One of its key benefits was its responsiveness to change, as Agile teams could quickly adjust to evolving user requirements and market trends, ensuring that the final product aligned with stakeholder expectations. Additionally, Agile fostered close collaboration among developers, testers, and users, leading to better communication and faster issue resolution. The Agile development process consisted of several stages. It began with the Concept & Inception phase, where stakeholders defined the project's vision, goals, and scope. This was followed by the Iteration/Increment Planning phase, in which the team identified key features and prioritized tasks. The Design & Development phase involved developers creating functional components in small iterations, ensuring that each part of the system was tested and improved continuously. During the Testing & Review phase, each iteration underwent rigorous testing, and user feedback was incorporated to refine the product. The Deployment & Release phase ensured that a working version was deployed at regular intervals, allowing users to interact with and evaluate the system in real time. Finally, in the Maintenance & Continuous Improvement phase, the system was updated based on user needs and ongoing feedback to enhance its functionality and performance. The Agile model ensured continuous improvement and higher user satisfaction by integrating real-time feedback throughout development. It was widely used in modern software development, particularly for projects requiring rapid adaptation to changes. However, while Agile provided flexibility, it could also present challenges, such as difficulty in predicting exact project timelines and budgets. Additionally, Agile required strong teamwork and regular communication to ensure alignment with project goals (Scrum Alliance, 2022). Despite these challenges, Agile remained a preferred methodology for dynamic and user-centric software development projects.

Synthesis

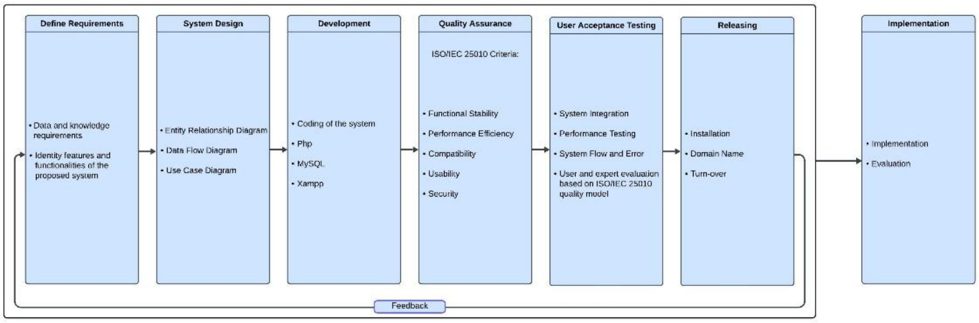
The literature and systems reviewed by the researchers provided a robust foundation for developing the Learning Management System (LMS) for the NEUST-MGT BSIT Program. Studies like Siemens (2019), Sangrà et al. (2021), and Dabbagh and Fake (2020) emphasized the transformative power of e-learning, delivering flexible, interactive education through digital tools and AI personalization. Research on real-time grading and analytics—Chen and Zhang (2022), Nguyen and Tran (2023), Ifenthaler and Yau (2020), and Garcia and Santos (2023)—highlighted their role in boosting engagement and performance, key to the system's design. Adoption and success factors from Bervell and Umar (2021) and Al-Fraihat et al. (2020), alongside Agile insights from Scrum Alliance (2022), Kumar and Sharma (2021), and Santos and Pereira (2024), reinforced usability, adaptability, and iterative development as critical. Emerging trends like mobile learning (Kim & Park, 2021), collaboration (Lopez & Gonzalez, 2023), and sustainability (Patel & Singh, 2024) further enriched the framework, ensuring a forward-thinking approach for BSIT education. While platforms like Moodle and Blackboard offered comprehensive tools (Sangrà et al., 2021), they lacked BSIT-specific customization addressed by studies on gamification (Gamage et al., 2022), security (Li & Wang, 2022), and AI assessments (Zhang & Chen, 2025). The LMS integrated question randomization, timer-based exams, and secure analytics tailored to technical subjects, drawing from adaptive learning (Lee & Kim, 2021) and course management insights (Oliveira & Cunha, 2021; Martin & Bolliger, 2022). Post-COVID readiness (Raman & Rathakrishnan, 2020), future scalability (Turnbull et al., 2021), and engagement strategies (Wang & Liu, 2023) ensured relevance, while collaborative and mobile features enhanced BSIT interaction and access. By synthesizing these findings, the researchers' Agile-developed LMS bridged traditional and modern education at NEUST-MGT BSIT Department, delivering a secure, data-driven, and interactive platform that empowered BSIT students and faculty, meeting CHED's standards and the digital age's demands.

Conceptual Framework

This conceptual framework provided a structured guideline for developing the

Learning Management System (LMS) for the NEUST-MGT BSIT Program, illustrating the relationships and interactions between its components. It served as a system design, development, and evaluation roadmap, ensuring alignment with academic objectives while enhancing student learning and faculty engagement.

Figure 1. *Conceptual Framework*



The conceptual framework for this study was based on the Agile Development Model, which ensured an iterative and adaptive approach in designing, developing, and testing the LMS. This model followed six essential stages, each critical for achieving a high-quality and functional system.

Requirements Definition: This initial phase involved identifying and analyzing user needs, system functionalities, and technical requirements. Feedback from faculty and students played a crucial role in shaping the system's core features, ensuring usability, performance, and scalability feasibility.

System Design: In this stage, the overall structure of the LMS was outlined, including the database schema, user interface components, and system architecture. The design phase ensured that the LMS was user-friendly, efficient, and capable of handling academic activities such as lesson uploads, quiz creation, and real-time grading.

Development: The development phase focused on coding and implementing the system components. The LMS was built incrementally using appropriate programming languages and frameworks, incorporating core functionalities such as student enrollment, quiz administration, and data analytics. Agile principles ensured continuous improvement through sprint-based development.

Quality Assurance: This stage involved rigorous testing to verify that the LMS met functional and non-functional requirements. Various test cases, including usability testing, performance evaluation, and security assessments, were conducted to ensure the system operated efficiently and securely.

User Acceptance Testing (UAT): Faculty members, students, and IT experts participated in this phase to evaluate the system's usability, effectiveness, and overall experience. Their feedback was essential for refining features, addressing issues, and ensuring the LMS met real-world academic needs.

Deployment and Continuous Improvement: Once validated, the LMS was deployed for actual use. However, Agile development emphasized continuous monitoring, feedback integration, and iterative updates to enhance system performance, usability, and adaptability based on user needs.

By following the Agile Development Model, this framework ensured a flexible and user-centered approach, allowing the LMS to evolve in response to changing educational requirements and technological advancements.

Statement of the Problem

This project aimed to develop a Learning Management System (LMS) for the NEUST-MGT

BSIT Program, designed to enhance online learning, real-time grading, and faculty-student interaction. The system would support faculty in managing lessons, quizzes, and exams while allowing students to track their academic progress effectively.

Specifically, this study sought to answer the following questions:

1. **How may the current system be described in terms of the following constructs?**

   1.1 Security of files and records;

   1.2 Management of data;

   1.3 Reliability of output;

   1.4 Efficiency of processes; and

   1.5 Compliance with regulations and standards?

2. **How are the features and functionalities of the Learning Management System for the NEUST-MGT BSIT Department iteratively developed and refined through Agile sprints, and how does continuous end-user feedback enhance their utilization by BSIT students and faculty?**

   2.1 Requirements;

   2.2 Design;

2.3 Development;

2.4 Testing;

2.5 Deployment;

2.6 Maintenance;

3. How may the elected IT consultants evaluate the developed applications in terms of the following software characteristics:

3.1 Functional Suitability;

3.2 Performance Efficiency;

3.3 Compatibility;

3.4 Usability; and

3.5 Security?

4. How may the end-users evaluate the developed application based on the System Usability Scale (SUS) in terms of the following usability characteristics:

4.1 Ease of Use;

4.2 Efficiency;

4.3 Learnability; and

4.4 User Satisfaction?

Addressing these challenges required designing a comprehensive Learning Management System that integrated various educational tools, automated assessment processes, enhanced data security and privacy measures, and provided in-depth analytics for student performance tracking. This system would empower educators to improve teaching strategies, monitor student engagement, and support the academic success of BSIT students at NEUST-MGT.

Scope and Delimitations

This study focused on developing an efficient Learning Management System (LMS) for the NEUST-MGT BSIT Program, designed to enhance digital learning, streamline assessment processes, and improve student performance tracking. The system provided a structured platform for managing subjects, administering quizzes and exams, automating grading, and facilitating real-time faculty-student interaction. The LMS ensured that students could access learning materials anytime while allowing faculty members to monitor progress effectively.

This study was conducted to address several challenges in traditional classroom setups, including manual assessment procedures, difficulty in tracking student performance, and limited accessibility to learning resources. The LMS was developed exclusively for the NEUST-MGT BSIT Program, meaning it did not extend to other departments or institutions. The study was limited to evaluating the system's functionality, usability, and effectiveness in supporting online learning based on faculty and student feedback.

This study was conducted to improve student engagement, assessment efficiency, and digital course management within the NEUST-MGT BSIT Program. The researchers developed the system at NEUST-MGT in Talavera, Nueva Ecija, from January to November in the year two thousand twenty-five. The study followed the Agile Development Model, ensuring iterative progress and continuous system enhancements while adhering to the project timeline set by the faculty adviser.

Significance of the Study

The results of this study would help the following:

Students: The system was beneficial as it provided a centralized platform for accessing learning materials, tracking progress, and engaging in digital assessments. Students benefited from it because they were the primary users of the LMS, enabling them to have a more structured and interactive learning experience. The system also enhanced accessibility to academic resources and improved student engagement through digital tools.

Faculty Members: The LMS assisted faculty members by automating grading, streamlining lesson delivery, and providing real-time student performance tracking. They benefited from it because it reduced the workload associated with manual assessment and record-keeping. Faculty could also use the system's analytics and feedback tools to identify student learning gaps and adjust teaching strategies accordingly.

Future Researchers: This study could be used as a reference for future research on learning management systems, online education, and digital assessment tools. It provided insights into the challenges and advantages of LMS implementation, which could help future researchers enhance and innovate similar systems.

Present Researchers: Conducting this study allowed the researchers to gain practical experience in software development, system evaluation, and educational research. They benefited from it because it strengthened their problem-solving and technical skills while contributing to advancing digital learning solutions.

Schools: This study was a valuable resource for adopting technology-driven learning. Schools benefited from it by gaining insights into digital education solutions and how they could improve the quality and efficiency of course management. The findings could encourage institutions to integrate LMS platforms to enhance teaching and learning experiences.

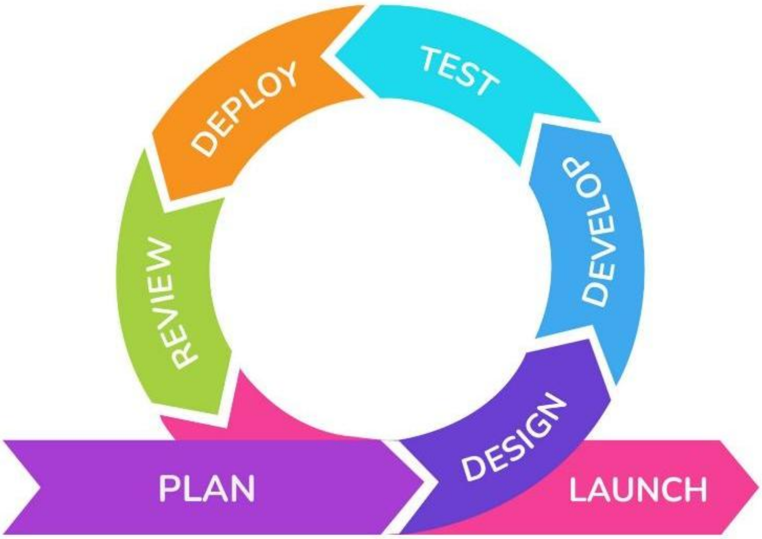
Administrators: The LMS enabled school administrators to oversee academic performance, monitor faculty efficiency, and improve institutional decision-making. They benefited from it because it provided real-time reports and analytics on student and faculty activities, supporting data-driven improvements in academic management.

Chapter II
Methodology

This research section presents the systematic process for the successful development and employment of the study. It includes the research design, locale, respondents, sampling technique, research instruments, data gathering procedures, data analysis technique, considerations.

Research Design

Figure 2. *Agile Model*



Developmental research design systematically organizes, designs, implements, and evaluates educational processes and products to ensure efficacy and consistency. This study adopted a quantitative research design to assess and refine the Learning

Management System (LMS) for the Nueva Ecija University of Science and Technology—Municipality Government of Talavera (NEUST-MGT) BSIT Program (Siemens, 2019). Quantitative data, including system performance metrics and respondent ratings, were collected to evaluate the LMS's technical quality and usability, providing precise, measurable insights into its effectiveness (George, n.d.).

The researchers utilized the Agile methodology within the Software Development Life Cycle (SDLC) to develop the LMS. Agile's iterative sprints—covering requirement definition, design, development, testing, and deployment—enabled continuous refinement based on numerical performance indicators and structured respondent evaluations (Scrum Alliance, 2022). This structured approach supports the project's goal of creating an LMS with features like real-time grading and analytics, ensuring consistent and quantifiable improvements (Kumar & Sharma, 2021).

Agile's adaptability justifies its use in this quantitative design, focusing on measurable outcomes such as system efficiency, security, and user satisfaction ratings (Al-Fraihat et al., 2020). It ensures the LMS meets predefined technical and educational benchmarks, bridging traditional and modern learning at NEUST-MGT through data-driven development (Bervell & Umar, 2021). A similar quantitative, iterative approach was effectively applied in Haruna et al. (2017), reinforcing Agile's suitability for this developmental study.

Research Locale

This study was conducted in Talavera, Nueva Ecija, specifically at Nueva Ecija University of Science and Technology Municipality Government of Talavera (NEUST - MGT).

Figure 3. *Map of Nueva Ecija University of Science and Technology Municipality Government of Talavera*

Figure 3 shows the topographic map of NEUST-MGT. It is located at 400 Diaz St, Talavera, Nueva Ecija.

Respondents of the Study

The researchers have identified three (3) sets of respondents in this study. They are (1) BSIT students who assessed the current manual system and evaluated the developed system’s usability based on the System Usability Scale (SUS); (2) BSIT faculty members who assessed the current manual system and evaluated the system’s usability for teaching and assessment purposes using SUS; and (3) IT consultants who evaluated the system’s technical quality based on specific software characteristics.

The first set of respondents, one-hundred (100) BSIT students, assessed the current manual system in terms of security, data management, reliability, efficiency, and compliance, and provided feedback on the Learning Management System (LMS) regarding ease of use, efficiency, learnability, and user satisfaction to enhance their learning experience. The second set of respondents, ten (10) BSIT faculty members, evaluated the manual system across the same constructs and assessed the LMS’s usability for managing lessons, quizzes, and student progress tracking using the same SUS criteria. The third set of respondents, three (3) IT consultants, reviewed the LMS's technical aspects, focusing on functional suitability, performance efficiency, compatibility, usability, and security to ensure alignment with software standards. Table 1 shows the distribution of respondents.

Table 1. *Distribution of Respondents*

Respondents	Frequency	Percentage
BSIT Students	100	88.50%
BSIT Faculty Members	10	8.85 %
IT Consultants	3	2.65%
Total	**113**	**100%**

As shown in Table 1 above, the distribution of respondents has a total frequency of forty-three (one-hundred thirteen).

Sample and Sampling Procedures

The researchers employed a purposive sampling technique to select the first group of respondents, consisting of one-hundred (100) BSIT students at Nueva Ecija University of Science and Technology—Municipality Government of Talavera (NEUST-MGT). These students served as evaluators of the manual classroom system and candidate end-users of the proposed Learning Management System (LMS). They were chosen based on the following criteria: (a) Enrolled in the BSIT Program during the 2024–2025 academic year. (b) Have experienced the existing manual system for course management, grading, and resource access.

In addition, a purposive sampling technique was used to select the second group of respondents, comprising ten (10) BSIT faculty members. These faculty members evaluated the manual system and the LMS's usability as end-users. They were selected based on the following criteria: (a) Currently teaching in the BSIT Program at NEUST-MGT. (b) Have utilized the manual system for grading, attendance tracking, and course delivery.

Furthermore, the researchers applied purposive sampling to identify the third group of respondents, consisting of three (3) IT consultants. These consultants conducted a technical evaluation of the LMS based on ISO 25010 standards. They were chosen based on the following criteria: (a) Hold a degree in Information Technology or a related field. (b) Possess professional experience in software development or system evaluation.

Research Instrument

The tools for this study are utilized to collect, measure, and assess data pertinent to the research interests outlined in the development of the Learning

Management System (LMS) for the NEUST-MGT BSIT Program. A survey instrument assessing the common problems encountered with traditional manual systems was employed to gather data based on the following categories: security of files and records, management of data, reliability of output, efficiency of processes, and compliance with regulations and standards. These categories reflect the inefficiencies identified in the current educational setup at NEUST-MGT, such as manual grading delays and limited resource access, which the proposed LMS aims to address.

To evaluate the quality of the LMS and its technical aspects within software characteristics quality standards, the researchers utilized questionnaires to gather information from two respondent groups: elected IT consultants and candidate end-users. The questionnaires comprised closed-ended questions designed to provide structured feedback. The questionnaire for three (3) IT consultants is divided into five (5) components aligned with the ISO 25010 standard: functional suitability, performance efficiency, compatibility, usability, and security. These components ensure a focused technical assessment of the LMS’s capabilities, from real-time grading accuracy to data security and system interoperability.

The end-user questionnaire, targeting 100 BSIT students and 10 BSIT faculty members, is based on the System Usability Scale (SUS) and divided into four (4) components: ease of use, efficiency, learnability, and user satisfaction. These focus areas assess the LMS's practical usability in delivering educational features like automated assessments, resource access, and performance tracking, ensuring it is user-friendly and effective in daily use. By collecting data from both IT consultants and end-users, the instruments ensure a balanced evaluation of the system's technical quality and usability, supporting the study's goal of enhancing digital learning and faculty-student interaction at NEUST-MGT.

Data Gathering Procedures

The following phases formed the basis for the procedures of this study.

Phase I: Data Gathering for Manual System Assessment

The researchers seek authorization from the respondents and their schools before handing them the survey questionnaire. The data needed to assess the manual system was gathered from one-hundred (100) BSIT students and ten (10) BSIT faculty members.

Phase II: Designing and Developing the Web-based System

The researchers developed both front-end and back-end aspects of the system. In this phase, all the inputs in Figure 1 will be incorporated through planning and requirements gathering, design and prototyping, development and implementation, and testing and deployment. Several programming languages were used by the researchers to meet the criteria for the system's overall functionality. The valuable assistance of Mr. Michael Bensi of the BSIT Department in making the evaluation test that will be used in the system will serve as the main feature of the system.

Phase III: Web-based System Pre-Testing

The researchers will conduct a peer review to assess the system's functionality, design, and security. The initial results will be tabulated for deliberation before analysis and will be enhanced as necessary.

Phase IV: Testing and Data Gathering

The researchers will seek authorization from the respondents and their schools if they are students or IT consultants before handing them the survey questionnaire. The system will undergo assessment following the selected guidelines of the software characteristics standard for IT consultants and the

System Usability Scale (SUS) for end-users. The data will be gathered from one-hundred (100) BSIT students, ten (10) BSIT faculty members, and three (3) IT consultants. The researchers will remain open to any additional concerns, questions, and suggestions from the respondents.

Data Analysis Technique

While gathering the needed data, the researchers will employ the following data analysis techniques:

1. To answer the first statement of the problem related to the security of files and records, data management, output reliability, process efficiency, and compliance with regulations and standards, frequency and weighted mean will be used to describe the set and the results of the evaluation made by the group of respondents to the manual system.

The scoring rubric in Table 2 will be used to interpret the results correctly.

Table 2. *Scoring Rubric for Level of Agreement*

Range	Qualitative Remarks	Verbal Description
3.26 - 4.00	Strongly Agree	The range of 3.26 to 4.00 is considered "Strongly Agree" in the qualitative description. This means the respondent has a high level of agreement with the statement or question. It indicates that the respondent strongly supports or approves of the presented

		idea, concept, or statement.
Range	**Qualitative Remarks**	**Verbal Description**
2.51 - 3.25	Agree	The range of 2.51 to 3.25 is classified as "Agree," indicating that the respondent has a favorable view of the statement but not as strong as those in the "Strongly Agree" category.
Range	**Qualitative Remarks**	**Verbal Description**
1.76 - 2.50	Disagree	The range of 1.76 to 2.50 is classified as

		"Disagree," which indicates that the respondent has an unfavorable view of the statement or question being asked.
Range	**Qualitative Remarks**	**Verbal Description**
1.00 - 1.75	Strongly Disagree	The range of 1.00 to 1.75 is classified as "Strongly Disagree," which indicates that the respondent has a high negative or unfavorable view of the statement or

question being asked.

Table 2 above illustrates the scoring rubric for the level of agreement, detailing the range, qualitative comments, and verbal interpretation of the four-point scale used to assess responses.

2. To address the second statement of the problem, the researchers will outline the iterative tasks and continuous feedback processes within Agile sprints to provide meaningful insights into the development and refinement of the Learning Management System (LMS) features and functionalities for the NEUST-MGT BSIT Department, and how these enhance end-user utilization by BSIT students and faculty.

    This study employs the Agile model as the framework for the system's development, characterized by iterative sprints that integrate continuous end-user feedback to refine features and ensure their effective utilization. The Agile process is structured around the following iterative phases, with tasks adapted dynamically based on feedback from BSIT students and faculty:

Requirement Definition: In this phase, the researchers initiate each sprint by collecting data on the LMS's necessary functionalities through interviews with stakeholders (BSIT students, faculty, and administrators), reviewing existing literature, and analyzing the current manual system's limitations. User needs (e.g., real-time grading, resource access) and functional (e.g., analytics, question randomization) and non-functional requirements (e.g., usability, security) are identified and prioritized for the sprint, ensuring alignment with end-user expectations. Feedback from previous sprints informs subsequent requirement updates, fostering iterative refinement.

System Design: During each sprint, the researchers convert prioritized requirements into a technical plan tailored to the sprint's goals. This includes designing system architecture components, creating user interface (UI) prototypes for features like timer-based exams, and defining database structures to support analytics or resource uploads. End-user feedback from prior sprints is incorporated to refine UI designs and ensure intuitive navigation, enhancing feature utilization by BSIT students and faculty.

Development: Within each sprint, the researchers implement and code the LMS features based on the sprint's design specifications. Suitable programming languages and tools are used to develop functionalities such as student enrollment, real-time grading, and performance analytics, with coding standards verified to ensure reliability. Incremental deliverables (e.g., a working quiz module) are produced per sprint, allowing end-users to test and provide feedback, which guides further refinements in subsequent sprints.

Testing: In this phase, each sprint's deliverables undergo rigorous testing to detect errors and validate alignment with user needs. Usability testing assesses ease of use for features like resource access, performance testing evaluates system efficiency (e.g., grading speed), and user acceptance testing (UAT) involves BSIT students and faculty evaluating feature functionality and utilization. Feedback from these tests, collected via System Usability Scale (SUS) questionnaires and open-ended surveys, drives iterative improvements in the next sprint, ensuring features are refined to enhance end-user engagement.

3. To address the third statement of the problem, regarding the IT consultant's evaluation of the system's technical quality, the researchers will apply the following software characteristics guidelines to gain a deeper understanding of the enhanced evaluation results.

Table 3 presents the scoring rubric for Functional Suitability.

Table 3. *Scoring Rubric for Functional Suitability*

Range	**Qualitative Remarks**	**Verbal Description**
3.26 - 4.00	Highly suitable	The system meets or exceeds all functional requirements with high accuracy, completeness, and efficiency.
Range	**Qualitative Remarks**	**Verbal Description**
2.51 - 3.25	Suitable	The system meets

		most of the functional requirements with satisfactory accuracy, completeness, and efficiency.
Range	**Qualitative Remarks**	**Verbal Description**
1.76 - 2.50	Unsuitable	The system meets some of the functional requirements but with low accuracy, completeness, and efficiency.
Range	**Qualitative Remarks**	**Verbal Description**
1.0 - 1.75	Highly unsuitable	The system fails to meet most or all of the

		functional requirements and has a deficient accuracy, completeness, and efficiency.

As presented in Table 3 above, the scoring rubric was used to measure the functionality of a system when utilized under certain circumstances.

Table 4 depicts the scoring rubric for Performance Efficiency.

Table 4. *Scoring Rubric for Performance Efficiency*

Range	**Qualitative Remarks**	**Verbal Description**
3.26 - 4.00	Excellent	The system's performance is exceptional, meeting or exceeding all efficiency

		requirements.

Range	**Qualitative Remarks**	**Verbal Description**
3.26 - 4.00	Excellent	Users experience no delays or errors, and the system performs consistently well under all operating conditions.
2.51 - 3.25	Good	Users experience minimal delays and errors, and the system performs well under normal operating conditions.

Range	**Qualitative Remarks**	**Verbal Description**
1.76 - 2.50	Fair	The system's performance is average, but

		there is room for improvement. Users may experience occasional delays or errors, but the system functions adequately overall.
Range	**Qualitative Remarks**	**Verbal Description**
1.00 - 1.75	Poor	The system's performance could improve and meet the minimum requirements for efficient operation. Users may experience delays, errors, and other issues negatively

		affecting productivity.

Depicted in Table 4 above is the rubric used to determine the capacity of a system to carry out its intended function within a predetermined time.

Table 5 shows the scoring rubric for Compatibility.

Table 5. *Scoring Rubric for Compatibility*

Range	**Qualitative Remarks**	**Verbal Description**
3.26 - 4.00	Highly compatible	The system or software is fully compatible with other hardware or software and can seamlessly integrate.

Range	Qualitative Remarks	Verbal Description
2.51 - 3.25	Compatible	The system or software can work with hardware or software with minor adjustments or configurations.
Range	**Qualitative Remarks**	**Verbal Description**
1.76 - 2.50	Somewhat compatible	The system or software can work with other hardware or software with significant adjustments or configurations.
Range	**Qualitative Remarks**	**Verbal Description**
1.00 - 1.75	Not compatible	The system or software cannot work with other hardware or

		software and needs a complete overhaul or replacement.

Shown in Table 5 above is the rubric used to measure the ability of a system to share resources and an environment with another system.

Table 6 presents the scoring rubric for Usability.

Table 6. *Scoring Rubric for Usability*

Range	**Qualitative Remarks**	**Verbal Description**
3.26 - 4.00	Excellent	The system is easy to use and navigate, and users can easily accomplish their tasks. The system is well-designed,

		intuitive, and efficient. Users rarely encounter errors or obstacles and are highly satisfied with the system.
Range	**Qualitative Remarks**	**Verbal Description**
2.51 - 3.25	Good	The system is generally easy to use, but there may be some minor usability issues or areas where improvement is needed. Users may encounter occasional

		obstacles or errors, but these do not significantly impede their ability to complete tasks. Users are generally satisfied with the system.

Range	**Qualitative Remarks**	**Verbal Description**
1.76 - 2.50	Fair	The system has significant usability issues that make it difficult or frustrating for users to accomplish tasks. Users may Encounter frequent errors

		or obstacles that impede their progress. The system may need to be better designed or require significant effort. Users are generally dissatisfied with the system.

Range	**Qualitative Remarks**	**Verbal Description**
1.00 - 1.75	Poor	The system is difficult or impossible to use, and users cannot accomplish their tasks. The system may be unusable due to

		significant errors or obstacles. The system is poorly designed or requires significant effort to use. Users are highly dissatisfied with the system.

Presented in Table 6 above is the rubric used to analyze the usability of a system by being able to communicate and transfer data via the user interface between a user and a system.

In table 7, the scoring rubric for Security is shown.

Table 7. *Scoring Rubric for Security*

Range	**Qualitative Remarks**	**Verbal Description**

3.26 - 4.00	Strong Security	The system has multiple security measures that provide comprehensive protection against unauthorized access, data breaches, and other security threats. The security features are regularly updated and tested to ensure that they effectively prevent and detect potential security breaches.
Range	**Qualitative Remarks**	**Verbal Description**

2.51 - 3.25	Adequate Security	The system has basic security measures in place that are effective in preventing most. However, there may be some areas of vulnerability that additional security measures may need to be implemented.
Range	**Qualitative Remarks**	**Verbal Description**
1.76 - 2.50	Weak Security	The system has significant security vulnerabilities that need to be addressed. There may be gaps in

		the security measures that allow unauthorized access, data breaches, and other security threats. Immediate action is needed to improve.
Range	**Qualitative Remarks**	**Verbal Description**
1.00 - 1.75	No Security	The system has no security measures, or the existing measures could be more effective. The system is highly vulnerable to

	security breaches and data.

As depicted in Table 7 above, the rubric to determine a system's security was used to assess its capacity to guard against malicious actor attack patterns and safeguard data.

4. To address the fourth statement of the problem, regarding the end-users' evaluation of the Learning Management System (LMS) usability based on the System Usability Scale (SUS), the researchers will analyze quantitative data collected iteratively during Agile sprints to assess the system's quality in terms of ease of use, efficiency, learnability, and user satisfaction. This analysis ensures that continuous end-user feedback from BSIT students and faculty enhances the LMS's usability, aligning with the iterative development process (Scrum Alliance, 2022).

   The researchers will use the following statistical methods to treat the SUS data collected from 100 BSIT students and 10 BSIT faculty members:

Percentage. It is used to determine the rate of the responses. To compute the percentage, the following formula will be used:

$$P = \frac{F}{N} \times 100$$

Where: P = Percentage

F = Frequency

N = Total population

Weighted Arithmetic Mean. It is an average computed by allotting different weights to some individual values (Admin, 2021). The ranges of the weighted mean and its verbal interpretation is shown in Table 8 below.

Table 8 shows the range of weighted mean and its verbal interpretation

Table 8. *Ranges of Weighted Arithmetic Mean and its Verbal Interpretation*

Range of Weighted Mean	Verbal Interpretation
3.26 - 4.00	Strongly Agree
2.51 - 3.25	Agree
1.76 - 2.50	Disagree
1.00 - 1.75	Strongly Disagree

Above is Table 8 showing the ranges of weighted arithmetic means and its corresponding verbal interpretation.

In identifying the weighted mean, the researchers will apply the formula below:

$Weighted\ Mean$: $\frac{\sum wx}{\sum w}$

Where: Σ = denotes the sum

w = weight

x = value

Through this method, the researchers can analyze and interpret all the responses and feedback from the respondents effectively.

Ethical Considerations

To guarantee that the respondents' rights are upheld, the researchers followed a set of guidelines for carrying out this investigation. In order to achieve this, the following moral guidelines are followed:

Informed Consent: A letter outlining the goals and objectives of the study, the procedures that will be followed, and the possible risks and benefits will be sent to the respondents. This includes explaining the purpose of assessing the current manual system and evaluating the Learning Management System (LMS) for the NEUST-MGT BSIT Program. The 100 BSIT students, 10 BSIT faculty members, and 3 IT consultants are allowed to ask questions and clarify any concerns before giving their consent to participate.

Anonymity and Confidentiality: The researchers will make sure that the respondents' identities are kept private and anonymous. Data collected from the survey questionnaires assessing the manual system and the LMS evaluations will be coded to protect the identities of the 100 BSIT students, 10 BSIT faculty members, and 3 IT consultants. Access to the data will be restricted to the researchers working on the project and will be securely preserved in a password-protected digital storage system.

Voluntary Participation: The respondents will not be coerced or pressured to participate in the study. The 100 BSIT students, 10 BSIT faculty members, and 3 IT consultants will be given the option to withdraw their participation at any time without facing any consequences, ensuring their involvement is entirely voluntary.

Protection of Vulnerable Populations: The researchers will make sure that groups that are particularly susceptible to harm, such as minors or individuals with disabilities, are not used unfairly or put in danger during the study. Since the respondents are BSIT students, faculty, and IT consultants, no vulnerable populations are involved, but the researchers will remain vigilant to avoid any unintended risks.

Researchers' Integrity: The researchers will maintain a high level of integrity throughout the study. They will ensure that the assessment of the manual system and the evaluation of the LMS will be conducted ethically, with results reported accurately and objectively, free from bias or fabrication, to uphold the credibility of the findings.

To summarize, the researchers will adhere to the ethical considerations of informed consent, anonymity and confidentiality, voluntary participation, protection of vulnerable populations, and the researchers' integrity in conducting this study.

Chapter III
RESULTS AND DISCUSSION

This chapter presents the results of the assessment of the manual system, the design and development of the Learning Management System (LMS) based on the phases of the Agile Development Model, the assessment of the system's technical qualities by elected IT experts in compliance with ISO 25010 software characteristics guidelines, and the assessment of the system's usability by candidate end-users based on the System Usability Scale (SUS).

1. Student's Assessment of the Manual System

This section presents the results of the assessment of the current manual system as evaluated by BSIT students. The assessment covers five areas: security of files and records, management of data, reliability of output, efficiency of processes, and compliance with regulations and standards. A total of 100 BSIT students served as respondents. Data were analyzed using frequency and weighted mean, with interpretation guided by the scoring rubric in Table 2 (Chapter II): 3.26–4.00 (Strongly Agree), 2.51–3.25 (Agree), 1.76–2.50 (Disagree), 1.00–1.75 (Strongly Disagree).

1.1 Security of Files and Records

Table 9 presents the results of the student's assessment on the security of files and records using the manual system.

Table 9. *Results of the Student's Assessment on the Security of Files and Records Using Manual System.*

Item Statement	Weighted Mean	Verbal Description
The traditional classroom setup occasionally results in lost or misplaced student records.	2.99	Agree
Unauthorized access to student grades and documents occurs due to poor control measures.	2.99	Agree
Physical records are vulnerable to damage (e.g., fire, flood, misplacement) with no safeguards. access to files and records.	3.10	Agree

There is no backup or recovery system for lost or damaged student files.	2.91	Agree
Missing or inaccessible teaching materials disrupt classroom activities.	2.96	Agree
Grand Mean	**2.99**	
Verbal	**Agree**	

As shown in Table 9, the respondents gave a grand mean of 2.99, interpreted as Agree, for the security of files and records in the manual system. This indicates that students observed issues such as misplaced records, unauthorized access to grades, and the lack of reliable backup or recovery measures. They also agreed that physical records are vulnerable to damage and that missing learning materials disrupt classroom activities.

Security of files and records is an important concern because it ensures the protection and accessibility of academic information. The findings suggest that the manual system lacks the safeguards needed to provide secure and dependable record-keeping for students.

These results emphasize the need for an LMS that centralizes storage, controls access, and includes backup and recovery functions, ensuring that students benefit from secure and accessible academic records.

1.2 Management of Data

Table 10 presents the results of the student's assessment on the management of data using the manual system.

Table 10. *Results of the Student's Assessment on the Management of Data Using Manual System*

Item Statement	Weighted Mean	Verbal Description
Manual grading produces inconsistent or unreliable results due to human error.	2.97	Agree
Attendance records are often inaccurate, affecting student performance tracking.	2.95	Agree
Paper-based systems fail to provide dependable output for large class sizes.	2.96	Agree
Manual record verification is prone to mistakes and delays.	2.97z	Agree

Lack of centralized coordination leads to unreliable scheduling and resource availability.	2.96	Agree
Grand Mean	**2.96**	
Verbal	**Agree**	

As shown in Table 10, the respondents gave a grand mean of **2.96**, interpreted as **Agree**, for the management of data in the manual system. This indicates that students observed issues such as errors in grading, inaccurate attendance records, and delays in verifying student information. They also agreed that the lack of centralized coordination leads to unreliable scheduling and resource availability.

Proper data management is vital for keeping records accurate and consistent. The findings suggest that the manual system introduces inconsistencies and delays that affect students' performance tracking and access to academic resources.

These results highlight the importance of adopting an LMS that automates grading, centralizes attendance and records, and provides reliable scheduling, ensuring more dependable academic management for students.

1.3 Reliability of Output

Table 11 presents the results of the student's assessment on the reliability of output using the manual system.

Table 11. *Results of the Student's Assessment on the Reliability of Output Using Manual System*

Item Statements	Weighted Mean	Verbal Description
Manual grading often produces inconsistent results due to human errors.	2.97	Agree
Attendance records are often unreliable, leading to disputes over student attendance data.	2.95	Agree
Paper-based systems struggle to provide reliable outputs for large class sizes.	2.96	Agree
Manual verification of grades and records is prone to mistakes and inconsistencies.	2.97	Agree
Lack of centralized coordination leads to	2.96	Agree

unreliable scheduling and resource availability.	
Grand Mean	**2.96**
Verbal	**Agree**

As shown in Table 11, the respondents gave a grand mean of **2.96** , interpreted as **Agree**, for the reliability of output in the manual system. This indicates that students recognized issues such as inconsistent grading results and errors in the manual verification of records. They agreed that the system does not always produce dependable outputs, especially when handling large volumes of academic data.

Reliability of output is important because it ensures that academic results are accurate and trustworthy. The findings suggest that the manual system creates doubts about the consistency of grades and records, which can affect student confidence and performance tracking.

These results highlight the need for an LMS that provides accurate, consistent, and dependable outputs, helping students trust the integrity of their academic records.

1.4 Efficiency of Processes

Table 12 presents the results of the student's assessment on the efficiency of processes using the manual system.

Table 12. *Results of the Student's Assessment on the Efficiency of Processes Using Manual System*

Item Statements	Weighted Mean	Verbal Description
Grading and feedback processes are slow and inefficient in the traditional setup.	2.97	Agree
Managing assessments manually delays the delivery of student results.	2.97	Agree
Teachers spend excessive time on administrative tasks rather than teaching.	3.05	Agree
Resource distribution (e.g., handouts, notes) is inefficient and inconsistent.	2.98	Agree
Handling large volumes of student data manually reduces overall process efficiency.	2.99	Agree

Grand Mean	**2.99**
Verbal	**Agree**

As shown in Table 12, the respondents gave a grand mean of **2.99**, interpreted as **Agree**, for the efficiency of processes in the manual system. This indicates that students experienced delays in grading and feedback, slow distribution of learning materials, and excessive time spent by teachers on administrative tasks rather than instruction. They agreed that handling large volumes of student data manually reduces the overall efficiency of academic processes.

Efficiency of processes is important because it affects how quickly and smoothly academic tasks are completed. The findings suggest that the manual system causes delays and inefficiencies that affect both teaching and learning.

These results emphasize the need for an LMS that automates grading, speeds up access to learning resources, and streamlines administrative work, ensuring faster and more efficient academic processes for students.

1.5 Compliance with Regulations and Standards

Table 13 presents the results of the student’s assessment on compliance with regulations and standards using the manual system.

Table 13. *Results of the Student’s Assessment on Compliance with Regulations and Standards Using Manual System*

Item Statements	Weighted Mean	Verbal Description
Student grades are not submitted on time due to manual processes, violating school deadlines.	2.93	Agree
The traditional setup fails to follow rules for keeping accurate attendance records.	2.92	Agree
Manual records do not meet requirements for secure storage of student information.	2.95	Agree
There is no system to ensure grades and reports follow institutional grading policies.	2.93	Agree
The setup lacks a way to track compliance with class attendance and reporting rules.	2.92	Agree

Grand Mean	**2.97**
Verbal	**Agree**

As shown in Table 13, the respondents gave a grand mean of **2.97,** interpreted as **Agree,** for compliance with regulations and standards in the manual system. This indicates that students observed issues such as late submission of grades, and the lack of secure storage for academic information. They also agreed that there is no systematic way to ensure adherence to institutional grading policies and reporting requirements.

Compliance is important because it ensures that academic processes follow institutional rules and standards. The findings suggest that the manual system makes it difficult to meet deadlines and maintain accurate, secure, and policy-compliant records.

These results highlight the importance of adopting an LMS that supports timely grade submission, secure record-keeping, and reliable monitoring of institutional requirements, ensuring that both students and faculty benefit from a more consistent and compliant academic system.

1. Faculty's Assessment of the Manual System

This section presents the results of the assessment of the current manual system as evaluated by BSIT faculty members. The assessment covers five areas: security of files and records, management of data, reliability of output, efficiency of processes, and compliance with regulations and standards. A total of 10 BSIT faculty members served as respondents. Data were analyzed using frequency and weighted mean, with interpretation guided by the scoring rubric in Table 2 (Chapter II): 3.26–4.00 (Strongly Agree), 2.51–3.25 (Agree), 1.76–2.50 (Disagree), 1.00–1.75 (Strongly Disagree).

1.1 Security of Files and Records

Table 14 presents the results of the faculty's assessment on the security of files and records using the manual system.

Table 14. *Results of the Faculty's Assessment on the Security of Files and Records Using Manual System*

Item Statement	Weighted Mean	Verbal Description
The traditional classroom setup occasionally	2.80	Agree

results in lost or misplaced student records.		
Unauthorized access to student grades and documents occurs due to poor control measures.	2.80	Agree
Physical records are vulnerable to damage (e.g., fire, flood, misplacement) with no safeguards. access to files and records.	3.00	Agree
There is no backup or recovery system for lost or damaged student files.	2.70	Agree
Missing or inaccessible teaching materials disrupt classroom activities.	2.60	
Grand Mean	**2.78**	
Verbal	**Agree**	

As shown in Table 14, the respondents gave a grand mean of **2.78,** interpreted as **Agree,** for the security of files and records in the manual system. This indicates that faculty recognized issues such as misplaced student records, unauthorized access to grades, and the lack of reliable backup or recovery measures. They also agreed that physical records are vulnerable to damage and that missing or inaccessible teaching materials disrupt classroom activities.

Security of files and records is a critical concern for faculty because it affects the accuracy and confidentiality of academic information. The findings suggest that the manual system does not provide adequate safeguards to ensure secure and dependable record-keeping.

These results emphasize the importance of adopting an LMS with centralized storage, access controls, and backup features, giving faculty a more reliable and secure way to manage academic records.

1.2 Management of Data

Table 15 presents the results of the faculty’s assessment on the management of data using the manual system.

Table 15. *Results of the Faculty's Assessment on the Management of Data Using Manual System*

Item Statement	Weighted Mean	Verbal Description
Manual grading produces inconsistent or unreliable results due to human error.	2.78	Agree
Attendance records are often inaccurate, affecting student performance tracking.	2.78	Agree
Paper-based systems fail to provide dependable output for large class sizes.	2.56	Agree
Manual record verification is prone to mistakes and delays.	2.56	Agree
Lack of centralized coordination leads to unreliable scheduling and resource availability.	2.22	Agree
Grand Mean	**2.58**	
Verbal	**Agree**	

As shown in Table 15, the respondents gave a grand mean of **2.58**, interpreted as **Agree**, for the management of data in the manual system. This indicates that faculty experienced challenges such as errors in manual grading, delays in verifying records, and the lack of dependable systems for tracking student performance. They also observed that the absence of centralized coordination leads to unreliable scheduling and resource allocation.

Management of data is important for ensuring accuracy and efficiency in both teaching and administrative tasks. The findings suggest that the manual system often results in inconsistencies and delays that hinder faculty in monitoring student outcomes and organizing academic activities.

These results highlight the need for an LMS that automates grading, streamlines record verification, and centralizes academic data, providing faculty with more reliable and efficient management tools.

1.3 Reliability of Output

Table 16 presents the results of the faculty’s assessment on the reliability of output using the manual system.

Table 16. *Results of the Faculty's Assessment on the Reliability of Output Using Manual System*

Item Statements	Weighted Mean	Verbal Description
Manual grading often produces inconsistent results due to human errors.	2.56	Agree
Attendance records are often unreliable, leading to disputes over student attendance data.	2.22	Disagree
Paper-based systems struggle to provide reliable outputs for large class sizes.	2.33	Disagree
Manual verification of grades and records is prone to mistakes and inconsistencies.	2.22	Disagree
Lack of centralized coordination leads to unreliable scheduling and resource availability.	2.22	Disagree
Grand Mean	**2.31**	
Verbal	Disagree	

As shown in Table 16, the respondents gave a grand mean of 2.31, interpreted as Disagree, for the reliability of output in the manual system. This indicates that faculty did not view the manual processes as consistently reliable. They observed issues such as inconsistent grading results, errors in verifying records, and difficulties in maintaining accuracy when handling large volumes of data.

Reliability of output is essential because it ensures that academic records and results are dependable and trustworthy. The findings suggest that the manual system fails to provide consistent outcomes, which can affect the integrity of academic reporting and evaluation.

These results underscore the need for an LMS that delivers accurate and consistent outputs, minimizes human error, and ensures that faculty can rely on the system for dependable academic results.

1.4 Efficiency of Processes

Table 17 presents the results of the faculty’s assessment on the efficiency of processes using the manual system.

Table 17. *Results of the Faculty's Assessment on the Efficiency of Processes Using Manual System*

Item Statements	**Weighted Mean**	**Verbal Description**
Grading and feedback processes are slow and inefficient in the traditional setup.	2.70	Agree
Managing assessments manually delays the delivery of student results.	2.60	Disagree
Teachers spend excessive time on administrative tasks rather than teaching.	2.90	Agree
Resource distribution (e.g., handouts, notes) is inefficient and inconsistent.	2.70	Disagree
Handling large volumes of student data manually reduces overall process efficiency.	2.70	Agree
Grand Mean	**2.72**	
Verbal	**Disagree**	

As shown in Table 17, the respondents gave a grand mean of **2.72**, interpreted as **Agree**, for the efficiency of processes in the manual system. This indicates that faculty experienced delays in grading and providing feedback, slow distribution of learning materials, and excessive time spent on administrative tasks rather than teaching. They also recognized that managing large volumes of student data manually reduces overall process efficiency.

Efficiency of processes is important because it directly affects the timely delivery of academic outputs and the ability of teachers to focus on instruction. The findings suggest that the manual system creates unnecessary delays and workload that hinder teaching effectiveness.

These results highlight the need for an LMS that automates grading, speeds up resource distribution, and streamlines routine administrative tasks, enabling faculty to work more efficiently and devote more time to instruction.

1.5 Compliance with Regulations and Standards

Table 18 presents the results of the student’s assessment on compliance with regulations and standards using the manual system.

Table 18. *Results of the Student's Assessment on Compliance with Regulations and Standards Using Manual System*

Item Statements	Weighted Mean	Verbal Description
Student grades are not submitted on time due to manual processes, violating school deadlines.	2.60	Agree
The traditional setup fails to follow rules for keeping accurate attendance records.	2.50	Disagree
Manual records do not meet requirements for secure storage of student information.	2.70	Agree
There is no system to ensure grades and reports follow institutional grading policies.	2.60	Agree
The setup lacks a way to track compliance with class attendance and reporting rules.	2.50	Disagree
Grand Mean	**2.58**	
Verbal	**Agree**	

As shown in Table 18, the respondents gave a grand mean of **2.58,** interpreted as **Agree,** for compliance with regulations and standards in the manual system. This indicates that faculty observed issues such as late submission of grades, incomplete or unsecured student records, and the lack of a systematic process for monitoring institutional requirements. They agreed that the manual system often fails to ensure timely and accurate compliance with school policies.

Compliance is essential in maintaining the integrity and accountability of academic operations. The findings suggest that the manual system does not consistently meet institutional standards for record-keeping, reporting, and grade submission.

These results emphasize the importance of adopting an LMS that enforces deadlines, secures academic data, and tracks compliance with institutional regulations, giving faculty greater assurance in meeting required standards.

2. The Design and Development of the System based on the Phases of the Agile Development Model

This section presents the design and development of the LMS based on the Agile Development Model phases: Requirement Definition, System Design, Development, Quality Assurance, User Acceptance Testing (UAT), and Deployment and Continuous Improvement.

2.1 Requirement

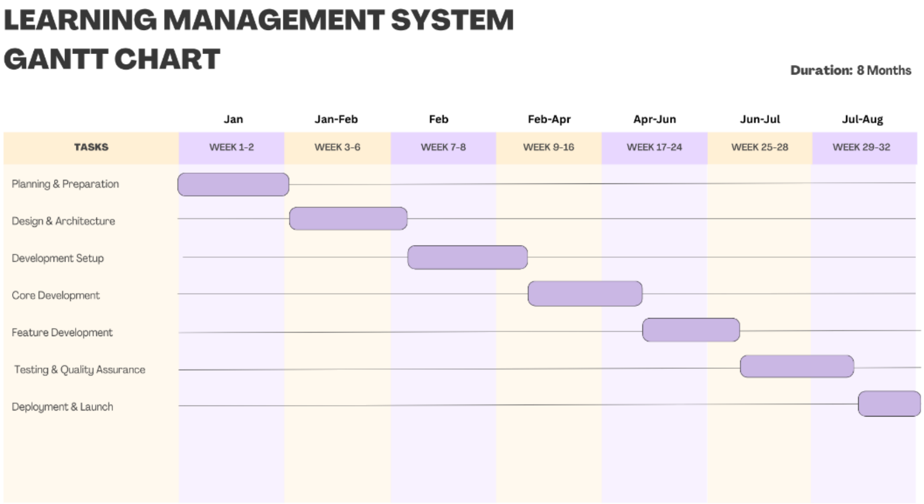
Requirements elicitation is the stage where the researchers prepare for the project. It outlines how the system will be developed by identifying the main activities and their timelines for implementing Learning Management System of NEUST-MGT, BSIT Department. Surveys were conducted to ensure the system aligns with the needs of the candidate end-users and selected IT experts. A Gantt chart was used to present the SDLC activities and serve as a guide in building the system.

Gantt Chart

A Gantt chart is a widely used tool for showing activities within their assigned timeframes. Tasks are displayed as bars on a timeline, making it easy to see when each activity starts and ends, how long it will take, the overall project duration, and where tasks overlap. It is useful for scheduling, monitoring progress, and ensuring activities are completed on time.

Figure 6 presents the Gantt chart of activities followed by the researchers in developing this research project.

Figure 4. *Gantt Chart*



The figure presents the activities with their schedules. It was used to monitor the project timeline for each task. This schedule guided the proponents in determining when to start and finish the project by outlining all the requirements that needed to be carried out.

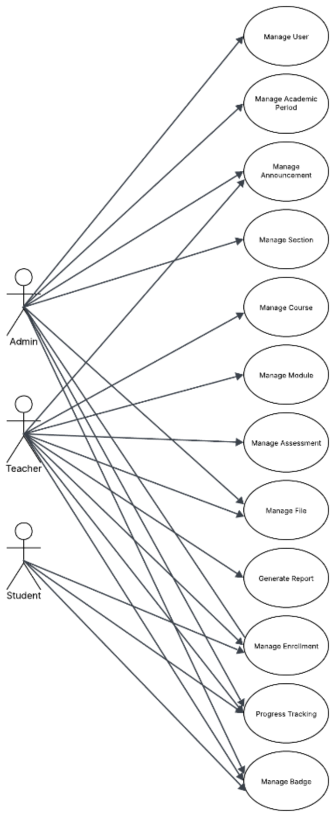
2.2 Design

The design phase includes important tasks such as creating the system’s wireframe and mockup and setting up the database. A use-case diagram provides a functional view of the system, showing how users or actors interact with it. Figure 7 presents the use-case diagram for the Online Auction for Artworks, which outlines the different features accessible to users.

Use-Case Diagram

A use-case diagram visually shows how users or actors interact with a system, highlighting the different ways they access and use its features.

Figure 5. *Use-Case Diagram*



The diagram illustrates how the different actors engage with the system. The admin is responsible for managing users, courses, and assessments, as well as overseeing system configurations and academic years. The Teacher has the ability to manage course content, create assessments, monitor student progress, and generate reports. Meanwhile, the Student takes part in course enrollment, accesses learning modules, takes assessments, and tracks their academic progress.

This visual representation offers a clear overview of the system's features, making it easy to understand the roles and interactions of each user.

Database Design

The database design phase focused on organizing data based on a database model. The designers defined data relationships to identify the essential information to be stored. Database normalization was then applied to clarify the dependencies between entities and their attributes.

Database Normalization

Database normalization organizes data into structured tables and defines their relationships. Its purpose is to remove redundancy, prevent anomalous dependencies, protect data integrity, improve adaptability, and enhance efficiency in queries and management tasks.

Figure 6. *Database Normalization*

Unnormalized Form

file_type
file_size
module_reference
uploaded_at
badge_id
badge_name
badge_description
icon_path
badge_type
criteria_json
badge_points
badge_teacher_id
badge_teacher_name
badge_course_id
badge_course_name
is_active
badge_created_at
achievement_id
earned_at
points_awarded
notification_id
notification_title
notification_message
notification_type
is_read
notification_created_at
related_course_id
related_assessment_id
related_badge_id
announcement_id
announcement_title
announcement_content
author_id
author_name
author_role
target_audience_json
is_global
priority
expires_at
read_by_json
announcement_created_at
request_id
request_status
requested_at
processed_at
processed_by
attempt_ip
ip_address
user_agent
success

attempted_at
location
token_id
token
token_expires_at
token_used
token_created_at
section_id
section_name
section_year_level
section_description
section_is_active
section_students_json
section_teachers_json
section_created_at
section_student_id
section_student_name
section_student_email
section_student_identifier
section_enrolled_at
section_teacher_id
section_teacher_name
section_teacher_email
section_teacher_department
section_assigned_at
course_section_id
course_section_assigned_at
academic_period_id
academic_year
semester_name
is_active
start_date
end_date
academic_period_created_at

Third Normal Form(3NF)

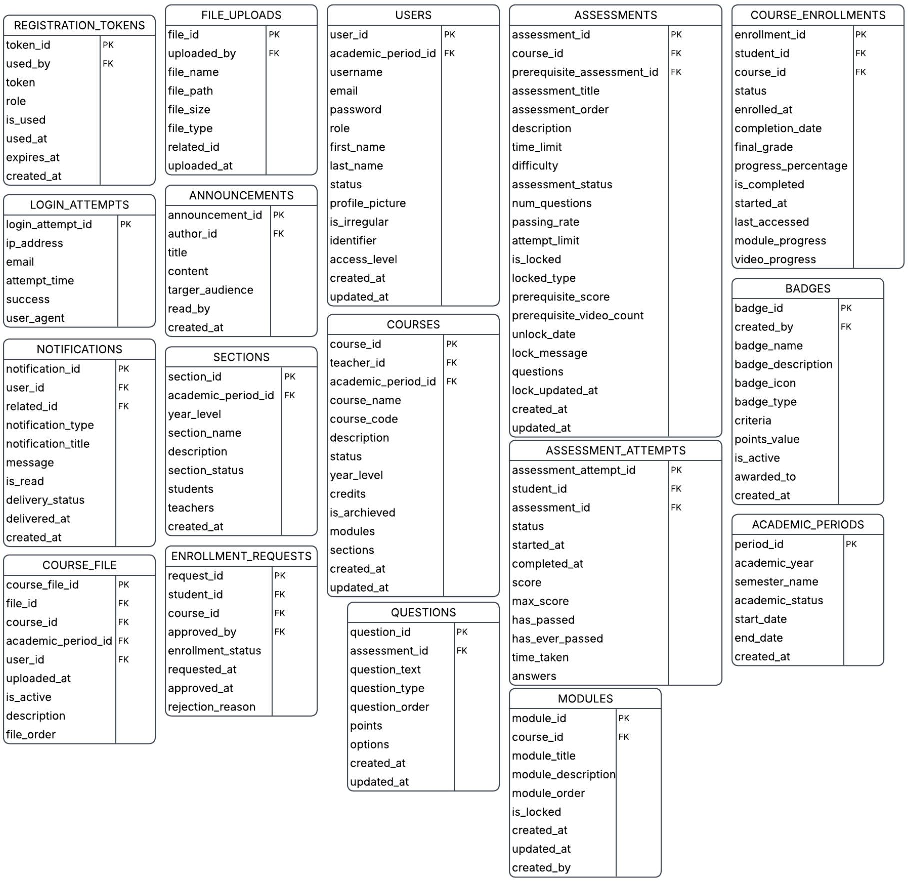


Figure 6 presents the database schema for the Learning Management System. Data concerning users, teachers, students, courses, and assessments are organized into distinct, normalized tables. This structure eliminates data redundancy by storing each piece of information such as a user's profile or a course's details in a single location. This design significantly enhances the database's manageability, integrity, and performance. As a result of this efficient data architecture, the platform can reliably handle complex operations like course enrollment, assessment processing, and personalized learning recommendations.

Entity-Relationship Diagram (ERD)

An Entity-Relationship Diagram (ERD) models the underlying data structure for the Learning Management System. It graphically represents key entities such as Users, Teachers, Students, Courses, and Assessments along with their attributes and the critical relationships that connect them. This blueprint defines how data interacts, illustrating connections like which student enrolls in a course or which teacher creates a specific assessment.

Figure 7. *Entity-Relationship Diagram*

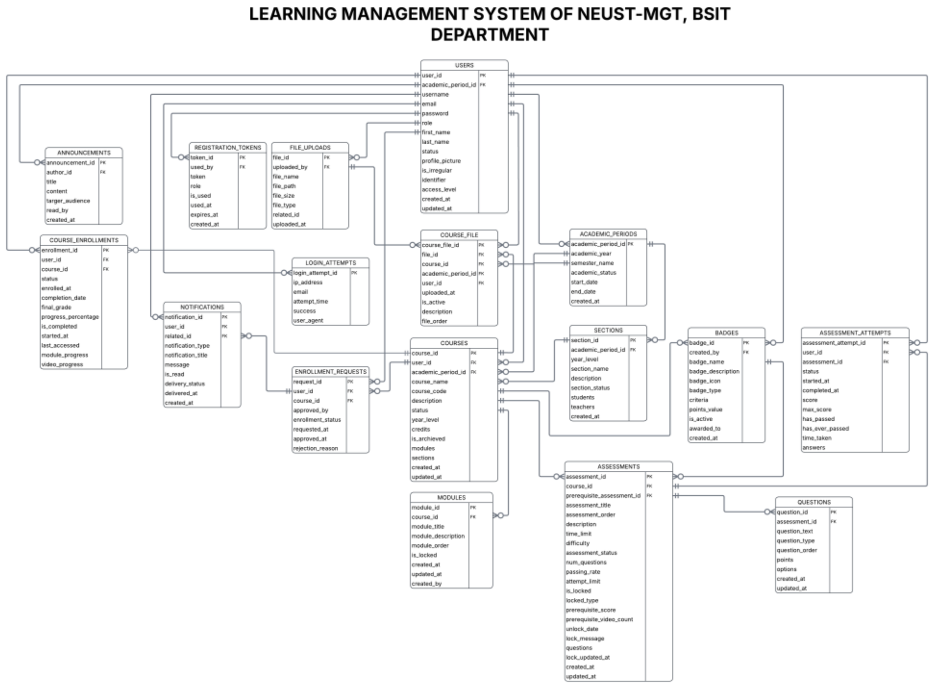


Figure 7 showcases the various entities and their connections within the database of the Learning Management System. It outlines how the different components of the system interact with one another. The diagram features essential tables that play a crucial role in the system's overall functionality.

The users table holds the information of all teachers and students who sign up to access the platform, including their profile details and login information. On the flip side, the students table contains the data of all users who register to enroll in courses and take assessments.

At the heart of the system is the courses table, where all course information submitted by teachers are kept. This table includes important details like the course's title, description, academic year, semester, and enrollment status. Its status is tracked through various fields which govern its journey from creation to active learning.

The assessments table records every assessment that takes place. It links a course to student attempts, capturing the score, completion time, and assessment results. This table is vital for identifying student performance and determining their academic progress.

This ERD provides a clear view of the database structure, highlighting the key elements related to user management, course management, and assessment features within the system.

Data Dictionary

A data dictionary is a central repository of metadata that explains the structure and meaning of data in a system. It helps in understanding the database schema, supports efficient database management, and ensures data integrity and consistency in the online art auction platform.

Table 19. *Data Dictionary of users*

Attribute	Data Type	Contraints	Field Size	Description
user_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	20	Unique identifier for a user
username	varchar	NOT NULL, UNIQUE	50	User's login username
email	varchar	NOT NULL, UNIQUE	100	User's email address
password	varchar	NOT NULL	255	Hashed password for authentication
first_name	varchar	NOT NULL	50	User's first name
last_name	varchar	NOT NULL	50	User's last name
role	enum	NOT NULL	-	User role
status	enum	NOT NULL, DEFAULT 'active'	-	Account status
profile_picture	varchar	NULL	255	Path to profile picture file
is_irregular	tinyint	DEFAULT 0	1	Flag for irregular students
identifier	varchar	NULL	20	Student ID, Teacher ID, or Admin ID based on role
department	varchar	NULL	100	Department name (for teachers only)
access_level	enum	NULL	-	Access level for admins
academic_period_id	int	NULL, Foreign Key	25	Academic period ID (for students only)

created_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Record creation timestamp
updated_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP ON UPDATE	19	Record last update timestamp

Table 20. *Data Dictionary of section*

Attribute	Data Type	Constraints	Field Size	Description
Section_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for section
year_level	int	NOT NULL, DEFAULT 1	11	Year level of the section
section_name	varchar	NOT NULL	100	Name of the section
description	text	NULL	65,535	Section description
is_active	tinyint	NOT NULL, DEFAULT 1	1	Flag indicating if section is active
academic_period_id	int	NOT NULL, Foreign Key	11	Academic period this section belongs to

students	long text	NULL	4,294,967,295	JSON array of student IDs
teachers	long text	NULL	4,294,967,295	JSON array of teacher IDs
created_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Record creation time stamp

Table 21. *Data Dictionary of registration_tokens*

Attribute	Data Type	Constraints	Field Size	Description
token_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for token
token	varchar	NOT NULL, UNIQUE	64	Registration token string
role	enum	NOT NULL	-	Role for registration
is_used	tinyint	DEFAULT 0	1	Flag indicating if token is used
used_by	int	NULL, Foreign Key	11	ID of user who used this token
used_at	timestamp	NULL	19	Timestamp when token was used
expires_at	timestamp	NOT NULL, DEFAULT	19	Token expiration timestamp

created_at	timestamp	CURRENT_TIMESTAMP ON UPDATE NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Token creation timestamp

Table 22. *Data Dictionary of questions*

Attribute	Data Type	Constraints	Field Size	Description
question_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for question
assessment_id	varchar	NOT NULL, Foreign Key	50	ID of the assessment this question belongs to
question_text	text	NOT NULL	65,535	The question content
question_type	varchar	NOT NULL, DEFAULT	32	Type of question
question_order	int	NOT NULL	11	Order of question in assessment

points	int	DEFAULT 1	11	Points awarded for correct answer
options	longtext	NULL	4,294,967,295	JSON array of question options
created_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Record creation timestamp
updated_at	timestamp	NULL, DEFAULT NULL ON UPDATE CURRENT_TIMESTAMP	19	Record last update timestamp

Table 23. *Data Dictionary of notifications*

Attribute	**Data Type**	**Constraints**	**Field Size**	**Description**
notification_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for notification
user_id	int	NOT NULL, Foreign Key	11	ID of the user receiving notification

title	varchar	NOT NULL	255	Title of the notification
message	text	NOT NULL	65,535	Content of the notification
Notification_type	enum	NOT NULL, DEFAULT	-	Type of notification
related_id	int	NULL	11	Related ID (e.g., enrollment_request_id)
is_read	tinyint	NOT NULL	1	Flag indicating if notification is read
delivery_status	enum	DEFAULT	-	Delivery status
delivered_at	timestamp	NULL	19	Delivery timestamp
created_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Record creation timestamp

Table 24. *Data Dictionary of login_attempts*

Attribute	Data Type	Constraints	Field Size	Description

login_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for login attempt
ip_address	varchar	NOT NULL	45	IP address of the attempt
email	varchar	NOT NULL	100	Email used in attempt
attempt_time	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Timestamp of attempt
success	tinyint	NOT NULL	1	Flag indicating if attempt was successful
user_agent	text	NULL	65,535	User agent string

Table 25. *Data Dictionary of file_uploads*

Attribute	Data Type	Constraints	Field Size	Description
file_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for file upload
file_name	varchar	NOT NULL	255	Original name of the file
file_path	varchar	NOT NULL	255	Path to stored file

file_size	bigi nt	NULL	20	File size in bytes
file_type	varc har	NULL	100	Name type of the file
uploaded_ by	int	NOT NULL, Foreign Key	11	ID of user who uploaded file
related_i d	int	NULL	11	ID of related entity
uploaded_ at	date time	DEFAULT CUR RENT_TIMEST AMP	19	Upload timestamp

Table 26. *Data Dictionary of enrollment_requests*

Attribute	Data Type	Constraints	Field Size	Description
request_id	int	Primary Key , NOT NULL, AUTO_INCREM ENT	11	Unique identifier for enrollment request
student_id	int	NOT NULL, Foreign Key	11	ID of the student makin g request
course_id	int	NOT NULL, Foreign Key	11	ID of the course being requested
enrollment _status	enum	NOT NULL, DEFAULT 'pending'	-	Request statu s

requested_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Request timestamp
approved_at	timestamp	NULL	19	Approval timestamp
approved_by	int	NULL, Foreign Key	11	ID of user who approved request
rejection_reason	text	NULL	65,535	Reason for rejection

Table 27. *Data Dictionary of course_enrollments*

Attribute	Data Type	Constraints	Field Size	Description
enrollment_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for enrollment
student_id	int	NOT NULL, Foreign Key	11	ID of the enrolled student
course_id	int	NOT NULL, Foreign Key	11	ID of the course
status	enum	NOT NULL, DEFAULT 'active'	-	Enrollment status

enrolled_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Enrollment timestamp
completion_date	timestamp	NULL	19	Course completion date
final_grade	decimal	NULL	5,2	Final grade for the course
progress_percentage	decimal	DEFAULT 0.00	5,2	Course completion percentage
is_completed	tinyint	DEFAULT 0	1	Flag indicating if course is completed
started_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Course start timestamp
last_accessed	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP ON UPDATE	19	Last access timestamp
module_progress	longtext	NULL	4,294,967,295	JSON tracking progress per module
video_progress	longtext	NULL	4,294,967,295	JSON tracking video completion

Table 28. *Data Dictionary of courses*

Attribute	Data Type	Constraints	Field Size	Description
course_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for course
course_name	varchar	NOT NULL	100	Full name of the course
course_code	varchar	NOT NULL	20	Course code
description	text	NULL	65,535	Course description
teacher_id	int	NOT NULL, Foreign Key	11	ID of the teacher assigned to course
status	enum	NOT NULL, DEFAULT 'active'	-	Course status
academic_period_id	int	NOT NULL, Foreign Key	30	Academic period this course belongs to
year_level	varchar	NULL	10	Target year level
credits	int	DEFAULT 3	11	Number of credit units

is_archived	tinyint(1)	DEFAULT 0	1	Flag indicating if course is archived
modules	longtext	NULL	4,294,967,295	JSON array of modules with videos
sections	longtext	NULL	4,294,967,295	JSON array of section IDs
created_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Record creation timestamp
updated_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP ON UPDATE	19	Record last update timestamp

Table 29. *Data Dictionary of badges*

Attribute	Data Type	Constraints	Field Size	Description
badge_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for badge
badge_name	varchar	NOT NULL	50	Name of the badge
badge_description	text	NULL	65,535	Description of the badge

badge_icon	varchar	NOT NULL	255	Path to badge icon file
badge_type	enum	DEFAULT 'participation'	-	Type of badge
criteria	longtext	NULL	4,294,967,295	JSON criteria for earning the badge
points_value	int	DEFAULT 0	11	Points awarded for earning this badge
created_by	int	NULL, Foreign Key	11	ID of user who created the badge
is_active	tinyint(1)	DEFAULT 1	1	Flag indicating if badge is active
awarded_to	longtext	NULL	4,294,967,295	ID of users who earned this badge
created_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Record creation timestamp

Table 30. *Data Dictionary of assessment_attempts*

Attribute	Data Type	Constraints	Field Size	Description

assessment_attempts_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for attempt
student_id	int	NOT NULL, Foreign Key	11	ID of the student taking assessment
assessment_id	varchar	NULL, Foreign Key	50	ID of the assessment
status	enum	DEFAULT 'in_progress'	-	Attempt status
started_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Attempt start timestamp
completed_at	timestamp	NULL	19	Attempt completion timestamp
score	decimal	NULL	5,2	Student's score
max_score	decimal	NULL	5,2	Maximum possible score
has_passed	tinyint(1)	DEFAULT 0	1	Flag indicating if attempt passed
has_ever_passed	tinyint	DEFAULT 0	1	Flag indicating

				if student ever passed
time_taken	int	NULL	11	Time taken in seconds
answers	longtext	NULL	4,294,967,295	JSON array of student answers

Table 31. *Data Dictionary of assessments*

Attribute	Data Type	Constraints	Field Size	Description
assessment_id	varchar	Primary Key, NOT NULL	50	Unique string identifier for assessment
course_id	int	NOT NULL, Foreign Key	11	ID of the course this assessment belongs to
assessment_title	varchar	NOT NULL	100	Title of the assessment
assessment_order	int	DEFAULT 1	11	Order of assessment within course
description	text	NULL	65,535	Assessment description

time_limit	int	NULL	11	Time limit in minutes
difficulty	enum	DEFAULT 'medium'	-	Difficulty level
assessment_status	enum	DEFAULT 'active'	-	Assessment status
num_questions	int	DEFAULT 10	11	Number of questions in assessment
passing_rate	decimal	DEFAULT 70.00	5,2	Minimum percentage to pass
attempt_limit	int	DEFAULT 3	11	Maximum number of attempts allowed
is_locked	tinyint	DEFAULT 0	1	Flag indicating if assessment is locked
lock_type	enum	DEFAULT 'manual'	-	Type of lock mechanism
prerequisite_assessment_id	varchar	NULL, Foreign Key	50	ID of prerequisite assessment

prerequisite_ score	deci mal	NULL	5,2	Required score on prerequisit e assessment
prerequisite_ video_count	int	NULL	11	Required number of videos to watch
unlock_date	date time	NULL	19	Date when assessment unlocks
lock_message	text	NULL	65,535	Message shown when assess ment is locked
questions	long text	DEFAULT '[]'	4,294, 967,29 5	JSON array of qu estions
lock_updated_ at	time stam p	NULL	19	Timestamp when lock was last update d
created_at	time stam p	NOT NULL, DEFAULT CURRENT_TIME STAMP	19	Record creation timestamp

updated_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP ON UPDATE	19	Record last update timestamp

Table 32. *Data Dictionary of announcements*

Attribute	Data Type	Constraints	Field Size	Description
announcement_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for announcement
title	varchar	NOT NULL	100	Title of the announcement
content	text	NOT NULL	65,535	Content of the announcement
author_id	int	NOT NULL, Foreign Key	11	ID of the author
target_audience	longtext	NULL	4,294,967,295	JSON array of target users/courses/sections
read_by	longtext	NULL	4,294,967,295	JSON array of users who read this
created_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Record creation timestamp

Table 33. *Data Dictionary of academic_periods*

Attribute	Data Type	Constraints	Field Size	Description
period_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for academic period
academic_year	varchar	NOT NULL	20	Academic year
semester_name	varchar	NOT NULL	50	Name of the semester
is_active	tinyint	DEFAULT 1	1	Flag indicating if period is active
start_date	date	NULL	10	Academic period start date
end_date	date	NULL	10	Academic period end date
created_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Record creation timestamp

Table 34. *Data Dictionary of course_files*

Attribute	Data Type	Constraints	Field Size	Description
course_file_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for course file

file_id	int	Foreign Key, NOT NULL	11	References FILE_UPLOADS table
course_id	int	Foreign Key, NOT NULL	11	References COURSES table
academic_period_id	int	Foreign Key, NOT NULL	11	References ACADEMIC_PERIODS table
user_id	int	Foreign Key, NOT NULL	11	References USERS table (uploader)
uploaded_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	File upload timestamp
is_active	tinyint	DEFAULT 1	1	Flag indicating if course file is active
description	text	NULL	65535	Description of the course file
file_order	int	DEFAULT 1	11	Display order of the file within course

Table 35. *Data Dictionary of modules*

Attribute	Data Type	Constraints	Field Size	Description
module_id	int	Primary Key, NOT NULL, AUTO_INCREMENT	11	Unique identifier for module
course_id	int	Foreign Key , NOT NULL	11	References COURSES table
module_title	varchar	NOT NULL	255	Title of the module
module_description	text	NULL	65535	Description of the module
module_order	int	DEFAULT 1	11	Display order of the module within course
is_locked	tinyint	DEFAULT 0	1	Flag indicating if module is locked
created_at	timestamp	NOT NULL, DEFAULT CURRENT_TIMESTAMP	19	Module creation timestamp

updated_at	timestamp	NULL, ON UPDATE CURRENT_TIMESTAMP	19	Module last update timestamp
created_by	int	Foreign Key, NOT NULL	11	References USERS table (creator)

2.3 Development

The development phase focuses on implementing the system's functional and non-functional requirements through programming. A Data Flow Diagram (DFD) shows how data moves within the system, using elements like external entities, processes, data flows, and data stores to illustrate interactions and points of storage.

Context Diagram

Context Diagram gives the top-level view of a Data Flow Diagram (DFD). It presents the system as a single process and shows the data exchanged between the system and external entities, without displaying the internal details.

Figure 8. *Context Diagram*

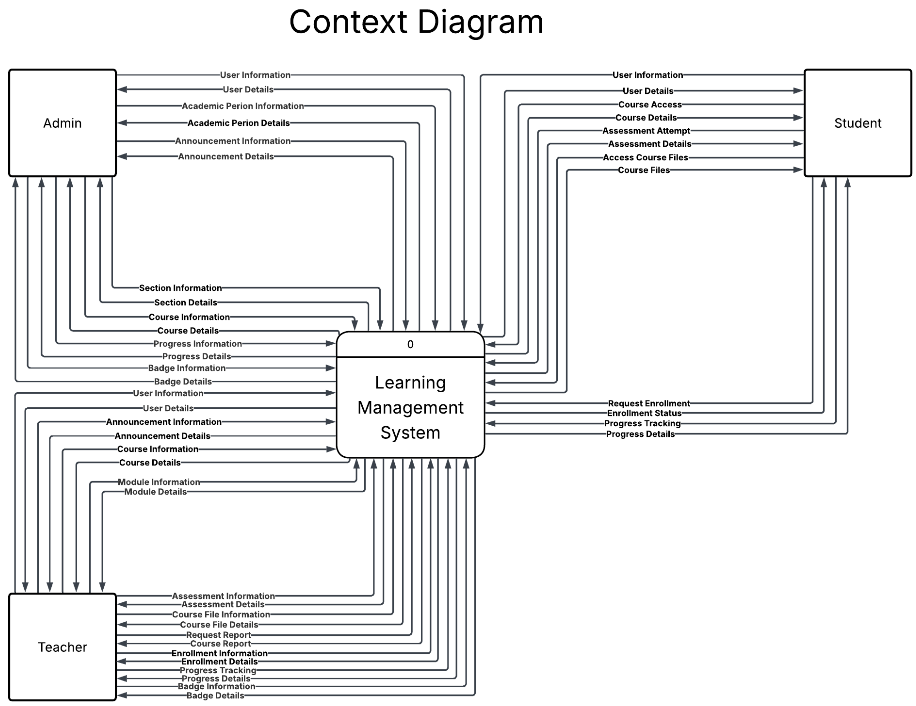


Figure 10 provides a high-level overview of the Learning Management System's interactions. The main participants include Teachers, Students and Admins.

Teachers and Students engage with the platform to perform key functions such as user registration, account authentication, browsing and enrolling in courses, and managing assessments. Admins oversee the listing of courses, coordinate academic events, validate enrollments, manage the technical infrastructure, monitor system performance, and generate operational reports.

The diagram outlines the scope of the platform, emphasizing core features like course listing, student enrollment, assessment processing, and user management all central to the Learning Management System presented in this study.

Data Flow Diagram - Level 0

A Level 0 Data Flow Diagram (DFD) shows the main processes of a system and their interactions with external entities and data stores. It offers a high-level view of data flow, emphasizing key functions and their connections to inputs, outputs, and storage within the system.

Figure 9. *Data Flow Diagram - Level 0*

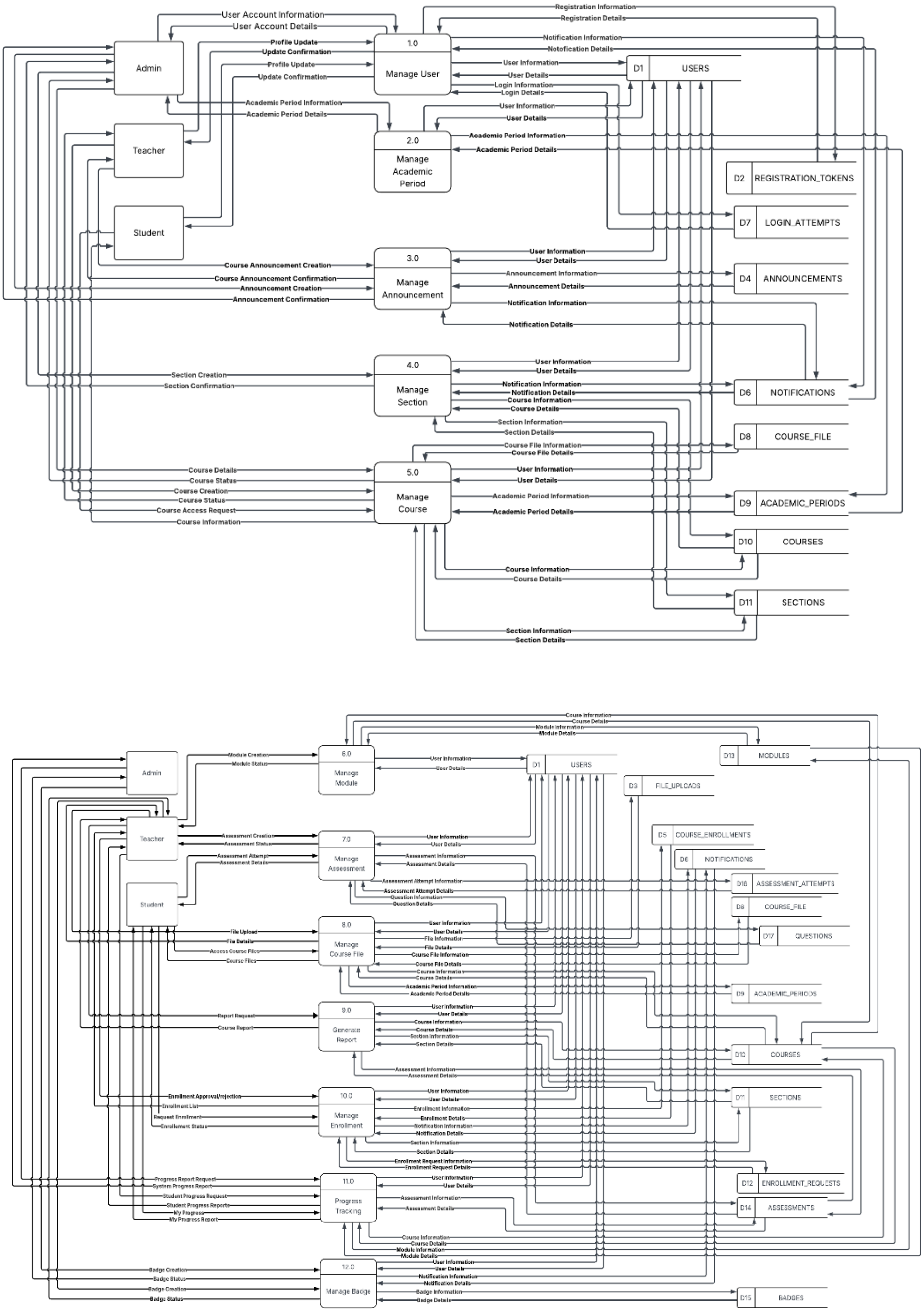
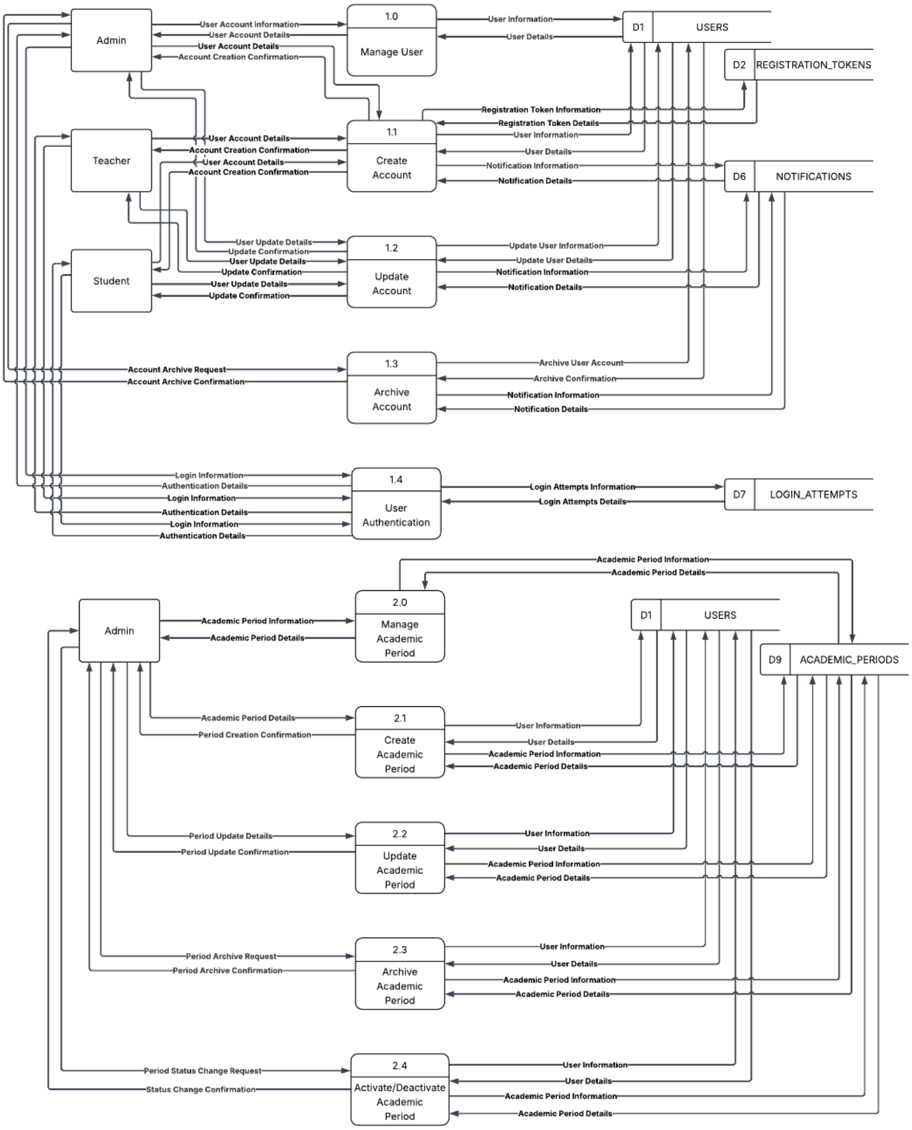


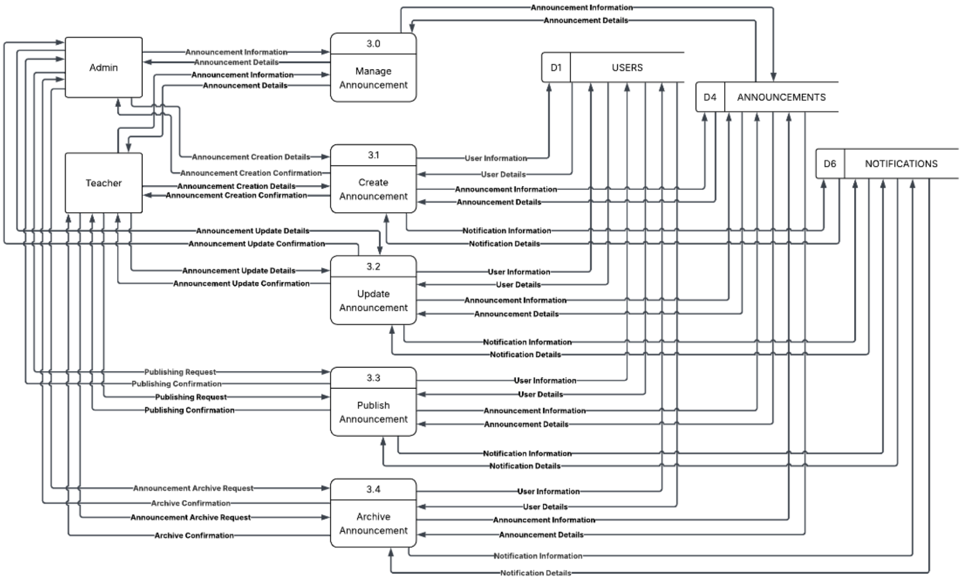
Figure 9 shows how data flows through the system. In DFD level 0, we can see several key processes: Login, Manage Users, Manage Courses, Manage Assessments, Manage Enrollments, Manage Modules, and Manage Reports. The diagram above highlights how data interacts between these functions and the relevant data stores, laying the groundwork for the Learning Management System.

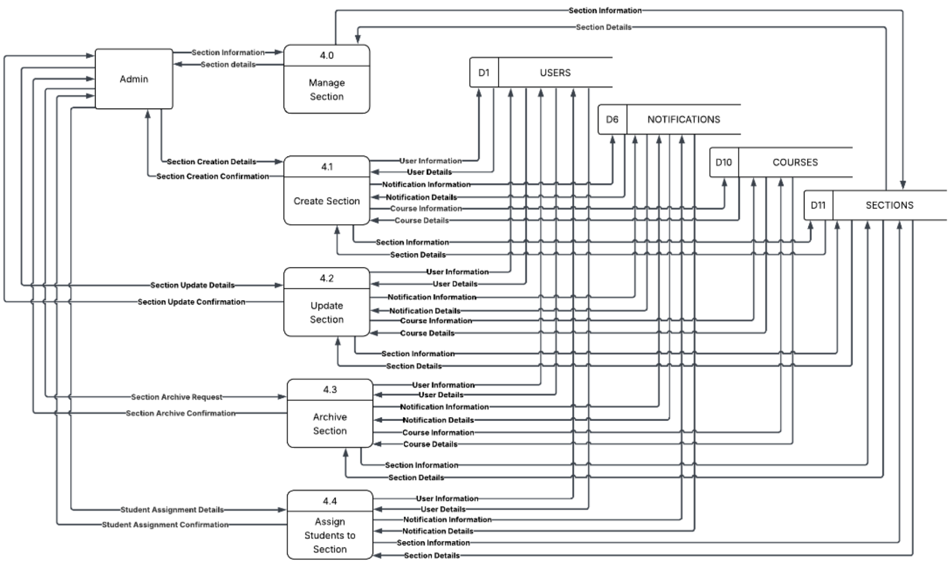
Data Flow Diagram - Level 1

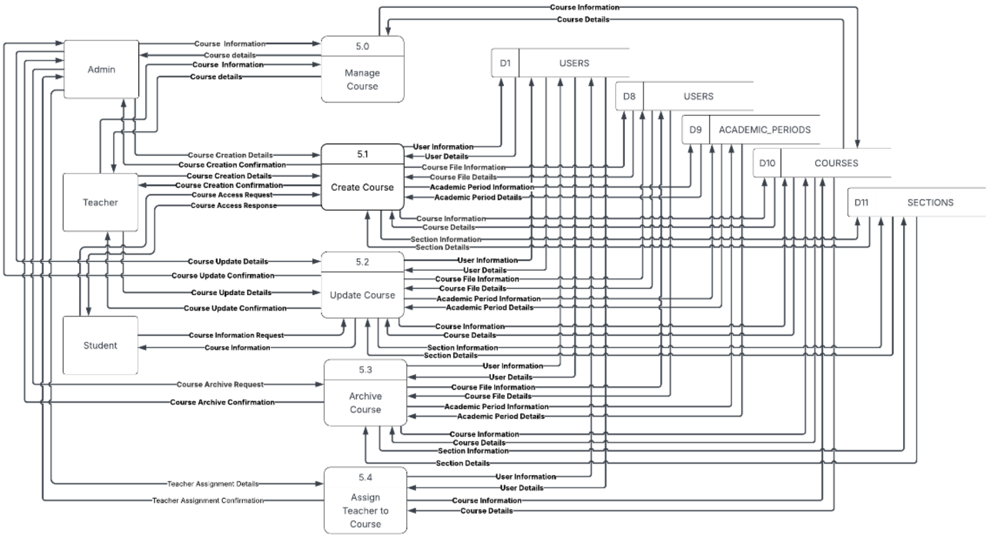
A Level 1 Data Flow Diagram (DFD) breaks down the main processes from Level 0 into detailed sub-processes. It gives a closer view of the system’s internal functions, data flows, and interactions, showing how each core activity is organized and carried out within the overall system.

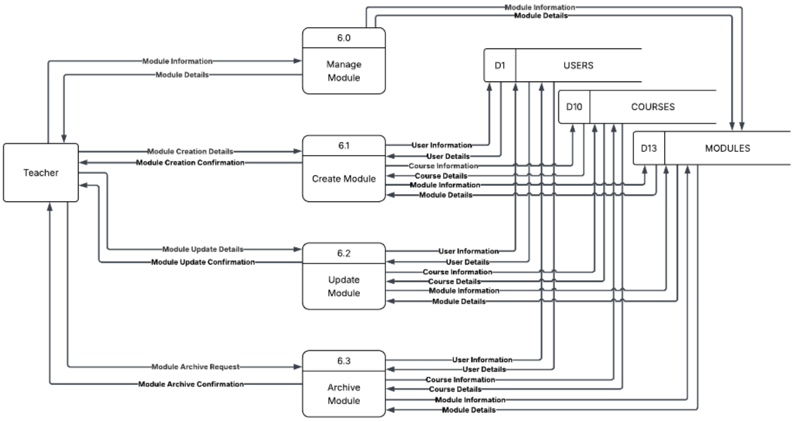
Figure 10. *Data Flow Diagram – Level 1*

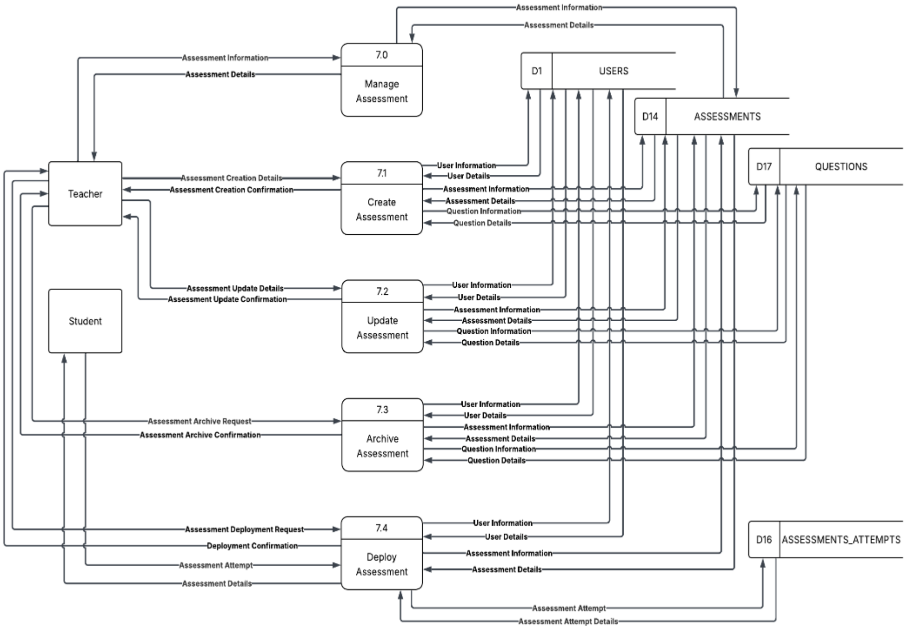



Admin
Teacher
3.0
Manage Announcement
3.1
Create Announcement
3.2
Update Announcement
3.3
Publish Announcement
3.4
Archive Announcement
D1 USERS
D4 ANNOUNCEMENTS
D6 NOTIFICATIONS
Announcement Information
Announcement Details
Announcement Creation Details
Announcement Creation Confirmation
Announcement Update Details
Announcement Update Confirmation
Publishing Request
Publishing Confirmation
Announcement Archive Request
Archive Confirmation
User Information
User Details
Notification Information
Notification Details

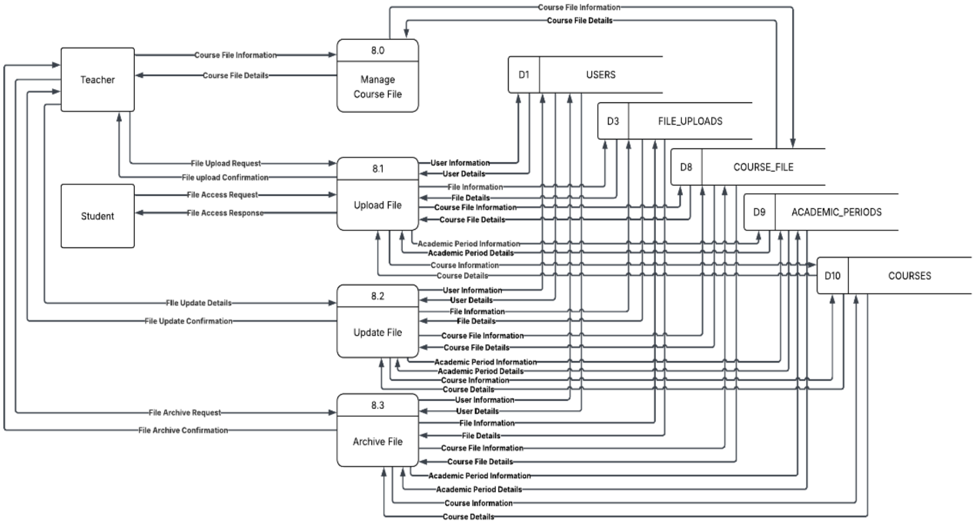

Admin
4.0
Manage Section
4.1
Create Section
4.2
Update Section
4.3
Archive Section
4.4
Assign Students to Section
D1 USERS
D6 NOTIFICATIONS
D10 COURSES
D11 SECTIONS
Section Information
Section Details
Section details
Section Creation Details
Section Creation Confirmation
Section Update Details
Section Update Confirmation
Section Archive Request
Section Archive Confirmation
Student Assignment Details
Student Assignment Confirmation
User Information
User Details
Notification Information
Notification Details
Course Information
Course Details


Admin
Teacher
Student
5.0
Manage Course
5.1
Create Course
5.2
Update Course
5.3
Archive Course
5.4
Assign Teacher to Course
D1
USERS
D8
USERS
D9
ACADEMIC_PERIODS
D10
COURSES
D11
SECTIONS
Course Information
Course Details
Course details
Course Creation Details
Course Creation Confirmation
Course Access Request
Course Access Response
Course Update Details
Course Update Confirmation
Course Information Request
Course Archive Request
Course Archive Confirmation
Teacher Assignment Details
Teacher Assignment Confirmation
User Information
User Details
Course File Information
Course File Details
Academic Period Information
Academic Period Details
Section Information
Section Details

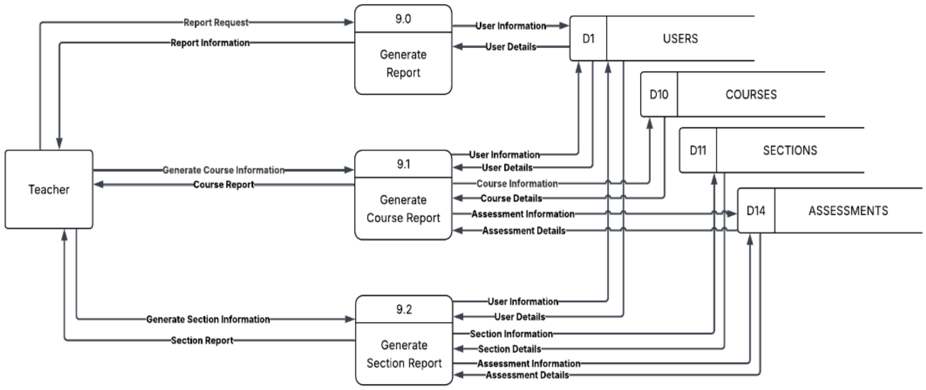

Teacher
6.0
Manage Module
6.1
Create Module
6.2
Update Module
6.3
Archive Module
D1
USERS
D10
COURSES
D13
MODULES
Module Information
Module Details
Module Creation Details
Module Creation Confirmation
Module Update Details
Module Update Confirmation
Module Archive Request
Module Archive Confirmation
User Information
User Details
Course Information
Course Details

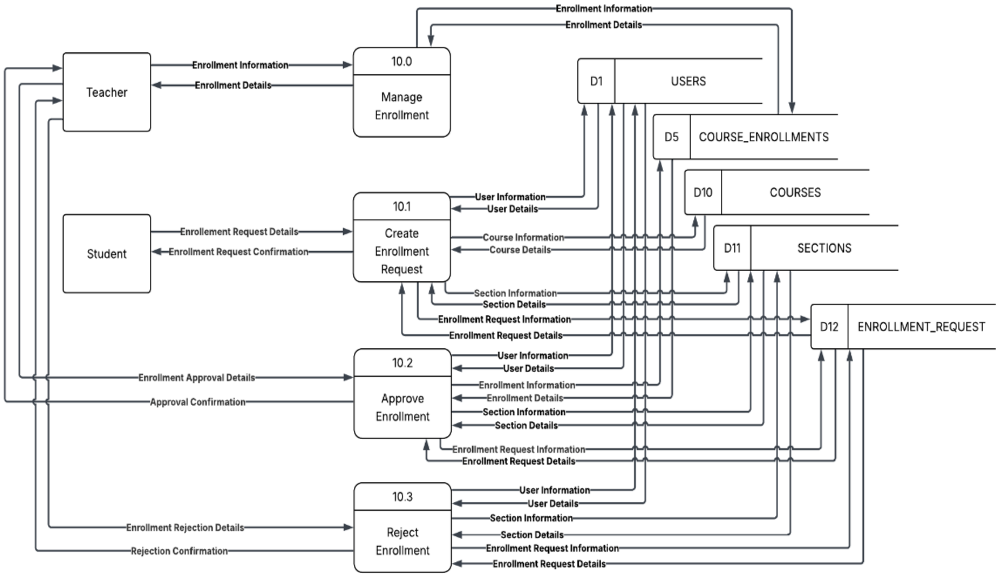


7.0
Manage Assessment
7.1
Create Assessment
7.2
Update Assessment
7.3
Archive Assessment
7.4
Deploy Assessment
Teacher
Student
D1
USERS
D14
ASSESSMENTS
D17
QUESTIONS
D16
ASSESSMENTS_ATTEMPTS
Assessment Information
Assessment Details
Assessment Creation Details
Assessment Creation Confirmation
Assessment Update Details
Assessment Update Confirmation
Assessment Archive Request
Assessment Archive Confirmation
Assessment Deployment Request
Deployment Confirmation
Assessment Attempt
Assessment Attempt Details
User Information
User Details
Question Information
Question Details



8.0
Manage Course File
8.1
Upload File
8.2
Update File
8.3
Archive File
Teacher
Student
D1
USERS
D3
FILE_UPLOADS
D8
COURSE_FILE
D9
ACADEMIC_PERIODS
D10
COURSES
Course File Information
Course File Details
File Upload Request
File upload Confirmation
File Access Request
File Access Response
File Update Details
File Update Confirmation
File Archive Request
File Archive Confirmation
User Information
User Details
File Information
File Details
Academic Period Information
Academic Period Details
Course Information
Course Details


9.0
Generate Report
Report Request
Report Information
User Information
User Details
D1
USERS
D10
COURSES
D11
SECTIONS
D14
ASSESSMENTS
Teacher
9.1
Generate Course Report
Generate Course Information
Course Report
Course Information
Course Details
Assessment Information
Assessment Details
9.2
Generate Section Report
Generate Section Information
Section Report
Section Information
Section Details


10.0
Manage Enrollment
Enrollment Information
Enrollment Details
Teacher
Student
D1
USERS
D5
COURSE_ENROLLMENTS
D10
COURSES
D11
SECTIONS
D12
ENROLLMENT_REQUEST
10.1
Create Enrollment Request
Enrollement Request Details
Enrollment Request Confirmation
User Information
User Details
Course Information
Course Details
Section Information
Section Details
Enrollment Request Information
Enrollment Request Details
10.2
Approve Enrollment
Enrollment Approval Details
Approval Confirmation
10.3
Reject Enrollment
Enrollment Rejection Details
Rejection Confirmation

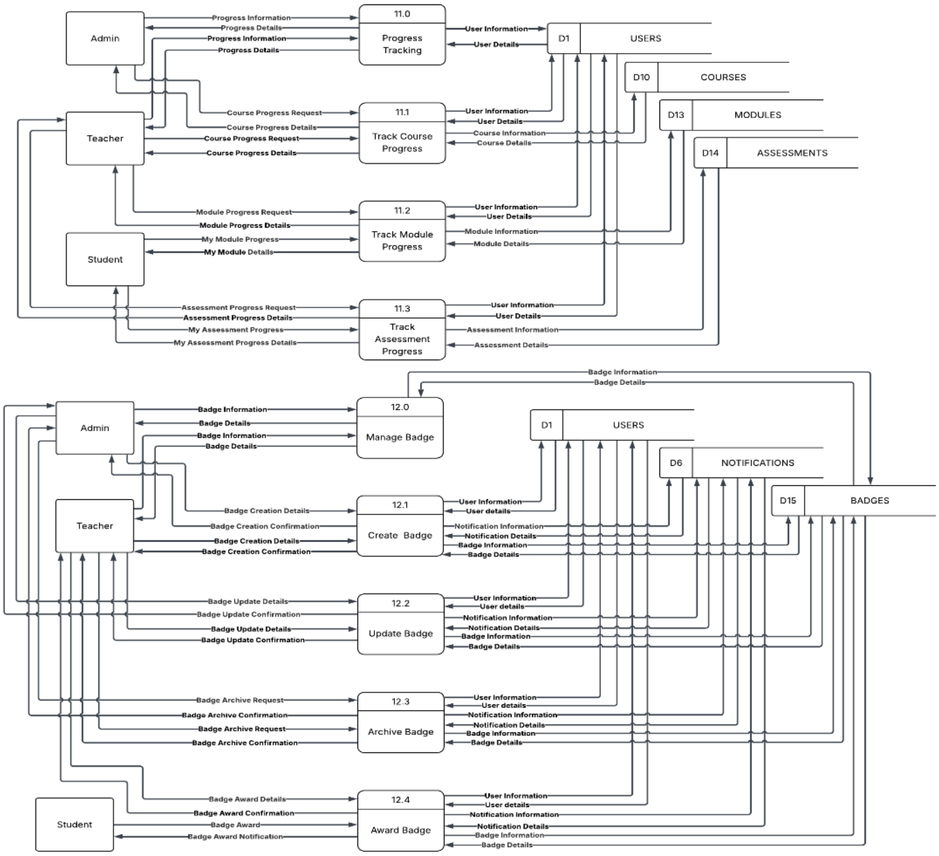


Figure 10 *illustrates how data flows through the various sub-processes involved in managing users, courses, assessments, enrollments, and modules.*

In the user management section, we break it down into tasks like adding users, editing user details, and overseeing user accounts. When adding users, we gather new account details and information from the Admin, Teacher, and Student. Editing users involves collecting updated information from these same groups. The Manage User Account process (1.0) takes care of account details and interacts with the User data through data store D1 User Details.

Course management follows a similar structure, with tasks for adding new courses, editing existing ones, and managing the current collection. Adding courses means collecting new details from the Teacher, while editing requires gathering updated course information from them. The Manage Courses and Content process (2.0) manages course details and exchanges information with the Course data store (D2) and File Storage (D7) for learning materials and video content.

In assessment management, we have processes for adding an assessment, starting it, monitoring its progress, and concluding it. The Teacher supplies the necessary assessment information to add (3.1) and configure assessment settings (3.2). Students attempt assessments (3.3) and receive results. The Manage Assessment process (3.0) oversees assessment details and exchanges information with the Assessment data store (D3).

Enrollment management encompasses placing enrollments, updating them, and viewing enrollment details. The student submits new enrollment information to place an enrollment (4.1) and the system handles approval/rejection (4.2). The Admin and Teacher manage section assignments (4.3). The Manage Enrollments process (4.0) handles enrollment details and manages the flow of enrollment information with the Enrollment data store (D4).

Additionally, the system includes progress tracking and badge management functionality. The Track Progress and Badges process (5.0) monitors student progress, generates reports (5.1), tracks video completion (5.2), and manages badge creation and assignment (5.3). This interacts with the Progress data store (D5) and Badge data store (D6).

2.4 Testing

During the testing phase, the Learning Management System was rigorously evaluated to ensure it met all specified functional and non-functional requirements. IT experts conducted systematic technical tests to assess system integration, security protocols, and performance under load, ensuring the platform's infrastructure was robust and scalable.

Concurrently, acceptance testing was carried out with end-users including teachers, students, and system administrators. Their feedback focused on real-world usability, evaluating critical features such as the course enrollment interface, assessment processing, module submission, and learning management. Insights from both IT consultants and end-users were synthesized to refine the platform, enhancing its reliability, security, and user experience before deployment.

2.5 Deployment

After testing, the Learning Management System will be deployed to a live production environment. This step includes setting up servers, databases, and security measures, as well as securing final approval from stakeholders and end-users. Deployment will take place only once all test cases are passed, ensuring the system's functionality, accuracy, and compliance with software standards.

2.6 Maintenance

The maintenance phase ensures the long-term performance and relevance of the Learning Management System through systematic updates and improvements. This stage involves four key maintenance approaches:

1. Predictive Maintenance:

Leveraging historical system data and monitoring tools to anticipate potential issues such as server overload during high-traffic enrollment periods or vulnerabilities in assessment processing enabling proactive resolution before problems affect users.

2. Perfective Maintenance:

Implementing enhancements to align with evolving user needs and technological trends, such as introducing new learning features, improving mobile responsiveness, or optimizing video loading for course modules.

3. Adaptive Maintenance:

Modifying the system to accommodate changes in external environments, such as updates to third-party authentication services, new security regulations, or compatibility requirements with evolving web standards.

4. Preventive Maintenance:

Proactively addressing latent issues such as code refactoring, database optimization, or security patching to reduce the risk of future failures and ensure system stability.

Together, these strategies ensure the platform remains efficient, secure, and aligned with user expectations while minimizing disruptions and extending the software's lifecycle.

3. How may the elected IT consultants evaluate the developed applications in terms of the following software characteristics:

Results of the Assessment Made by elected IT Consultants on the Systems’ Technical Qualities

3.1 Functional Suitability

Table 36 shows the result of the assessment made by the elected IT Experts regarding the functional suitability of the system.

Table 36. *Results of the Assessment Made by elected IT Consultants on Suitability the Systems' Functional*

Item Statements	Weighted Mean	Verbal Description
The LMS accurately supports real-time grading for quizzes and exams.	4.00	Strongly Agree
The system allows seamless student enrollment and subject management.	3.33	Strongly Agree
Question randomization and timer-based exams function correctly.	4.00	Strongly Agree
The LMS provides reliable analytics for student performance tracking.	3.67	Strongly Agree
All core features (e.g., resource uploads, grading) operate without errors.	3.67	Strongly Agree
Grand Mean	**3.73**	
Verbal	**Strongly Agree**	

As shown in Table 36, the respondents gave a grand mean of **3.73**, interpreted as **Strong Agree**, for the functional suitability of the Learning Management System. This indicates that IT consultants strongly agreed that the LMS meets its intended requirements with accuracy and completeness. They noted that features such as real-time grading, seamless student enrollment, question randomization, and reliable analytics function as designed without errors.

Functional suitability is a key measure of software quality because it reflects how well the system delivers on its intended purpose. The positive evaluation implies that the LMS supports academic tasks effectively, streamlines processes, and produces accurate results that align with institutional objectives.

These results suggest that the LMS is highly functional and capable of supporting both teaching and learning requirements. To maintain this level of performance, ongoing system monitoring and updates are recommended to ensure that the LMS continues to operate according to institutional needs and user expectations.

3.2 Performance Efficiency

Table 37 shows the result of the assessment made by the elected IT Consultants regarding the performance efficiency of the system.

Table 37. *Results of the Assessment Made by elected IT Consultants on the Systems' Performance Efficiency.*

Item Statements	Weighted Mean	Verbal Description
The LMS processes real-time grading quickly with no noticeable delays.	4.00	Strongly Agree
The system handles multiple simultaneous users (e.g., 100 students) efficiently.	3.33	Strongly Agree
Resource uploads (e.g., PDFs, videos) are processed without performance issues.	4.00	Strongly Agree
The LMS performs consistently under varying workloads.	3.67	Strongly Agree
The system uses server resources efficiently with minimal lag.	3.67	Strongly Agree
Grand Mean	**3.73**	
Verbal	**Strongly Agree**	

As shown in Table 37, the respondents gave a grand mean of **3.73,** interpreted as **Strongly Agree,** for the performance efficiency of the Learning Management System. This indicates that IT consultants strongly agreed that the LMS processes tasks quickly, handles multiple users effectively, and manages resource uploads without noticeable delays. They also confirmed that the system performs consistently under varying workloads while using server resources efficiently.

Performance efficiency is a critical aspect of software quality, as it ensures that the system remains responsive and reliable even with increased user demand. The positive evaluation suggests that the LMS is well-optimized to support real-time grading, content delivery, and simultaneous access by students and faculty.

These findings imply that the LMS delivers stable and efficient performance, minimizing disruptions in teaching and learning activities. To sustain this level of efficiency, regular monitoring of server load and system optimization is recommended as the number of users grows.

3.3 Compatibility

Table 38 shows the result of the assessment made by the elected IT Consultants regarding the compatibility of the system.

Table 38. *Results of the Assessment Made by elected IT Consultants on the System's Compatibility*

Item Statements	Weighted Mean	Verbal Description
The LMS integrates seamlessly with common browsers (e.g., Chrome, Firefox).	3.67	Strongly Agree
The system is compatible with standard desktop browsers (e.g., Chrome, Firefox) for student and faculty access.	4.00	Strongly Agree
The LMS supports various file formats (e.g., PDF, PPT, MP4) for resource uploads.	3.67	Strongly Agree
The system is compatible with standard operating systems (e.g., Windows, macOS) for student and faculty access.	3.67	Strongly Agree
The LMS is fully responsive and compatible with mobile devices (e.g., smartphones, tablets) for student and faculty access.	3.33	Strongly Agree

Grand Mean	**3.67**
Verbal	**Strongly Agree**

As shown in Table 38, the respondents gave a grand mean of **3.67,** interpreted as **Strongly Agree,** for the compatibility of the Learning Management System. This indicates that IT consultants strongly agreed that the LMS integrates well with common browsers, supports various file formats, and is accessible across different operating systems and devices, including desktops and mobile platforms.

Compatibility is an important factor in software quality because it ensures that the system functions smoothly across diverse environments and user setups. The positive rating suggests that the LMS provides flexibility for both students and faculty, allowing access through multiple devices and formats without technical issues.

These results imply that the LMS is capable of supporting a wide range of academic tasks and user preferences, increasing accessibility and convenience. Continued updates and testing across platforms are recommended to maintain this high level of compatibility as technologies evolve.

3.4 Usability

Table 39 shows the result of the assessment made by the elected IT Consultants regarding the usability of the system.

Table 39. *Results of the Assessment Made by elected IT Consultants on the System's Usability*

Item Statements	Weighted Mean	Verbal Description
The LMS interface is intuitive and easy to navigate for all users.	3.67	Strongly Agree
Faculty can easily create and manage quizzes and exams.	3.33	Strongly Agree
Students can access learning resources without technical difficulties	4.00	Strongly Agree
The system provides clear instructions and minimal errors during use.	3.67	Strongly Agree
The LMS design supports efficient task completion for both students and faculty.	3.67	Strongly Agree

Grand Mean	**3.67**
Verbal	**Strongly Agree**

As shown in Table 39, the respondents gave a grand mean of **3.67,** interpreted as **Strongly Agree,** for the usability of the Learning Management System. This indicates that IT consultants strongly agreed that the LMS interface is intuitive, easy to navigate, and supports efficient task completion. They also confirmed that faculty can easily create and manage quizzes, students can access learning resources without difficulty, and the system provides clear instructions with minimal errors.

Usability is a vital factor in software quality because it directly affects user adoption and effectiveness in carrying out tasks. The positive evaluation suggests that the LMS minimizes learning curves, reduces errors, and enables both students and faculty to complete academic activities smoothly.

These findings imply that the LMS promotes a user-friendly environment that supports teaching and learning efficiency. To maintain this high level of usability, periodic user testing and interface refinement are recommended to align with evolving academic needs.

3.5 Security

Table 40 shows the result of the assessment made by the elected IT Consultants regarding the security of the system.

Table 40. *Results of the Assessment Made by elected IT Consultants on the System's Security*

Item Statements	Weighted Mean	Verbal Description
The LMS uses encryption to secure student grades and personal data.	3.67	Strongly Agree
Access controls (e.g., user authentication) prevent unauthorized access.	3.33	Strongly Agree
The system regularly updates security measures to address vulnerabilities.	3.67	Strongly Agree
The LMS protects against data breaches and external threats.	3.33	Strongly Agree

The LMS uses CAPTCHA to prevent unauthorized access or automated threats.	3.67	Strongly Agree
Grand Mean	**3.53**	
Verbal	**Strongly Agree**	

As shown in Table 40, the respondents gave a grand mean of **3.53**, interpreted as **Strongly Agree**, for the security of the Learning Management System. This indicates that IT consultants strongly agreed that the LMS uses encryption to protect student data, applies access controls to prevent unauthorized use, and includes measures such as regular updates and CAPTCHA to safeguard against threats. They also recognized that the system is capable of preventing data breaches and ensuring the confidentiality of sensitive information.

Security is a fundamental aspect of software quality because it ensures the protection and integrity of academic records and user data. The positive evaluation suggests that the LMS provides strong safeguards against both internal and external risks, giving assurance to institutions, faculty, and students.

These findings imply that the LMS has a reliable security framework that supports trust and compliance in academic operations. To sustain this level of protection, continuous monitoring, timely updates, and vulnerability assessments are recommended.

data security, suggesting the LMS is highly secure for handling sensitive student information.

5. 4. How may the end-users evaluate the developed application based on the System Usability Scale (SUS) in terms of the following usability characteristics:

1. **How do faculty evaluate the system in terms of ease of use, efficiency, learnability, and user satisfaction?**
2. **How do students evaluate the system in terms of ease of use, efficiency, learnability, and user satisfaction?**

The following tables present the results of end-user assessment on the quality of using the system.

4.1 Ease of Use (Faculty's Assessment)

Table 41 presents the assessment results provided by the candidate end-users regarding the Ease of Use of the Learning Management System.

Table 41. *Results of the Assessment Made by Candidate End-Users on the System's Ease of Use*

Item Statements	**Weighted Mean**	**Verbal Description**
The LMS interface is easy to navigate using desktop browsers (e.g., Chrome, Firefox).	3.80	Strongly Agree
I can quickly find features like quizzes and resources on the LMS.	3.80	Agree
The system's layout is clear and user-friendly for desktop use.	3.50	Strongly Agree
I rarely encounter errors or confusion while using the LMS on a desktop.	3.40	Agree
The LMS requires minimal effort to access course materials on a desktop.	3.60	Strongly Agree
Grand Mean	**3.62**	
Verbal	**Strongly Agree**	

As shown in Table 41, the respondents gave a grand mean of **3.62,** interpreted as **Strongly Agree,** for the ease of use of the Learning Management

System. This indicates that faculty generally found the LMS interface easy to navigate, its features accessible, and the layout clear and user-friendly. They also agreed that the system requires minimal effort to use and reduces errors or confusion when managing academic tasks.

Ease of use is a critical aspect of system usability, as it directly affects how quickly and effectively faculty can perform instructional and administrative duties. The positive rating implies that the LMS supports teachers in creating lessons, administering assessments, and monitoring student performance with minimal difficulty.

These findings suggest that the LMS enhances faculty satisfaction by simplifying routine tasks and improving accessibility. To maintain this level of usability, continuous refinement of the interface and faculty support features is recommended.

4.2 Efficiency (Faculty's Assessment)

Table 42 shows the result of the assessment made by the candidate end-users regarding the efficiency of the system.

Table 42. *Results of the Assessment Made by candidate end-users on the Systems' Efficiency*

Item Statements	Weighted Mean	Verbal Description
The LMS allows me to complete tasks (e.g., taking quizzes) quickly on a desktop.	3.90	Strongly Agree
Real-time grading provides immediate feedback without delays on the LMS.	3.40	Agree
Accessing resources (e.g., PDFs, videos) is fast and efficient on a desktop.	3.70	Strongly Agree
The LMS supports efficient management of academic tasks (e.g., submitting assignments).	3.80	Strongly Agree
Faculty can manage lessons and assessments efficiently using the LMS on a desktop.	3.70	Strongly Agree
Grand Mean	**3.70**	
Verbal	**Strongly Agree**	

As shown in Table 42, the respondents gave a grand mean of **3.70,** interpreted as **Strongly Agree,** for the efficiency of the Learning Management System. This shows that faculty found the LMS effective in helping them complete tasks such as managing lessons, preparing quizzes, and uploading resources quickly. They also agreed that real-time grading provides immediate feedback without delays and that the system supports more efficient handling of assessments.

Efficiency is an important factor in system usability because it reflects how well the LMS supports the timely completion of teaching and administrative duties. The positive rating indicates that the system reduces delays common in manual methods and allows smoother academic workflows.

These results imply that the LMS contributes to better productivity for faculty, enabling them to focus more on instruction rather than routine administrative tasks. Regular updates and optimization are recommended to ensure that the system continues to support efficient faculty work, even as usage grows.

4.3 Learnability (Faculty’s Assessment)

Table 43 shows the result of the assessment made by the candidate end-users regarding the Learnability of the system.

Table 43. *Results of the Assessment Made by candidate end-users on the Systems' Learnability*

Item Statements	Weighted Mean	Verbal Description
I quickly learned how to use the LMS on a desktop without extensive training.	3.60	Strongly Agree
The LMS's features (e.g., quiz navigation, resource access) are easy to understand.	3.60	Strongly Agree
Instructions provided in the LMS are clear and helpful for desktop users.	3.40	Agree
I feel confident using the LMS on a desktop after a short period of use.	3.60	Strongly Agree
The LMS is easy to learn for both students and faculty using desktop browsers.	3.50	Strongly Agree
Grand Mean	**3.54**	
Verbal	**Strongly Agree**	

As shown in Table 43, the respondents gave a grand mean of **3.54**, interpreted as **Strongly Agree**, for the learnability of the Learning Management System. This indicates that faculty were able to learn how to use the LMS quickly without the need for extensive training. They also agreed that key features, such as creating quizzes, managing lessons, and uploading resources, are easy to understand and that the instructions provided are clear and helpful.

Learnability is an essential aspect of usability because it determines how fast new users adapt to the system. The positive result suggests that faculty can confidently operate the LMS after a short period, reducing the need for lengthy orientations or technical assistance.

These findings imply that the LMS supports ease of adoption for faculty, enabling them to focus more on teaching responsibilities rather than system operation. Continuous improvement of faculty guidance materials and training modules will further strengthen this advantage.

4.4 User Satisfaction (Faculty's Assessment)

Table 44 shows the result of the assessment made by the candidate end-users regarding the User Satisfaction of the system.

Table 44. *Results of the Assessment Made by candidate end-users on the System's User Satisfaction*

Item Statements	Weighted Mean	Verbal Description
I am satisfied with the overall experience of using the LMS on a desktop.	3.60	Strongly Agree
The LMS improves my ability to engage with academic tasks effectively.	3.60	Strongly Agree
The LMS meets my expectations for managing courses and assessments on a desktop.	3.70	Strongly Agree
I would recommend the LMS to other students or faculty for desktop use.	3.70	Strongly Agree
The LMS enhances my learning or teaching experience compared to the manual system.	3.70	Strongly Agree
Grand Mean	**3.66**	
Verbal	**Strongly Agree**	

As shown in Table 44, the respondents gave a grand mean of **3.66,** interpreted as **Strongly Agree,** for the user satisfaction of the Learning Management System. This shows that faculty were satisfied with their overall experience in using the LMS. They agreed that the system improves their ability to manage academic tasks, meets their expectations in handling courses and assessments, and enhances their teaching compared to the manual system.

User satisfaction is a key indicator of system acceptance and continued use. The positive evaluation implies that the LMS not only meets functional and usability needs but also supports a more efficient and engaging teaching environment.

These findings suggest that the LMS is well-received by faculty and has the potential to strengthen instructional delivery and academic management. To sustain this level of satisfaction, the system should be regularly updated and refined based on faculty feedback.

4.1 Ease of Use (Students' Assessment)

Table 45 presents the assessment results provided by the candidate end-users regarding the Ease of Use of the Learning Management System.

Table 45. *Results of the Assessment Made by Candidate End-Users on the System's Ease of Use*

Item Statements	Weighted Mean	Verbal Description
The LMS interface is easy to navigate using desktop browsers (e.g., Chrome, Firefox).	3.65	Strongly Agree
I can quickly find features like quizzes and resources on the LMS.	3.60	Agree
The system's layout is clear and user-friendly for desktop use.	3.62	Strongly Agree
I rarely encounter errors or confusion while using the LMS on a desktop.	3.48	Agree
The LMS requires minimal effort to access course materials on a desktop.	3.63	Strongly Agree
Grand Mean	**3.60**	
Verbal	**Strongly Agree**	

As shown in Table 45, the respondents gave a grand mean of **3.60**, interpreted as **Strongly Agree**, for the ease of use of the Learning Management System. This indicates that students generally found the LMS interface easy to navigate, its features accessible, and the layout clear and user-friendly. They also agreed that the system requires minimal effort to use and reduces errors or confusion when performing tasks.

Ease of use is a critical aspect of system usability, as it directly affects how quickly and effectively users can complete academic tasks. The positive rating implies that the LMS supports students in managing lessons, accessing resources, and taking assessments with minimal difficulty.

These findings suggest that the LMS enhances user satisfaction by simplifying tasks and improving accessibility. To maintain this level of usability, continuous refinement of the interface and user support features is recommended.

4.2 Efficiency (Student's Assessment)

Table 46 shows the result of the assessment made by the candidate end-users regarding the efficiency of the system.

Table 46. *Results of the Assessment Made by candidate end-users on the Systems' Efficiency*

Item Statements	Weighted Mean	Verbal Description

The LMS allows me to complete tasks (e.g., taking quizzes) quickly on a desktop.	3.59	Strongly Agree
Real-time grading provides immediate feedback without delays on the LMS.	3.45	Agree
Accessing resources (e.g., PDFs, videos) is fast and efficient on a desktop.	3.55	Strongly Agree
The LMS supports efficient management of academic tasks (e.g., submitting assignments).	3.59	Strongly Agree
Faculty can manage lessons and assessments efficiently using the LMS on a desktop.	3.61	Strongly Agree
Grand Mean	**3.56**	
Verbal	**Strongly Agree**	

As shown in Table 46, the respondents gave a grand mean of **3.56**, interpreted as **Strongly Agree**, for the efficiency of the Learning Management System. This shows that students found the LMS effective in helping them complete tasks such as taking quizzes, and accessing resources quickly. They also agreed that real-time grading provides immediate feedback without delays and that faculty can manage lessons and assessments more efficiently using the system.

Efficiency is an important factor in system usability because it reflects how well the LMS supports the timely completion of academic processes. The positive rating indicates that the system reduces delays common in manual methods and supports smoother academic workflows.

These results imply that the LMS contributes to better productivity for both students and faculty. Regular updates and optimization are recommended to ensure that the system continues to handle academic tasks efficiently, even as user demand increases.

4.3 Learnability (Student's Assessment)

Table 47 shows the result of the assessment made by the candidate end-users regarding the Learnability of the system.

Table 47. *Results of the Assessment Made by candidate end-users on the Systems' Learnability*

Item Statements	Weighted Mean	Verbal Description

I quickly learned how to use the LMS on a desktop without extensive training.	3.61	Strongly Agree
The LMS's features (e.g., quiz navigation, resource access) are easy to understand.	3.59	Strongly Agree
Instructions provided in the LMS are clear and helpful for desktop users.	3.53	Strongly Agree
I feel confident using the LMS on a desktop after a short period of use.	3.61	Strongly Agree
The LMS is easy to learn for both students and faculty using desktop browsers.	3.63	Strongly Agree
Grand Mean	**3.59**	
Verbal	**Strongly Agree**	

As shown in Table 47, the respondents gave a grand mean of **3.59**, interpreted as **Strongly Agree,** for the learnability of the Learning Management System. This indicates that students were able to learn how to use the LMS quickly without extensive training. They also agreed that the features, such as quiz navigation and resource access, are easy to understand and that the instructions provided are clear and helpful.

Learnability is an essential aspect of usability because it determines how fast new users adapt to the system. The positive result suggests that both students and faculty can confidently use the LMS after only a short period, minimizing the need for lengthy orientations or technical assistance.

These findings imply that the LMS supports ease of adoption, helping users focus more on academic tasks rather than struggling with system operations. Continuous improvement of user guidance and tutorials will further strengthen this advantage.

4.4 User Satisfaction (Student's Assessment)

shows the result of the assessment made by the candidate end-users regarding the User Satisfaction of the system.

Table 48. *Results of the Assessment Made by candidate end-users on the System's User Satisfaction*

Item Statements	Weighted Mean	Verbal Description

I am satisfied with the overall experience of using the LMS on a desktop.	3.61	Strongly Agree
The LMS improves my ability to engage with academic tasks effectively.	3.60	Strongly Agree
The LMS meets my expectations for managing courses and assessments on a desktop.	3.57	Strongly Agree
I would recommend the LMS to other students or faculty for desktop use.	3.58	Strongly Agree
The LMS enhances my learning or teaching experience compared to the manual system.	3.57	Strongly Agree
Grand Mean	**3.59**	
Verbal	**Strongly Agree**	

As shown in Table 48, the respondents gave a grand mean of **3.59**, interpreted as **Strongly Agree,** for the user satisfaction of the Learning Management System. This shows that students were satisfied with their overall experience in using the LMS. They agreed that the system improves their ability to complete academic tasks, meets their expectations in managing courses and assessments, and enhances their learning compared to the manual system.

User satisfaction is a key indicator of system acceptance and continued use. The positive evaluation implies that the LMS not only meets functional and usability needs but also contributes to a more engaging learning environment.

These findings suggest that the LMS is well-received by students and has the potential to strengthen teaching and learning outcomes. To sustain this level of satisfaction, the system should be regularly updated and refined based on user feedback.

Chapter IV
SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the study's summary, conclusions, and recommendations. It outlines the research's main outcomes, conclusions drawn from these findings, and insightful recommendations for further improvements.

Summary

The capstone project **"Learning Management System for NEUST-MGT BSIT Program"** was developed to automate and improve academic processes traditionally done through manual methods. The system's goal is to efficiently manage lessons, assessments, grading, and records by using the Agile development model, which emphasizes iterative planning, design, coding, testing, and review to minimize errors and reduce administrative workload. The research focused on short development cycles, allowing continuous feedback from stakeholders to ensure that the system remained adaptable and responsive to the needs of students and faculty.

This study has major benefits for students, faculty, and administrators. By automating classroom processes, the system improves accuracy, reduces delays in grading and feedback, and provides centralized access to resources and records.

Furthermore, the LMS was evaluated by IT consultants and end-users, who assessed its performance in terms of functionality, efficiency, usability, compatibility, and security—all of which received positive ratings.

Conclusion

Based on the findings, the following conclusions was made:

- The **Learning Management System (LMS)** was developed using a comprehensive **Agile methodology**, which included the phases of planning, designing, developing, testing, reviewing, deploying, and launching. This methodological approach ensured the dependability and adaptability of the software.

- The assessments of elected IT consultants confirm that the system performs well across the main software quality attributes of **functional suitability**, **performance efficiency**, **compatibility**, **usability**, **and security**. In terms of functional suitability, the system effectively supports all defined academic tasks, produces accurate results, and assists in achieving learning objectives. For performance efficiency, the system responds quickly to events, makes good use of resources, and has the capacity to support multiple users simultaneously.

In terms of compatibility, the system was commended for working smoothly across browsers, operating systems, and devices. Usability tests found the LMS intuitive and easy to navigate, allowing faculty and students to complete tasks efficiently. Security measures were also evaluated and found to be reliable in protecting academic data and preventing unauthorized access. Taken together, these findings demonstrate that the LMS is well-designed and capable of meeting both technical and user-centered requirements, making it a dependable academic tool.

- The usability findings indicate that the system meets end-user expectations in terms of **ease of use, efficiency, learnability, and satisfaction.** Students and faculty agreed that the LMS is clear and simple to use, allowing them to complete activities quickly and with minimal effort. They also found the system easy to learn, with features that are intuitive and straightforward to recall even after periods of inactivity. Clear instructions and feedback features enhanced the learning experience, while overall satisfaction ratings show that the system promotes confidence and is recommendable to others. These findings confirm that the LMS is highly usable, making it practical and reliable for daily academic use.

- The **Learning Management System** was designed as a valuable tool that benefits its users. It simplifies grading, lesson management, resource access, and record-keeping, reducing the burden on faculty and providing students with faster access to academic materials and results. By streamlining these processes, the LMS allows teachers to focus more on instruction while improving the learning experience for students.

Recommendations

In light of the study, the researchers make the following recommendations to improve the efficacy and utility of the Learning Management System.

- Host the LMS on the cloud to improve scalability and accessibility, allowing students and faculty to access the system anytime and from any device.

- Add advanced analytics and reporting features to help faculty track student performance, monitor progress, and identify areas where learners may need additional support.

- Include an integrated messaging or chat feature within the LMS so students and faculty can communicate directly about lessons, assessments, and feedback.

- Provide regular training and orientation for faculty and students to maximize the use of system features, ensuring consistent adoption and effective use.

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APPENDICES

Appendix A

Request Letter for Adviser

Republic of the Philippines
NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY
Cabanatuan City, Nueva Ecija, Philippines
ISO 9001:2015 CERTIFIED

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

February 8, 2024

LEONYLYN P. BENSI
Professor
College of Communication and Information Technology
Nueva Ecija of Science and Technology

Dear Ma'am:

Greetings for Peace!

The IT Capstone Project and Research is a culminating project as part of the requirements for the degree, Bachelor of Science in Information and Communications Technology. The undersigned are the third-year BSIT students enrolled in the IT-CAP01: Capstone Project and Research 1 during the 2nd Semester of Academic Year 2023-2024

In line with this, may the undersigned ask for your approval to become their Capstone and Research Project adviser for the study entitled: "Dormify: Streamlining Dormitory Living with an Advanced Management System" Your expertise in the field and guidance will truly contribute to the success of this academic endeavor.

The undersigned are looking forward for your approval and acceptance. Thank you.

Sincerely,

ICO, DAVE BRYAN

DIMACULANGAN, ELIJAH JOSHUA

FEDERICO, JOSHUA JAY

PALMA, MARIA JEMIA DANIELA

Noted:

RUTH G. LUCIANO, PH. D
Capstone Project and Research Instructor

Conforme:
LEONYLYN P. BENSI
Professor

Transforming Communities through Science and Technology

Approved Letter of Adviser

Appendix B

RESEARCH INSTRUMENTS

Survey Instrument of Assessing the Common

Republic of the Philippines
NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY
Cabanatuan City, Nueva Ecija, Philippines
ISO 9001:2015 CERTIFIED

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

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Professor
College of Communication and Information Technology
Nueva Ecija of Science and Technology

Dear Ma'am:

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Sincerely,

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DIMACULANGAN, ELIJAH JOSHUA

FEDERICO, JOSHUA JAY

PALMA, MARIA JEMIA DANIELA

Noted:

RUTH G. LUCIANO, PH. D
Capstone Project and Research Instructor

Conforme:
LEONYLYN P. BENSI
Professor

Transforming Communities through Science and Technology

Problems Encountered Using Manual System

2. Attendance records are often inaccurate, affecting student performance tracking.				
3. Paper-based systems fail to provide dependable output for large class sizes.				
4. Manual record verification is prone to mistakes and delays.				
5. Lack of centralized coordination leads to unreliable scheduling and resource availability.				

Reliability of Output				
Item	Strongly Agree (Lubos na Sumasang-Ayon) (4)	Agree (Sumasang-Ayon) (3)	Disagree (Hindi Sumasang-Ayon) (2)	Strongly Disagree (Lubos na hindi Sumasang-Ayon) (1)
1. Manual grading often produces inconsistent results due to human errors.				
2. Attendance records are often unreliable, leading to disputes over student attendance data.				
3. Paper-based systems struggle to provide reliable outputs for large class sizes.				
4. Manual verification of grades and records is prone to mistakes and inconsistencies.				
5. Lack of centralized coordination leads to unreliable scheduling and resource availability.				

EFFICIENCY OF PROCESSES				
Item	Strongly Agree (Lubos na Sumasang-Ayon)	Agree (Sumasang-Ayon)	Disagree (Hindi Sumasang-Ayon)	Strongly Disagree (Lubos na hindi Sumasang-Ayon) (1)

	(4)	(3)	(2)	
6. Grading and feedback processes are slow and inefficient in the traditional setup.				
7. Managing assessments manually delays the delivery of student results.				
8. Teachers spend excessive time on administrative tasks rather than teaching.				
9. Resource distribution (e.g., handouts, notes) is inefficient and inconsistent.				
10. Handling large volumes of student data manually reduces overall process efficiency.				

COMPLIANCE WITH REGULATIONS AND STANDARDS				
Item	**Strongly Agree (Lubos na Sumasang-Ayon) (4)**	**Agree (Sumasang-Ayon) (3)**	**Disagree (Hindi Sumasang-Ayon) (2)**	**Strongly Disagree (Lubos na hindi Sumasang-Ayon) (1)**
1. Student grades are not submitted on time due to manual processes, violating school deadlines.				
2. The traditional setup fails to follow rules for keeping accurate attendance records.				
3. Manual records do not meet requirements for secure storage of student information.				
4. There is no system to ensure grades and reports follow institutional grading policies.				
5. The setup lacks a way to track compliance with class attendance and reporting rules.				

Survey Instrument of Assessing the Technical Qualities of the System (For elected IT Consultants)

Name (optional):____________________

Age: ____

Gender:
- ☐ Male
- ☐ Female

IT Consultant Questionnaire for Learning Management System Technical Evaluation

General Instruction: Please evaluate the Learning Management System (LMS) developed for the NEUST-MGT BSIT Program by placing a checkmark in the box corresponding to your rating for each item. Use the following scale:

- **4:** Highly Suitable/Excellent/Strong
- **3:** Suitable/Good/Adequate
- **2:** Unsuitable/Fair/Weak
- **1:** Highly Unsuitable/Poor/No Security

Functional Suitability
Assesses whether the LMS meets its functional requirements with accuracy and completeness.

Item	(4)	(3)	(2)	(1)
1. The LMS accurately supports real-time grading for quizzes and exams.				
2. The system allows seamless student enrollment and subject management.				
3. Question randomization and timer-based exams function correctly.				
4. The LMS provides reliable analytics for student performance tracking.				
5. All core features (e.g., resource uploads, grading) operate without errors.				

Performance Efficiency
Evaluates the system's performance in terms of speed and resource usage.

Item	(4)	(3)	(2)	(1)
1. The LMS processes real-time grading quickly with no noticeable delays.				
2. The system handles multiple simultaneous users (e.g., 100 students) efficiently.				
3. Resource uploads (e.g., PDFs, videos) are processed without performance issues.				
4. The LMS performs consistently under varying workloads.				
5. The system uses server resources efficiently with minimal lag.				

Compatibility *Assesses the LMS's ability to integrate with other systems and devices.*				
Item	**(4)**	**(3)**	**(2)**	**(1)**
1. The LMS integrates seamlessly with common browsers (e.g., Chrome, Firefox).				
2. The system is compatible with standard desktop browsers (e.g., Chrome, Firefox) for student and faculty access.				
3. The LMS supports various file formats (e.g., PDF, PPT, MP4) for resource uploads.				
4. The system is compatible with standard operating systems (e.g., Windows, macOS) for student and faculty access.				
5. The LMS is fully responsive and compatible with mobile devices (e.g., smartphones, tablets) for student and faculty access.				

Usability *Evaluates the ease of use and intuitiveness of the LMS interface.*				
Item	**(4)**	**(3)**	**(2)**	**(1)**
1. The LMS interface is intuitive and easy to navigate for all users.				
2. Faculty can easily create and manage quizzes and exams.				
3. Students can access learning resources without technical difficulties.				
4. The system provides clear instructions and minimal errors during use.				
5. The LMS design supports efficient task completion for both students and faculty.				

Security *Assesses the LMS's ability to protect data and prevent unauthorized access.*				
Item	**(4)**	**(3)**	**(2)**	**(1)**
1. The LMS uses encryption to secure student grades and personal data.				
2. Access controls (e.g., user authentication) prevent unauthorized access.				

3. The system regularly updates security measures to address vulnerabilities.				
4. The LMS protects against data breaches and external threats.				
5. The LMS uses CAPTCHA to prevent unauthorized access or automated threats.				

Assessment on the Quality of Using the System (For candidate end-users)

Name (optional):____________________

Age: ____

Gender:
- ☐ Male
- ☐ Female

End-User Questionnaire for Learning Management System Usability

General Instruction: Please evaluate the usability of the LMS, designed for desktop use, by placing a checkmark in the box corresponding to your rating for each item. Use the following scale:

- **4:** Strongly Agree (Lubos na Sumasang-Ayon)
- **3:** Agree (Sumasang-Ayon)
- **2:** Disagree (Hindi Sumasang-Ayon)
- **1:** Strongly Disagree (Lubos na Hindi Sumasang-Ayon)

Ease of Use *Assesses how intuitive and straightforward the LMS is to use on desktop browsers.*				
Item	**(4)**	**(3)**	**(2)**	**(1)**
1. The LMS interface is easy to navigate using desktop browsers (e.g., Chrome, Firefox).				
2. I can quickly find features like quizzes and resources on the LMS.				
3. The system's layout is clear and user-friendly for desktop use.				
4. I rarely encounter errors or confusion while using the LMS on a desktop.				
5. The LMS requires minimal effort to access course materials on a desktop.				

Efficiency *Evaluates how quickly and effectively users can complete tasks on the LMS.*				
Item	**(4)**	**(3)**	**(2)**	**(1)**
1. The LMS allows me to complete tasks (e.g., taking quizzes) quickly on a desktop.				
2. Real-time grading provides immediate feedback without delays on the LMS.				
3. Accessing resources (e.g., PDFs, videos) is fast and efficient on a desktop.				
4. The LMS supports efficient management of academic tasks (e.g., submitting assignments).				

5. Faculty can manage lessons and assessments efficiently using the LMS on a desktop.				

Learnability *Assesses how easily users can learn to use the LMS.*				
Item	**(4)**	**(3)**	**(2)**	**(1)**
1. I quickly learned how to use the LMS on a desktop without extensive training.				
2. The LMS's features (e.g., quiz navigation, resource access) are easy to understand.				
3. Instructions provided in the LMS are clear and helpful for desktop users.				
4. I feel confident using the LMS on a desktop after a short period of use.				
5. The LMS is easy to learn for both students and faculty using desktop browsers.				

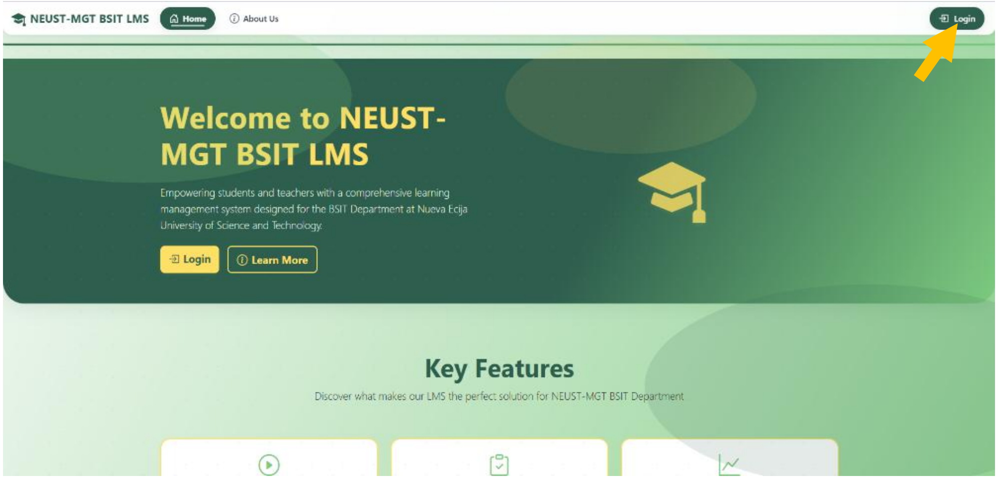
User Satisfaction *Evaluates overall satisfaction with the LMS experience on desktop browsers.*				
Item	**(4)**	**(3)**	**(2)**	**(1)**
1. I am satisfied with the overall experience of using the LMS on a desktop.				
2. The LMS improves my ability to engage with academic tasks effectively.				
3. The LMS meets my expectations for managing courses and assessments on a desktop.				
4. I would recommend the LMS to other students or faculty for desktop use.				
5. The LMS enhances my learning or teaching experience compared to the manual system.				

Appendix C

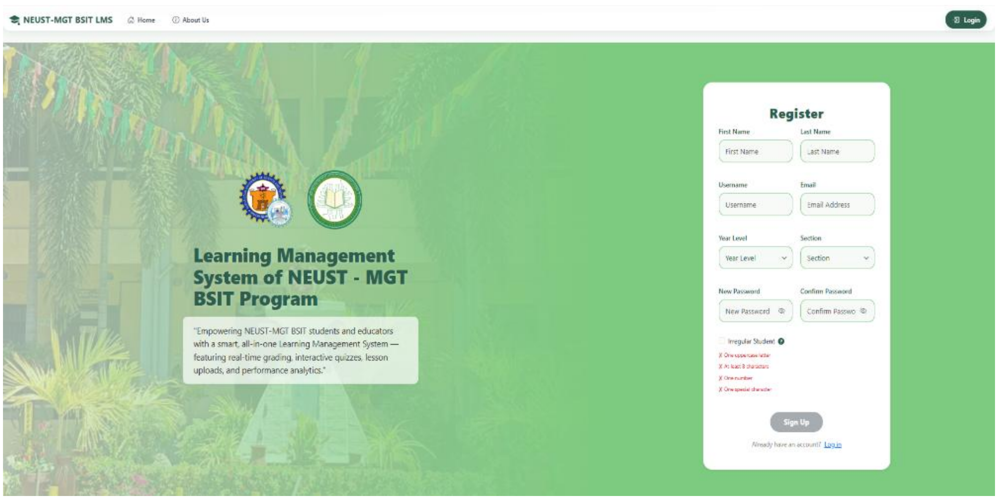
USER MANUAL

1 . How to Login / Sign up

1.1 On the landing page, click the **Login** button.



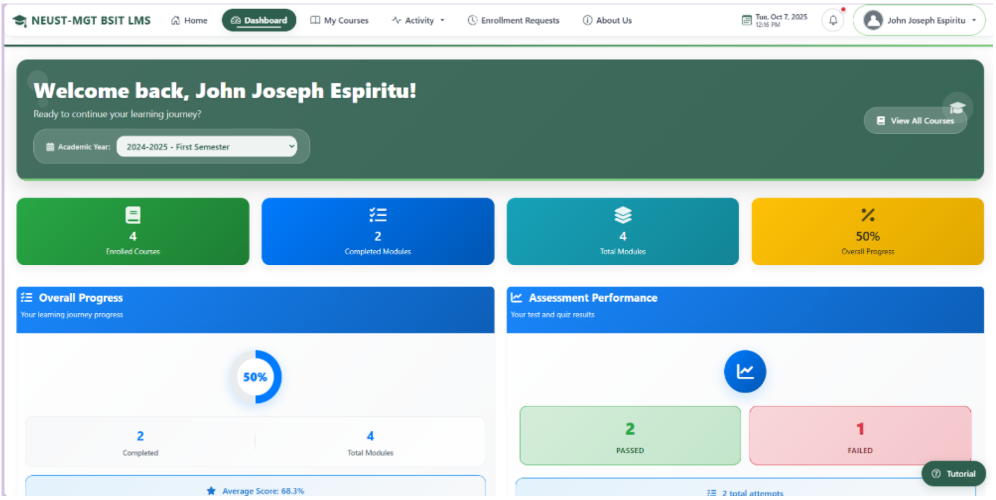
1.1.1 For new student users, click **Don't have an account? Register** and fill out the registration form.



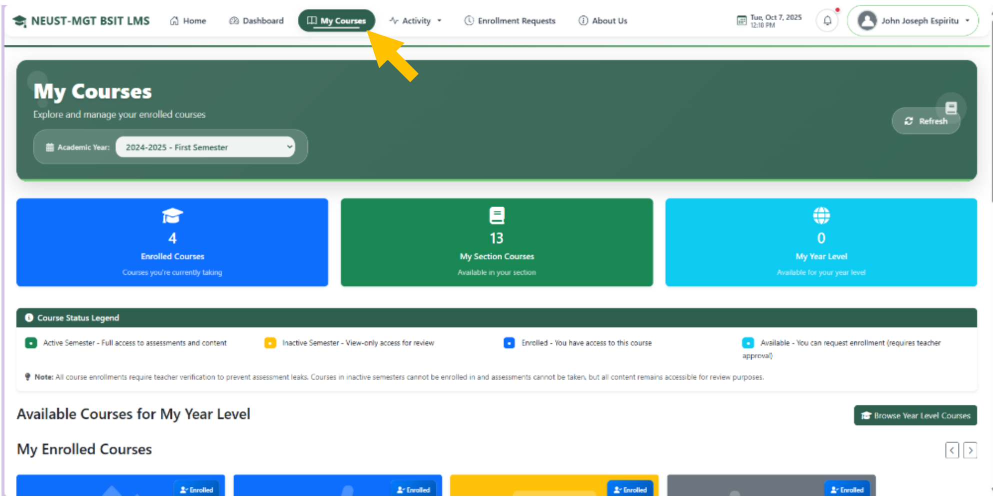
Student Side

1 . 2 . Navigating through the system as **Student.**

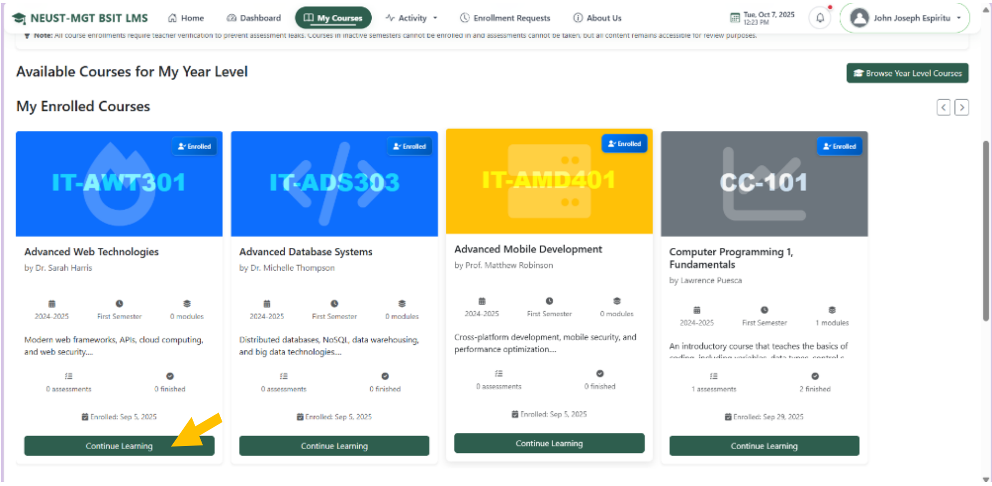
1.2.1. After logging in, the system redirects to your student dashboard.



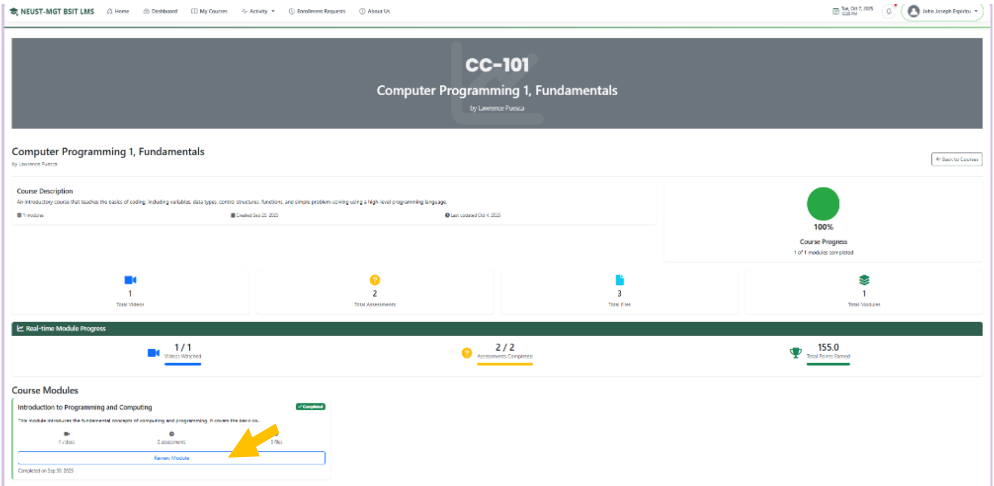
1.2.2. Navigate to My Courses from menu.



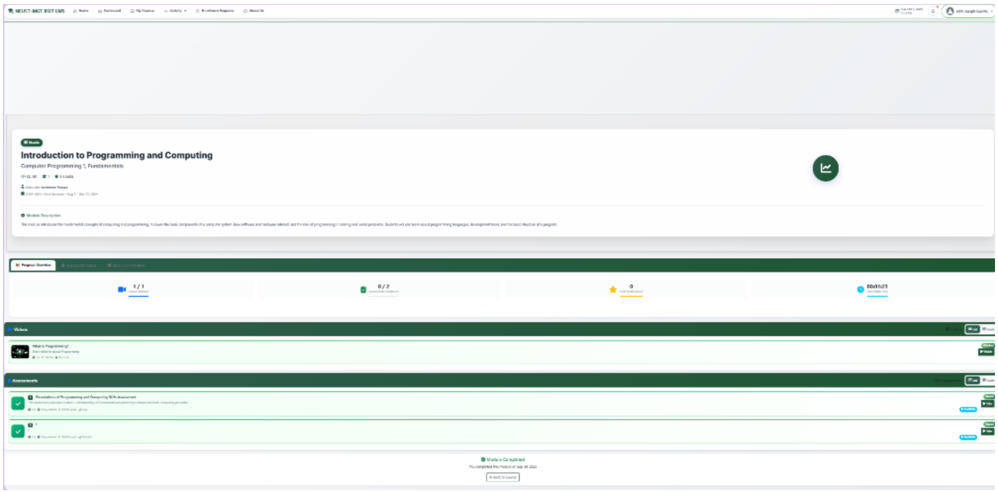
1.3. To take courses scroll down and click **Continue Learning** button.



1.3.1. This page displays the available modules and statistic from the course and click the review module button.

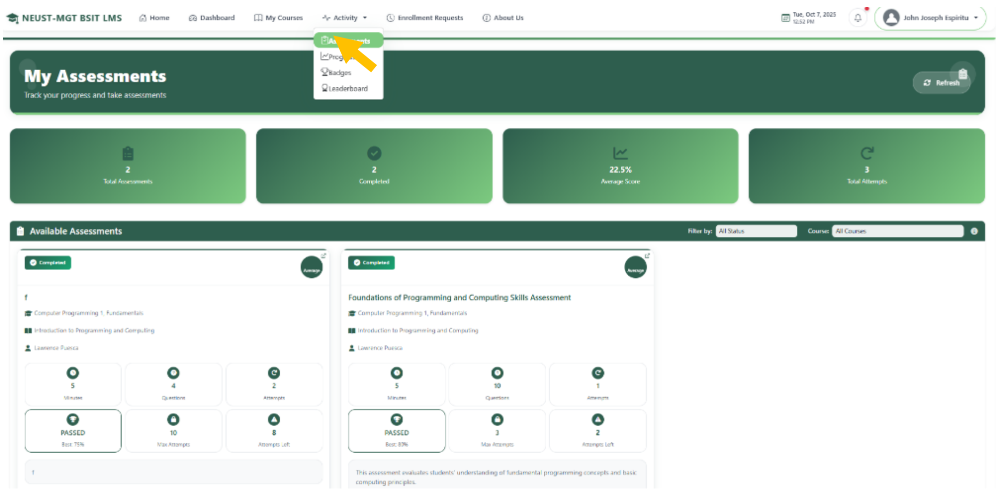


1.3.2. After clicked the review module button it will redirect to module assessments and videos list that students can take.

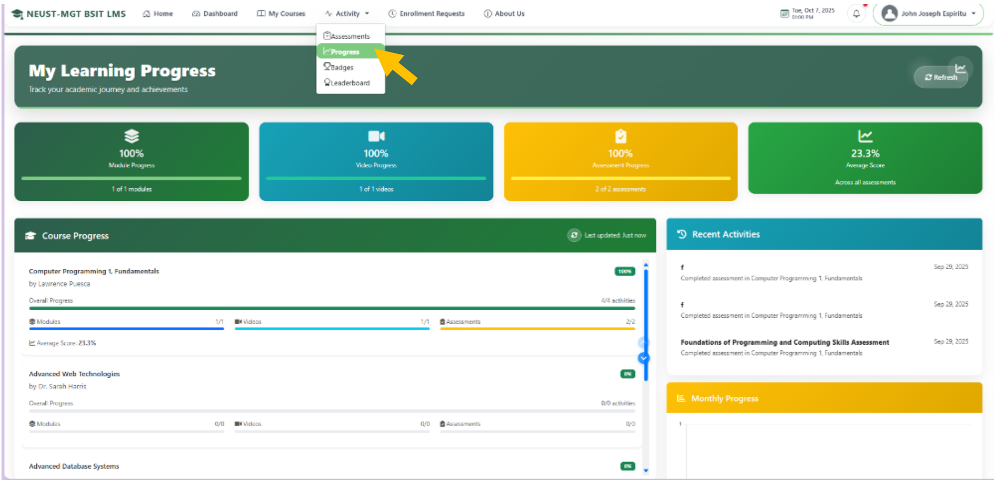


2. Navigating to **Activity** from menu **Assessments** link.

2.1. Click Assessment it redirect on this page, this page displayed list and progress of each assessments.

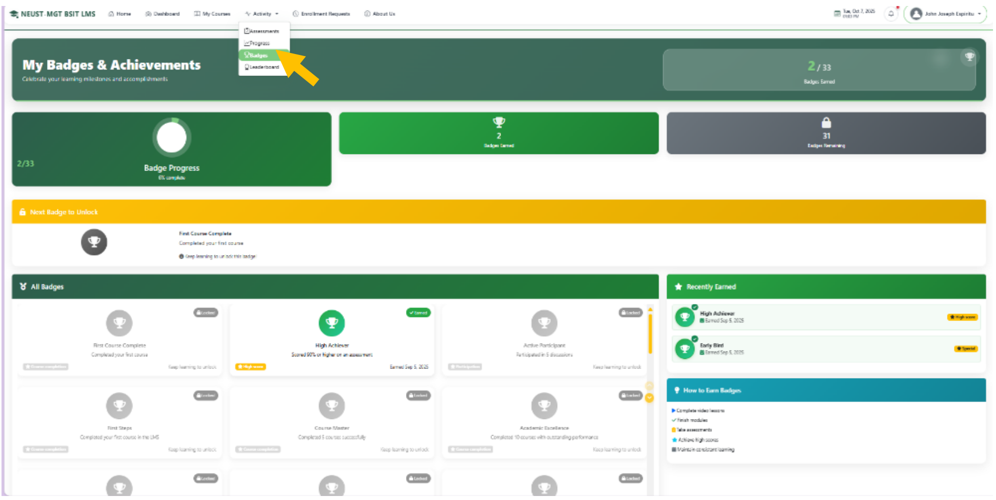


2.1 Click **Progress** it redirect on this page, this page displayed overall progress of each courses assigned to student.

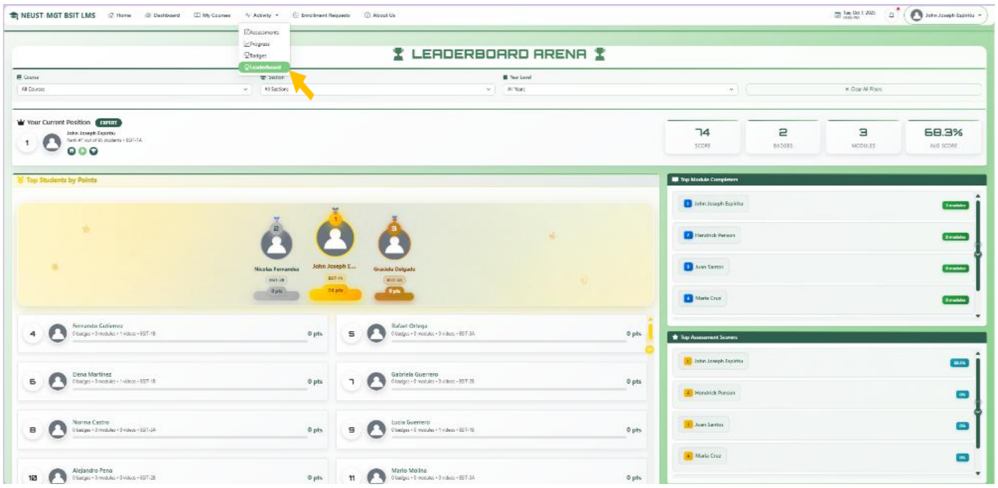


2.2. Click **Badges** it redirect on this page this page displayed earned badges of the

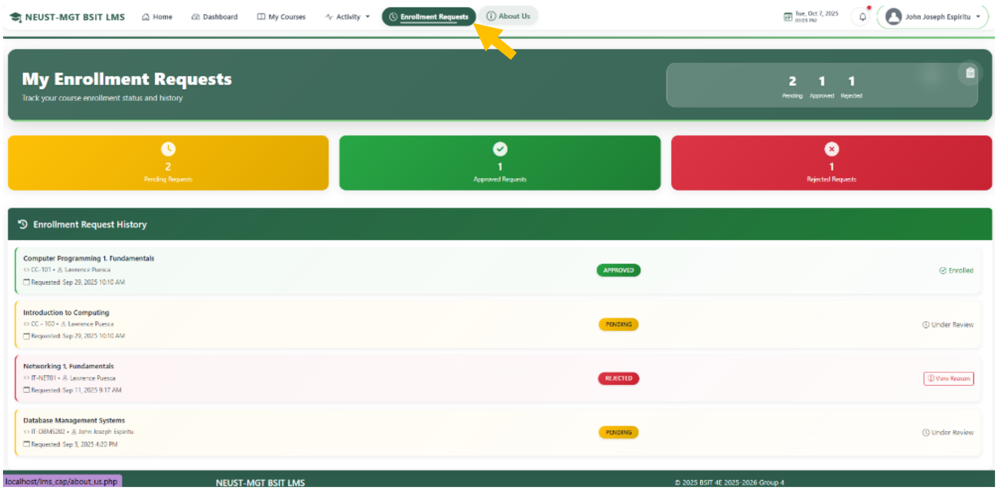
student.



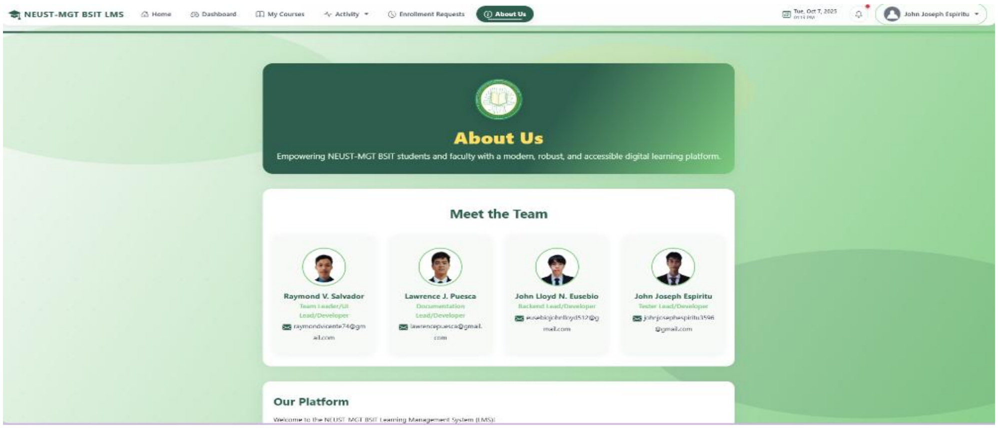
2.3. Click **Leaderboard** it redirect on this page this page displayed the overall ranking of the students base on their progress.



3. Navigating to Enrollment Requests.
3.1. Click Enrollment Requests from menu. This page show the status of student requests.

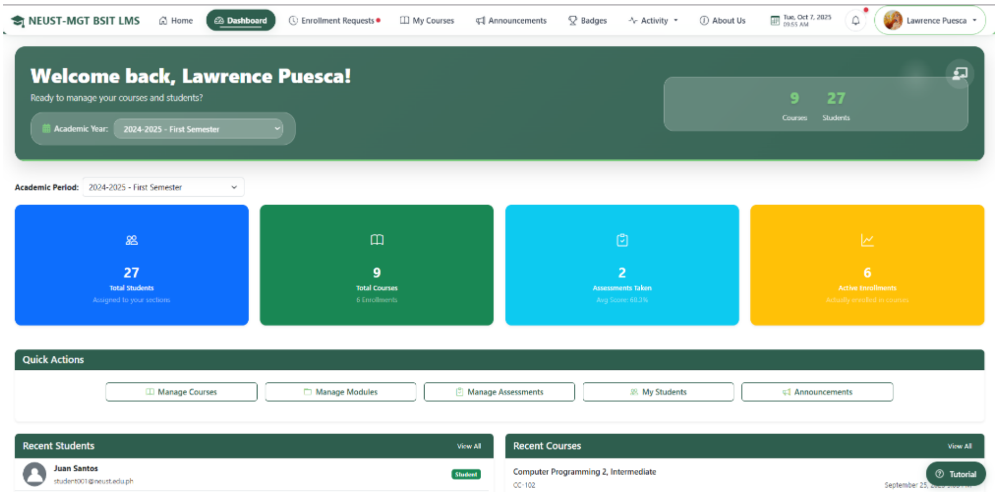


4. About Us page.
4.1. Click About Us from menu for all role to redirect to this page that displayed about us contents.

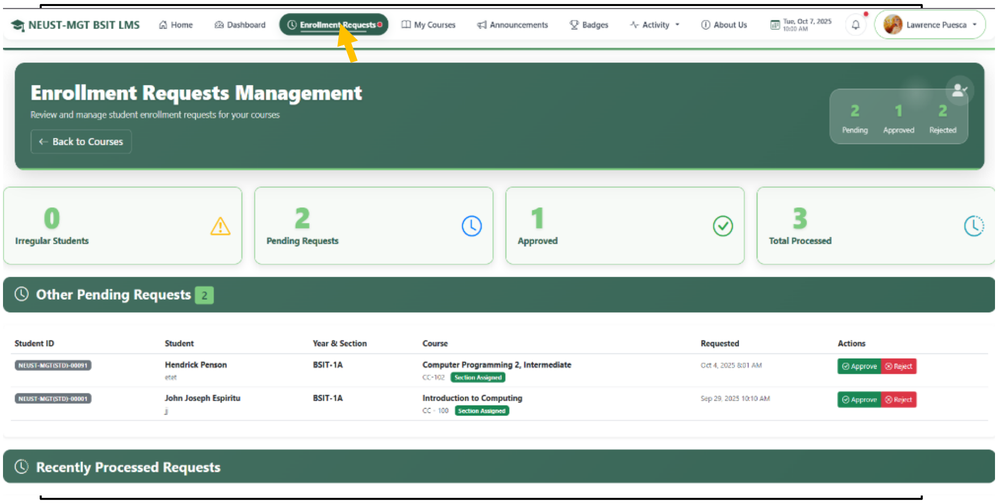


Teacher Side

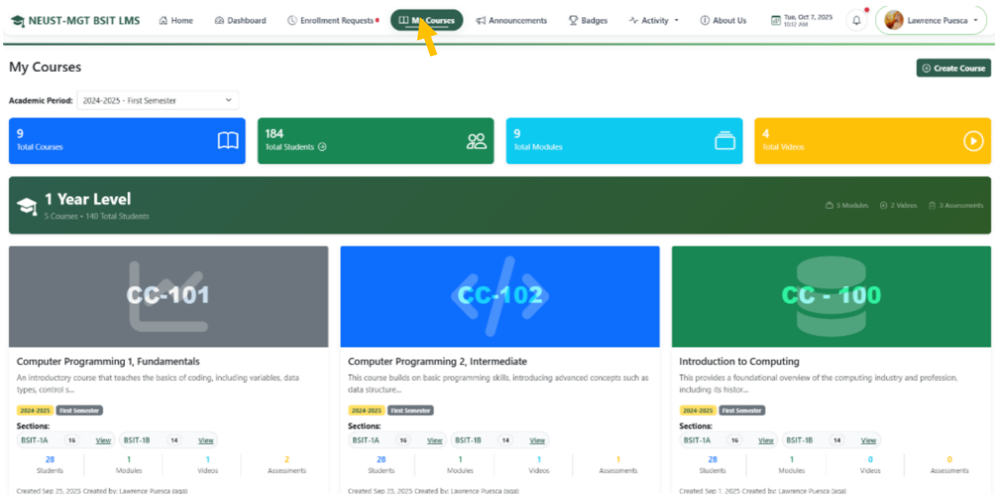
5. Navigating through the system as **Teacher**. 7.1. After logging in, the system redirects to your teacher dashboard.



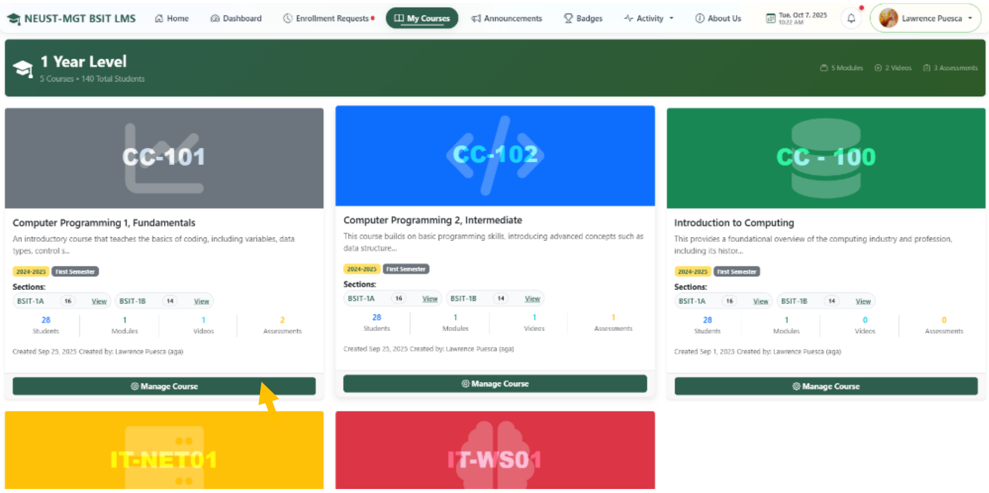
5.1. Navigate to Enrollment Requests page from menu on this page you manage request of student



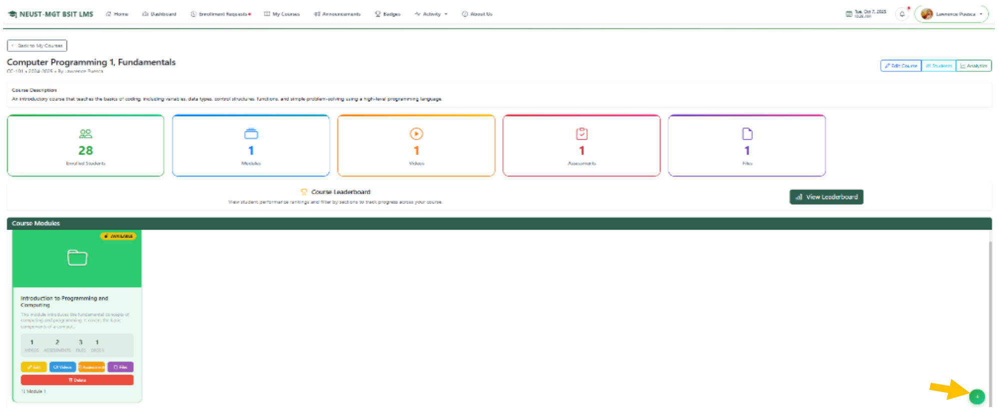
5.2. Navigate to Courses from menu you can scroll down.



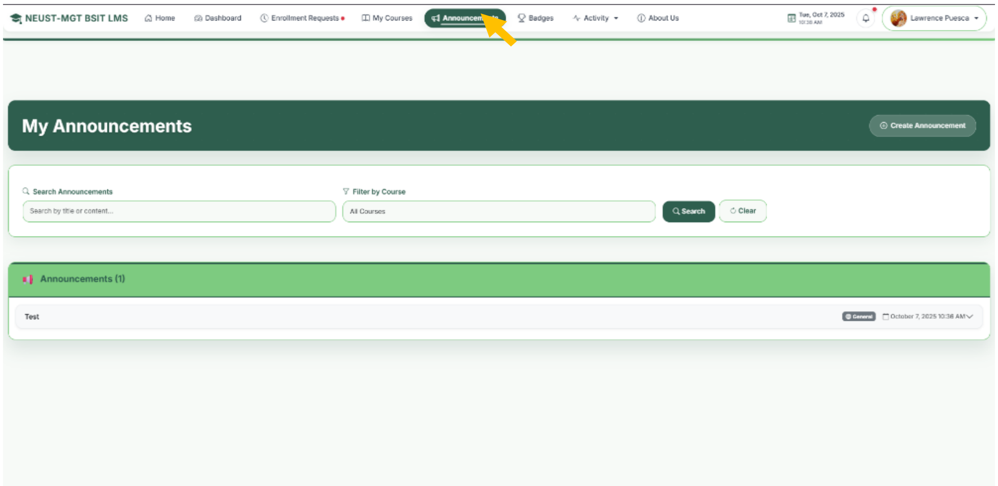
5.2.1. To manage specific courses, click Manage Courses



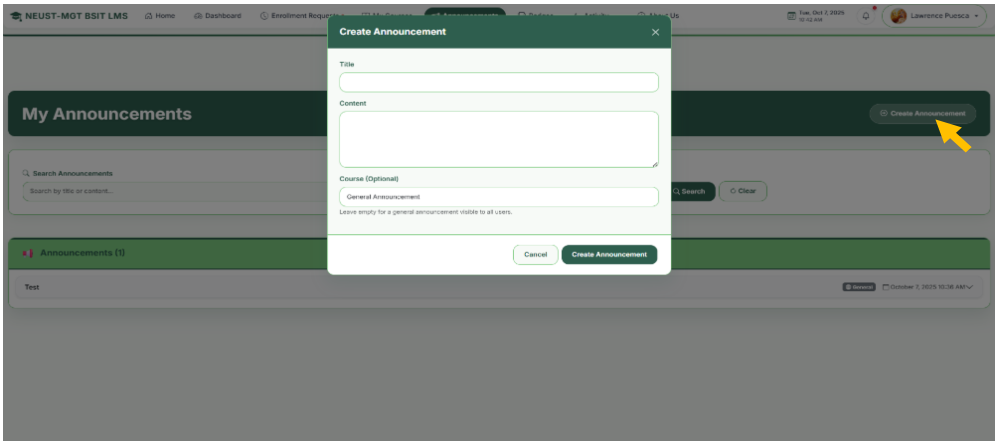
5.2.2. When clicked the Manage Course, you can Manage each module from courses from this page and can add module click the plus sign.



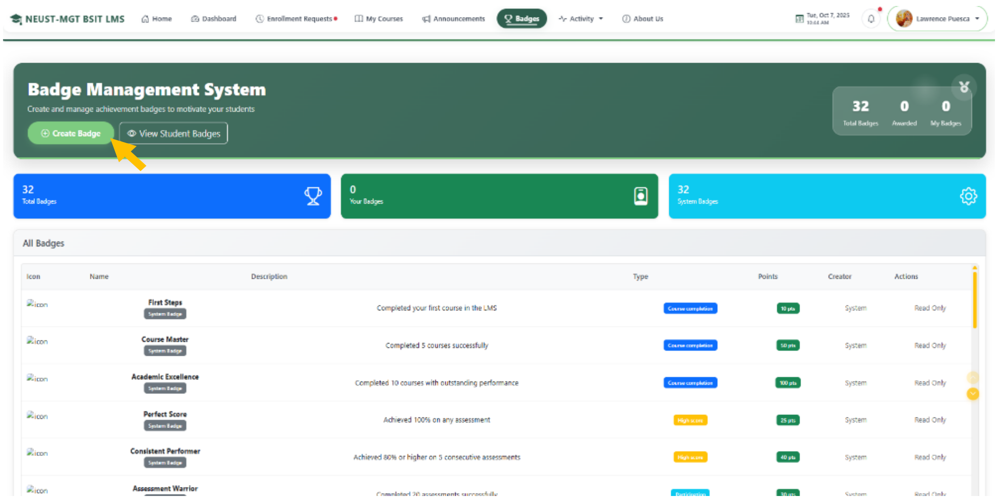
6.1. Navigating to Announcement from menu.



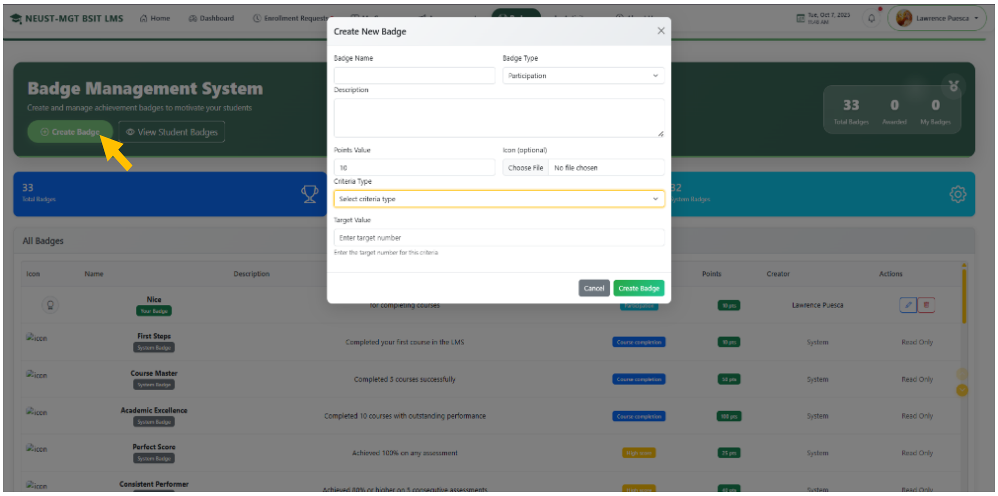
6.1.1. To add announcement, click the Create Announcement button.



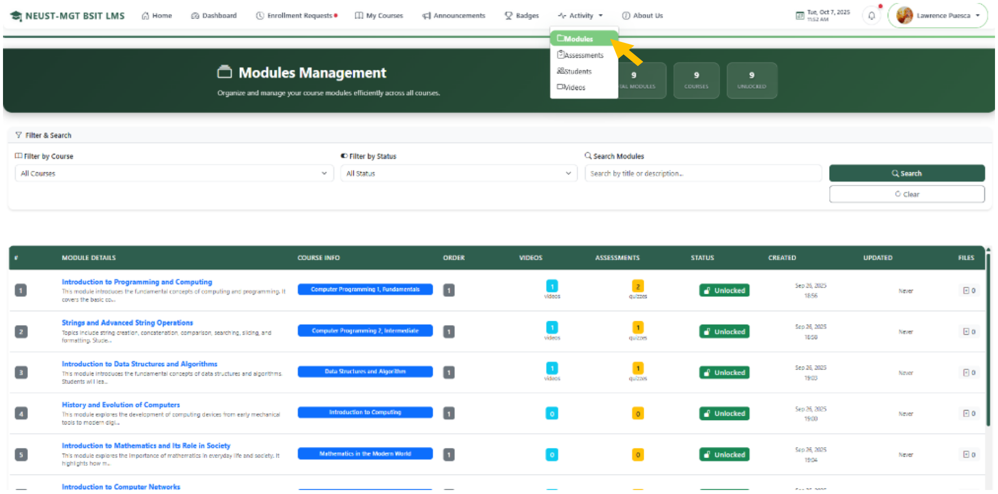
6.2. Navigate to Badge Page from menu.



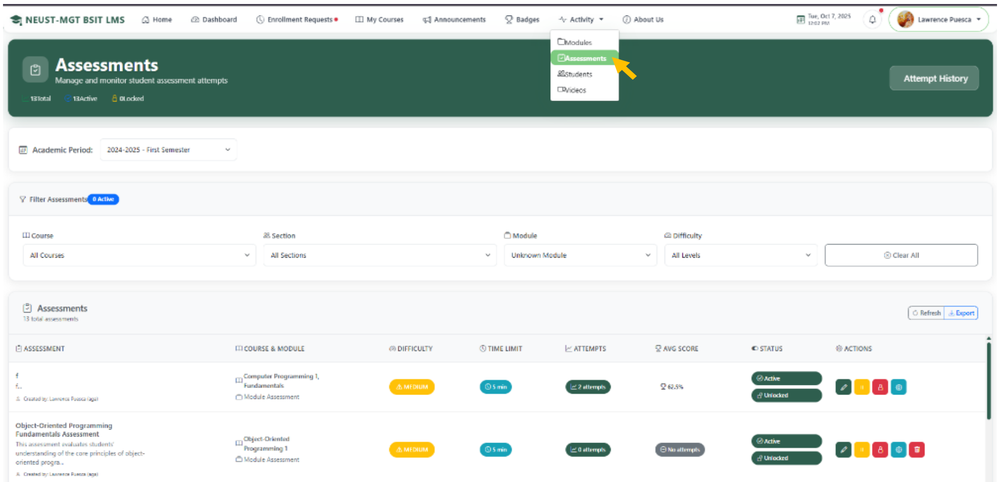
6.2.1. To create Badge from Badge page click **Create Badge and** fill up the form.



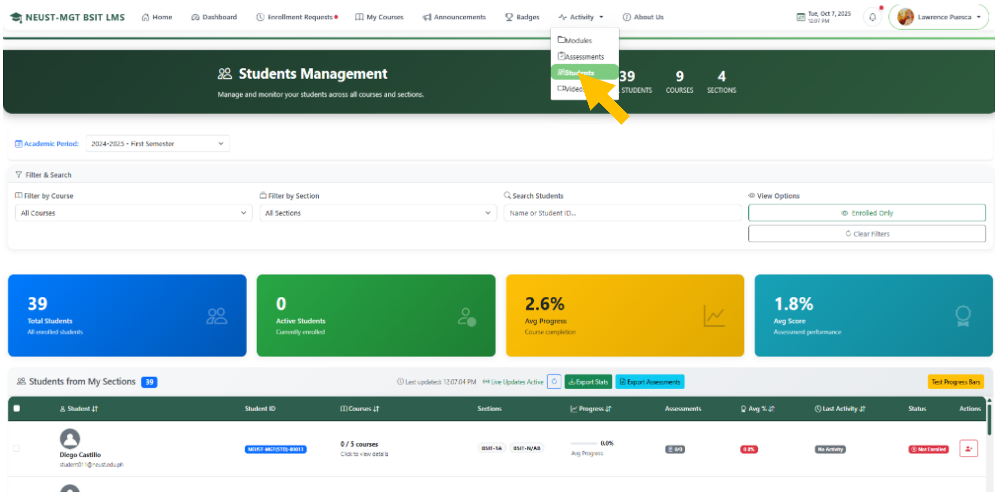
6.3. Navigate to **Activity** menu then **Modules on** this page display list of available modules.



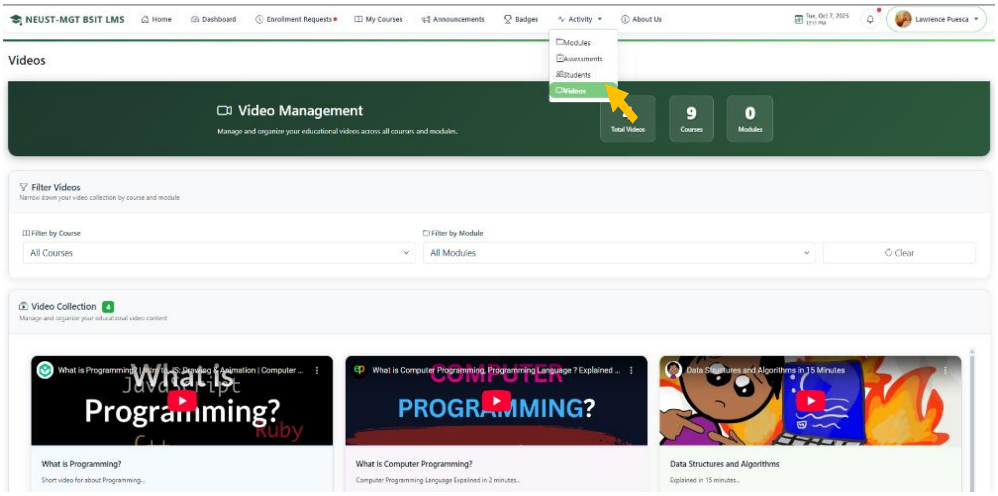
6.3.1. Next navigate to **Activity** menu the **Assessments** this page displayed list of available assessments.



6.3.2. Next navigate to **Activity** menu then **Students** this page displayed list of students.



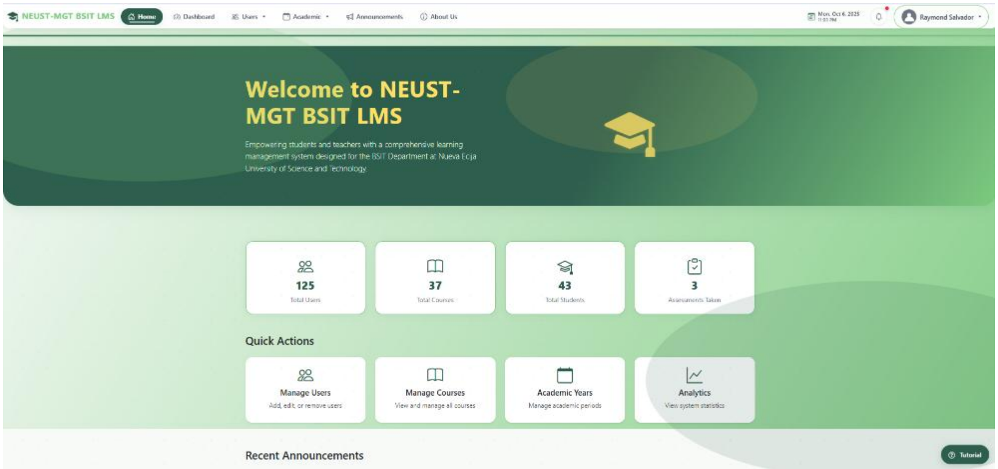
6.4. Next navigate to **Activity** menu then **Videos** this page displayed list of videos.



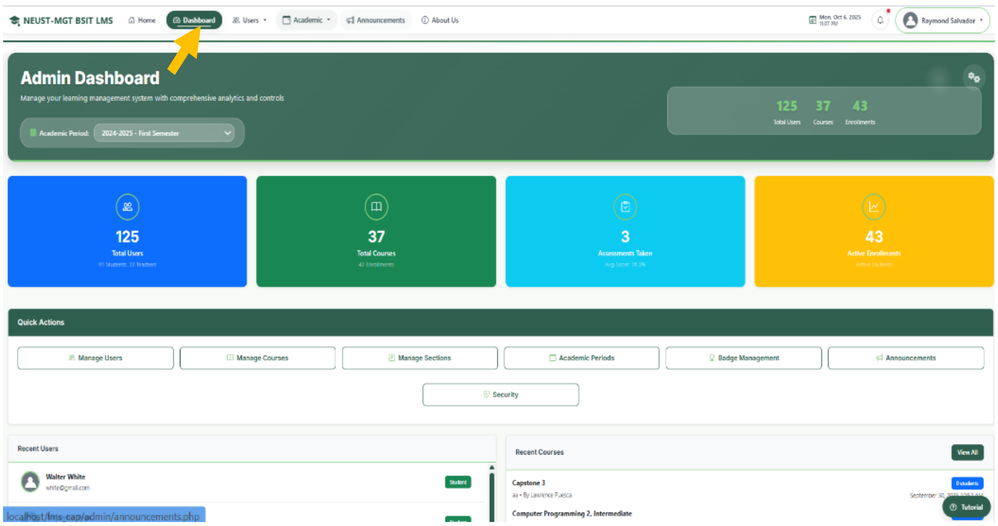
Admin Side

7. Navigating through the system as **admin.**

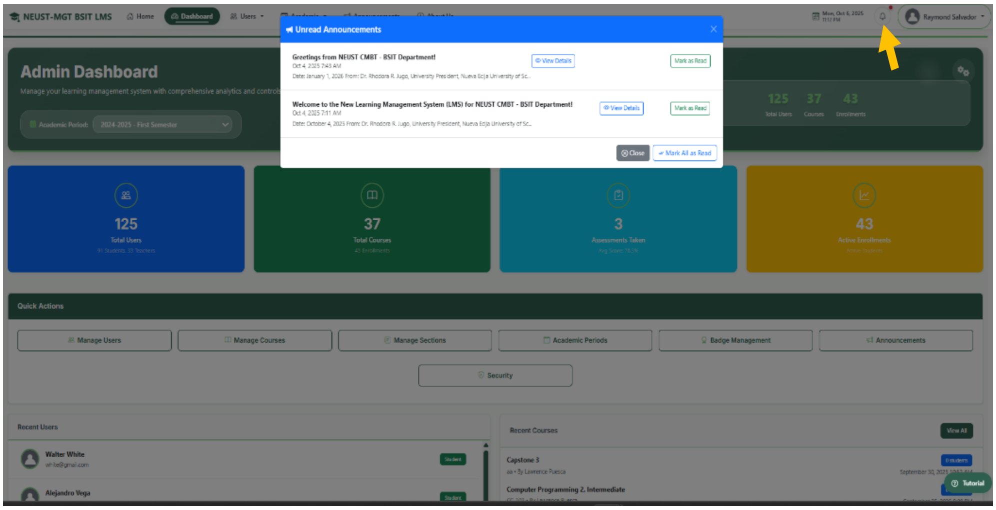
2.1 After logging in, the system redirects to your dashboard.



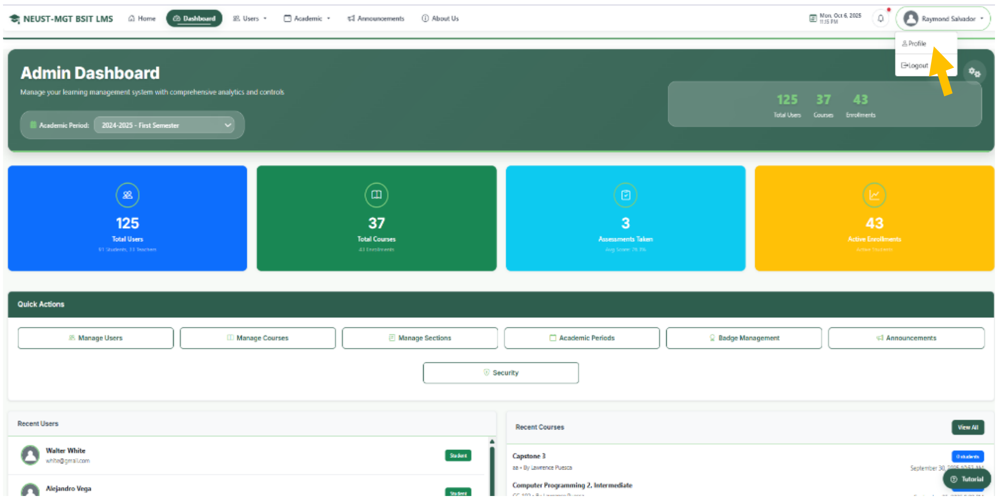
7.1. Use the navbar navigation to access system modules.



7.2 The notification bell at the top shows real-time updates.

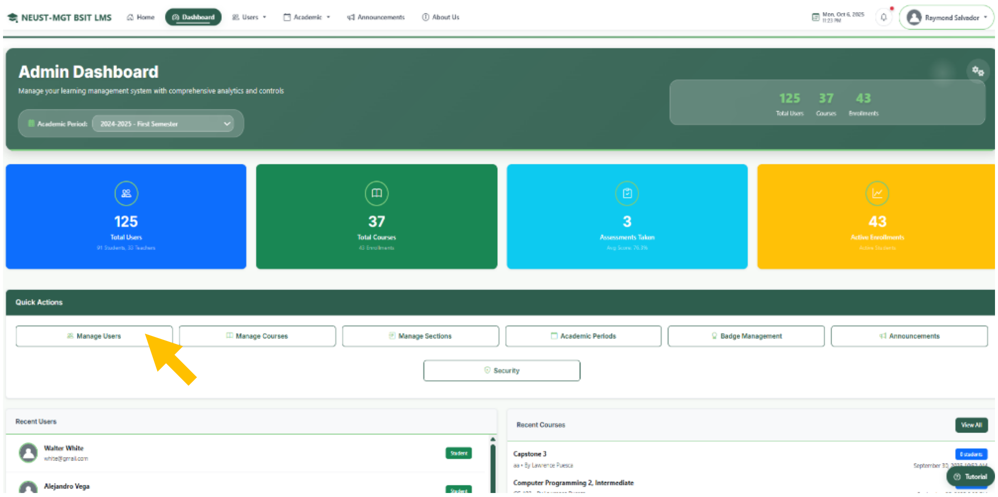


7.3 The profile icon with user name allows you to manage your account settings.

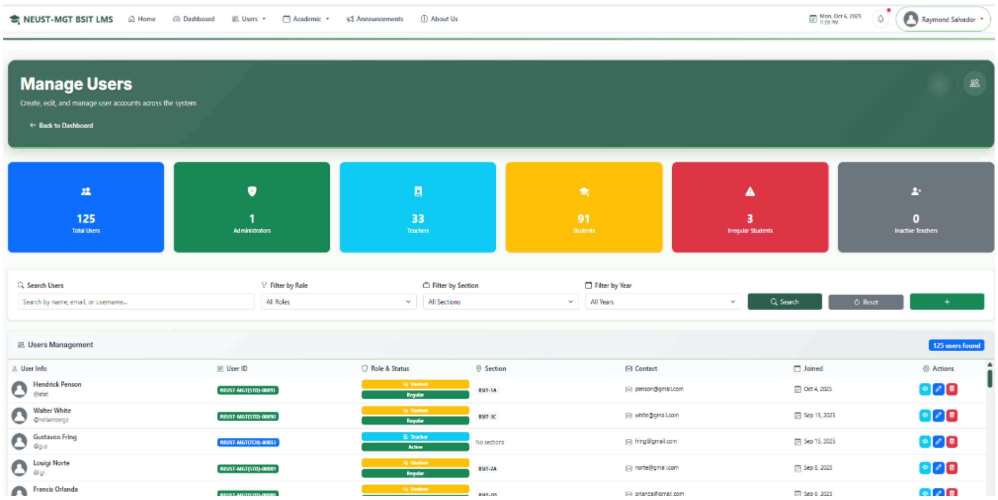


8. Admin Dashboard Functions

8.1 Manage users, click **Manage Users** in the sidebar.

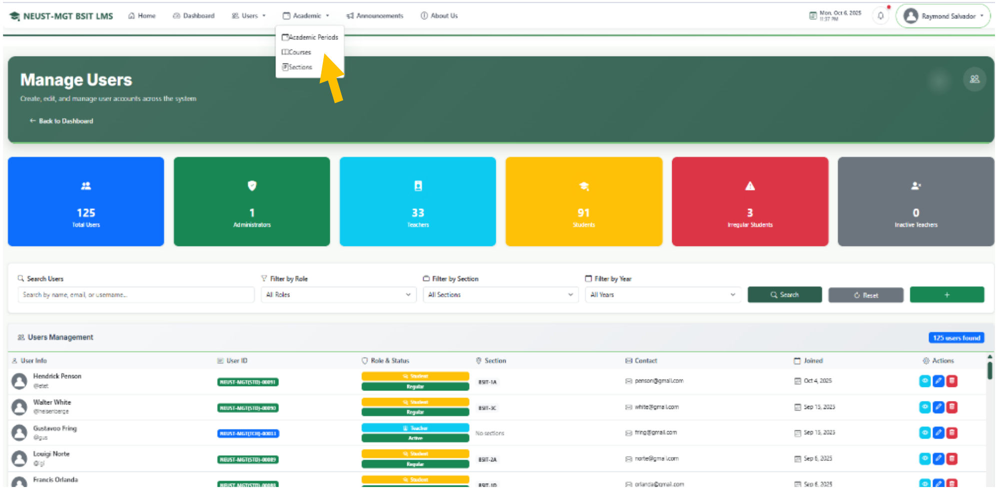


8.2. View, add, edit, or deactivate user accounts.

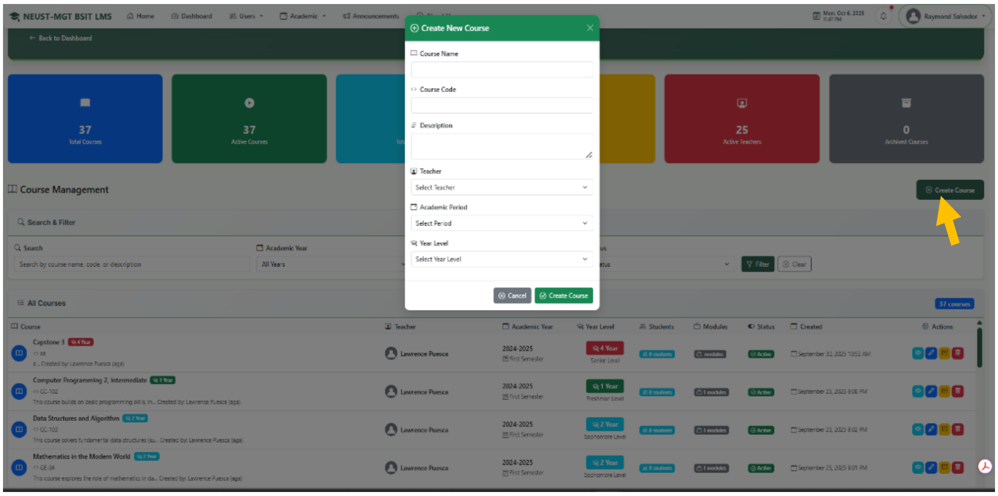


9. Manage Courses

9.1. Click Academic from menu and **Course.**

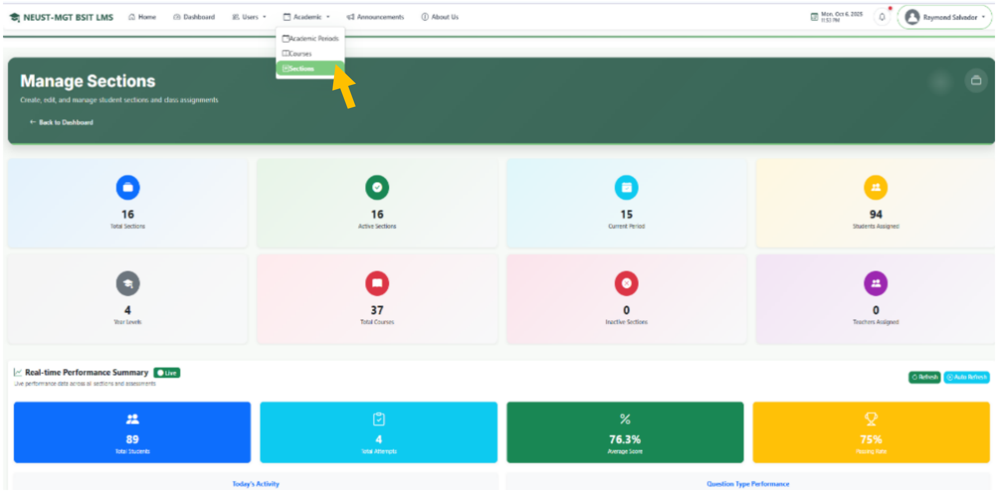


9.2. Click create course then Add new course and assign them to Teachers.

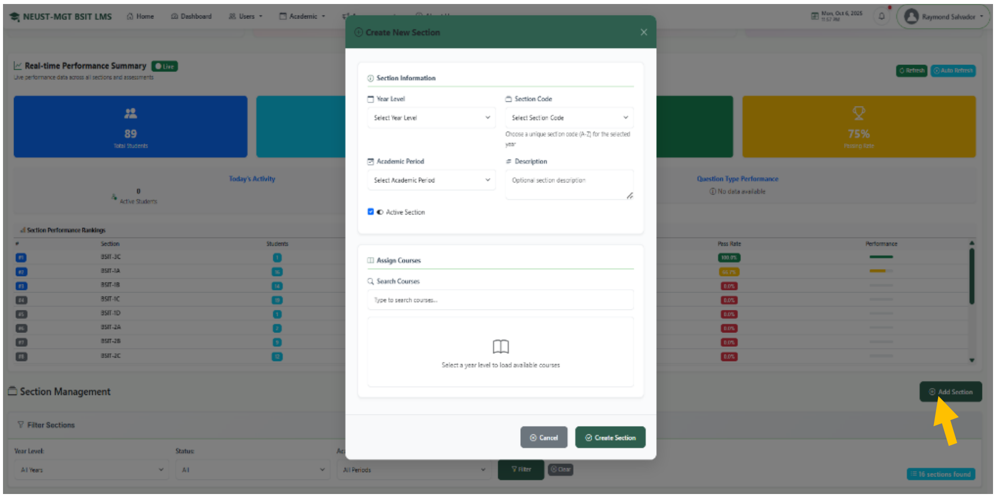


10. Manage Section

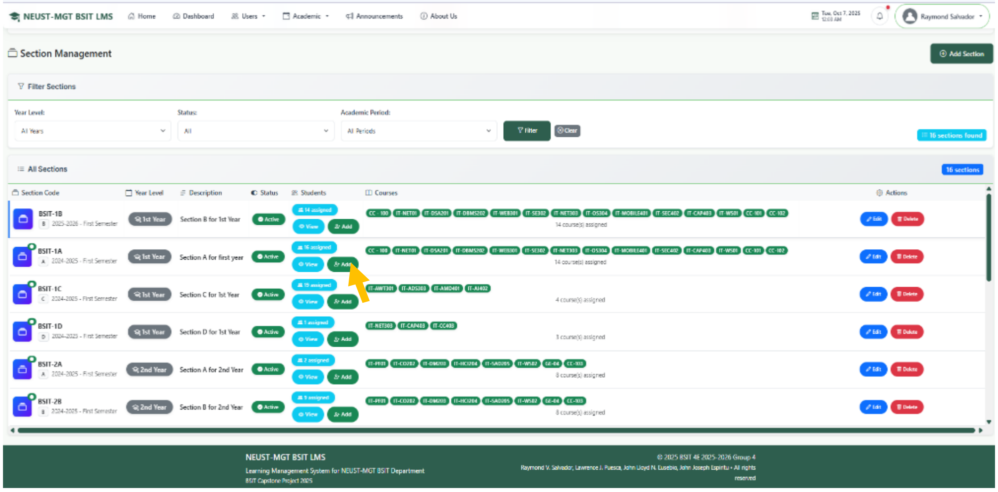
10.1. Click Academic from menu and **Section.**



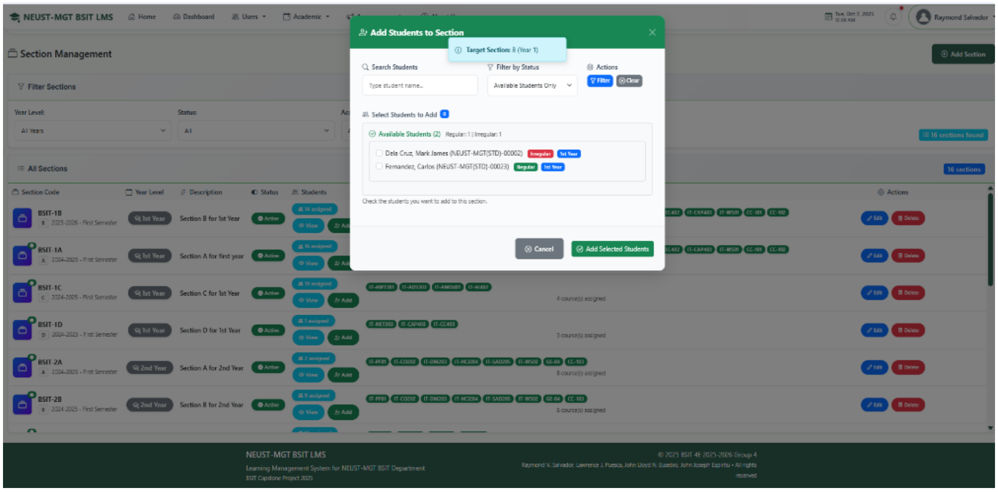
10.2. Click create section then Add new section and assign them to courses.



10.2.1. To add student to each section click add button.

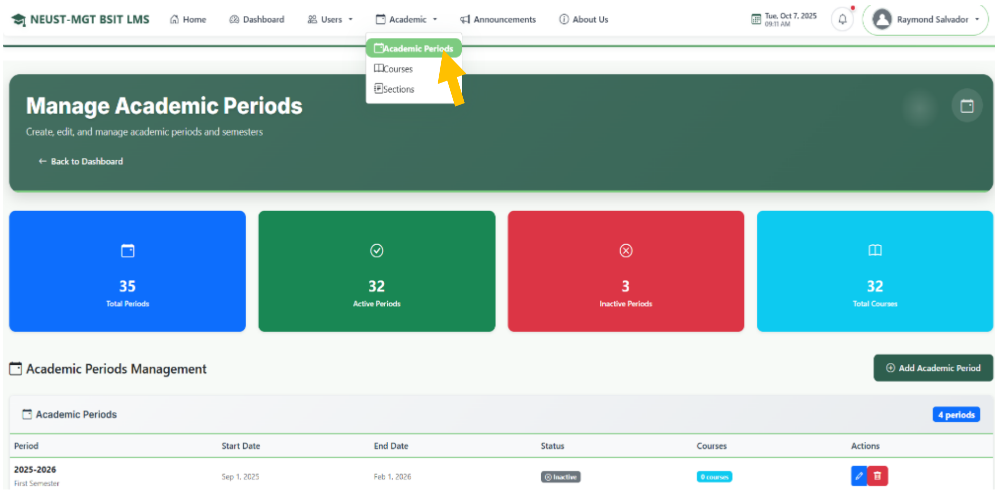


5.2.2. Click checkbox to add student to section.

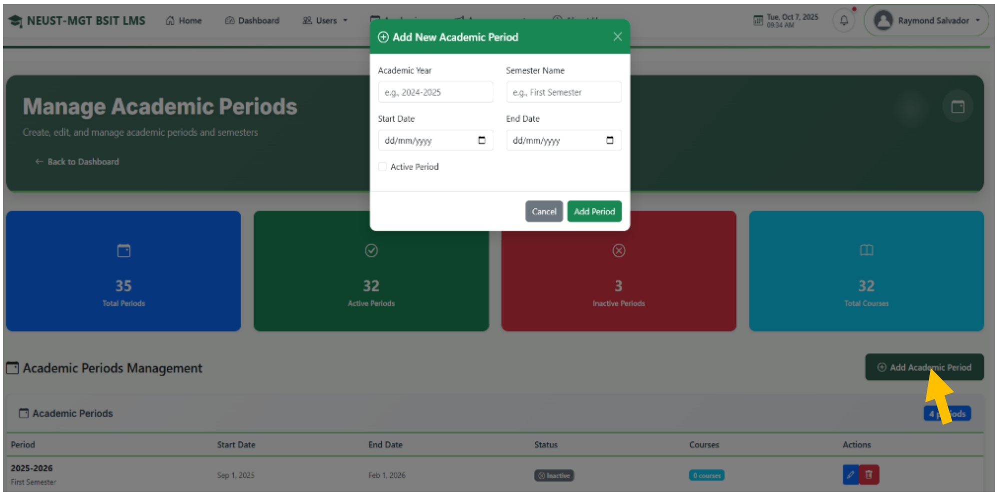


11. Add New Academic Period

11.1. Click Academic from menu and **Academic Periods.**

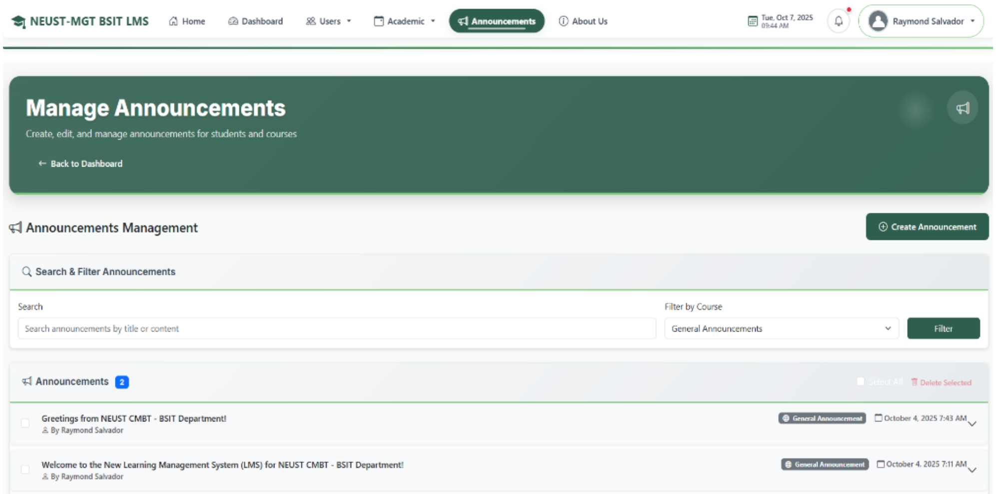


11.2. Fill out the form with information and click **Add Period.**

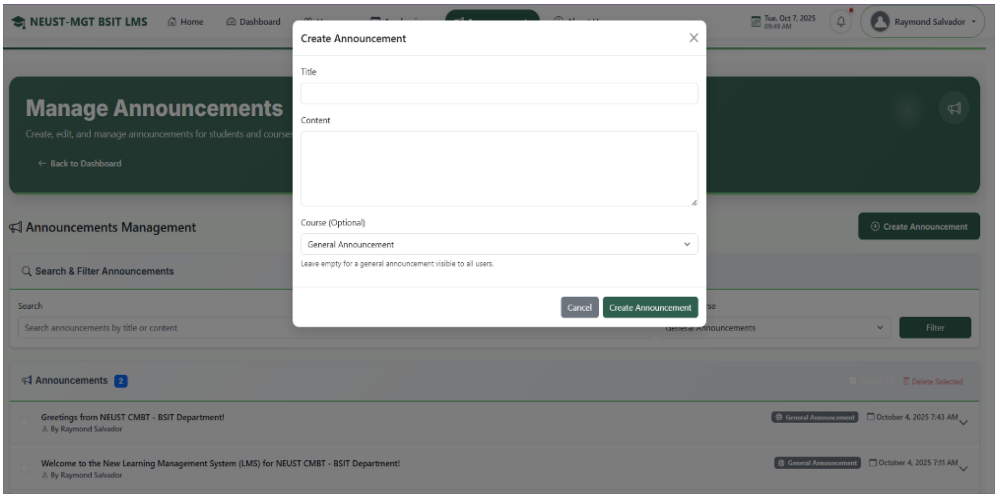


12. Navigate to Announcement page

12.1. Click **Announcements** from menu.

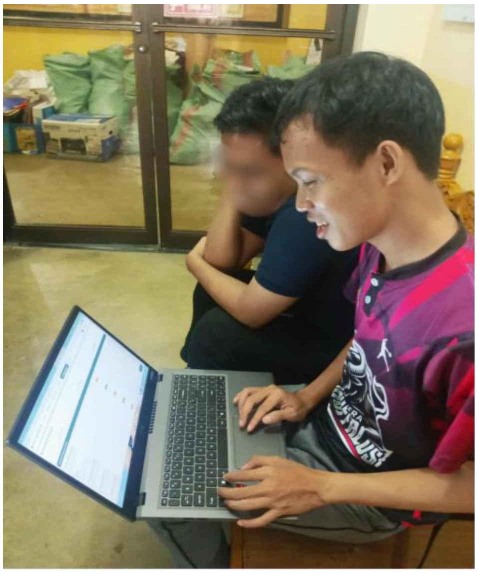


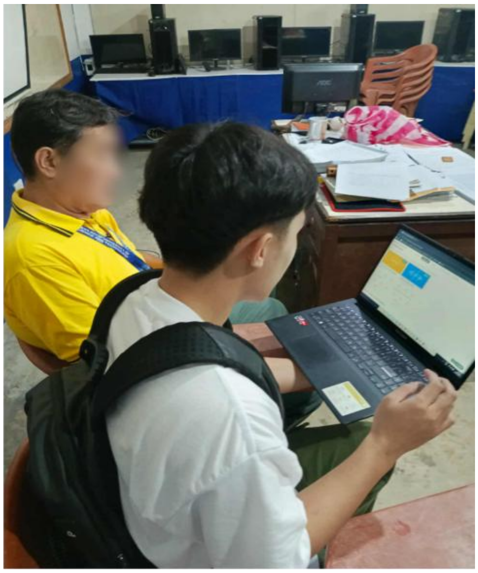
12.2. Fill out to create announcement, then click Create Announcement to publish.



Appendix D

DOCUMENTATION





Appendix E

CURRICULUM VITAE

Curriculum Vitae

I. Personal Information

Name: John Joseph Espiritu

Contact Number: 09927115196

Email: johnjosephespiritu3596@gmail.com

Age: 21

Sex: Male

Address: Pula, Talavera, Nueva Ecija

Birthdate: March 31, 2004

Citizenship: Filipino

II. Educational Attainment

Primary: Pula Elementary School

(Pula, Talavera, Nueva Ecija)

Secondary: San Ricardo National Highschool

(San Ricardo, Talavera, Nueva Ecija)

Course: Bachelor of Science in Information Technology

Major: Web Systems Technology

Curriculum Vitae

I. Personal Information

Name: John Lloyd Eusebio

Contact Number: 09273758896

Email: eusebiojohnlloyd512@gmail.com

Age: 21

Sex: Male

Address: Bagong Sikat, Talavera, Nueva Ecija

Birthdate: November 12, 2003

Citizenship: Filipino

II. Educational Attainment

Primary: Bagong Sikat Elementary School

(Bagong Sikat, Talavera, Nueva Ecija)

Secondary: San Ricardo National Highschool

(San Ricardo, Talavera, Nueva Ecija)

Course: Bachelor of Science in Information Technology

Major: Web Systems Technology

Curriculum Vitae

I. Personal Information

Name: Lawrence Puesca

Contact Number: 09772882516

Email: lawrencepuesca@gmail.com

Age: 21

Sex: Male

Address: San Ricardo, Talavera, Nueva Ecija

Birthdate: January 15, 2004

Citizenship: Filipino

II. Educational Attainment

Primary: San Ricardo Elementary School

(San Ricardo, Talavera, Nueva Ecija)

Secondary: San Ricardo National Highschool

(San Ricardo, Talavera, Nueva Ecija)

Course: Bachelor of Science in Information Technology

Major: Web Systems Technology

Curriculum Vitae

I. Personal Information

Name: Raymond Salvador

Contact Number: 09217214123

Email: raymondvicente74@gmail.com

Age: 23

Sex: Male

Address: Pinagpanaan, Talavera, Nueva Ecija

Birthdate: December 13, 2001

Citizenship: Filipino

II. Educational Attainment

Primary: Talavera South Central School

(Pinagpanaan, Talavera, Nueva Ecija)

Secondary: Talavera National Highschool

(Talavera, Nueva Ecija)

Course: Bachelor of Science in Information Technology

Major: Web Systems Technology